GEK-83768A

Instructions

DS3828 SILTRON™ Power Conversion Module Maintenance Manual

For Process Control Applications

DS3828 SILTRON™
Power Conversion Module
Maintenance Manual
FOR
PROCESS CONTROL APPLICATIONS

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, U.S.A.

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CHAPTER 1
INTRODUCTION

GENERAL INFORMATION

This manual is a guide to the installation, operation, maintenance, and troubleshooting of General Electric's DS3828 SILTRON Power Conversion Module (PCM). This particular manual covers the SILTRON PCM models used in process control applications. Chapter 1 of this manual is an introduction to the manual and the equipment. Chapter 2 is an in-depth description that tells you how the equipment operates. Chapter 3 contains setup and adjustment procedures for the equipment. Chapter 4, in conjunction with another GE publication, ***SILTRON Power Conversion Module Troubleshooting Guide (GEK-83815)***, contains information to help you find and correct problems that might occur. Appendix A contains a list of PCM test points and their values. Appendix B contains removal and replacement procedures for SCR cells.

We assume that a user of this manual has knowledge and experience in electrical and electronics technology as well as prior experience with industrial control systems.

In order to properly maintain SILTRON power conversion equipment, you will need the following additional documentation:

- System elementary diagram
- SILTRON Power Conversion Module Troubleshooting Guide

You will also need the following test equipment:

- Multimeter or digital voltmeter
- Hand tachometer
- Strip chart recorder
- Dual Trace Oscilloscope

EQUIPMENT SUMMARY

General Electric's SILTRON Power Conversion Module uses silicon controlled rectifiers, commonly called SCR's to convert three-phase AC power to adjustable DC power, which in turn is used to power a DC motor. The PCM also provides field excitation for the motor. The PCM can accurately control the speed and torque of the motor using feedback signals from the motor and commands from several possible sources, including an operator's control panel or a master programmable controller. The PCM can provide armature current to several motors at once, but can supply field excitation to only one motor. A Per Motor Module (PMM) is available which can supply field excitation for additional motors. Figure 1-1 is a block diagram of a simple drive system using the SILTRON Power Conversion Module.

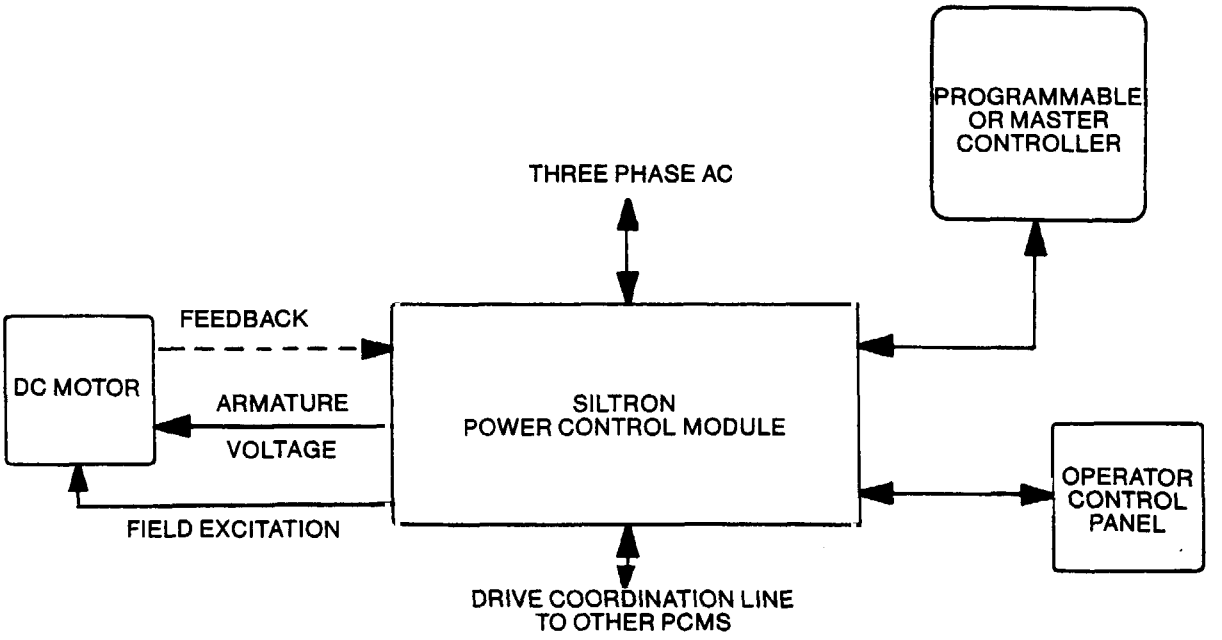


Figure 1-1.
Simple Drive System Using SILTRON Power Conversion Module

The SILTRON Power Conversion Module is available in the following models, which differ mainly in their current-handling capability:

LOW HORSEPOWER ("A" Version - DS3828A)
- Shown in Figure 1-2.
- Handles DC currents to 200A and DC voltages to 500V.
- Powers motors rated up to 125 HP.
- Operates in six-pulse non-reversing mode.

LOW HORSEPOWER ("D" Version - DS3828D)
- Shown in Figure 1-2.
- Handles DC currents to 100A and DC voltages to 500V.
- Powers motors rated up to 60 HP.
- Operates in six-pulse reversing mode.

MEDIUM HORSEPOWER ("B" Version - DS3828B)
- Shown in Figure 1-3.
- Handles DC currents to 500A and DC voltages to 500V.
- Powers motors rated up to 300HP.
- Operates in six-pulse reversing or non-reversing mode.

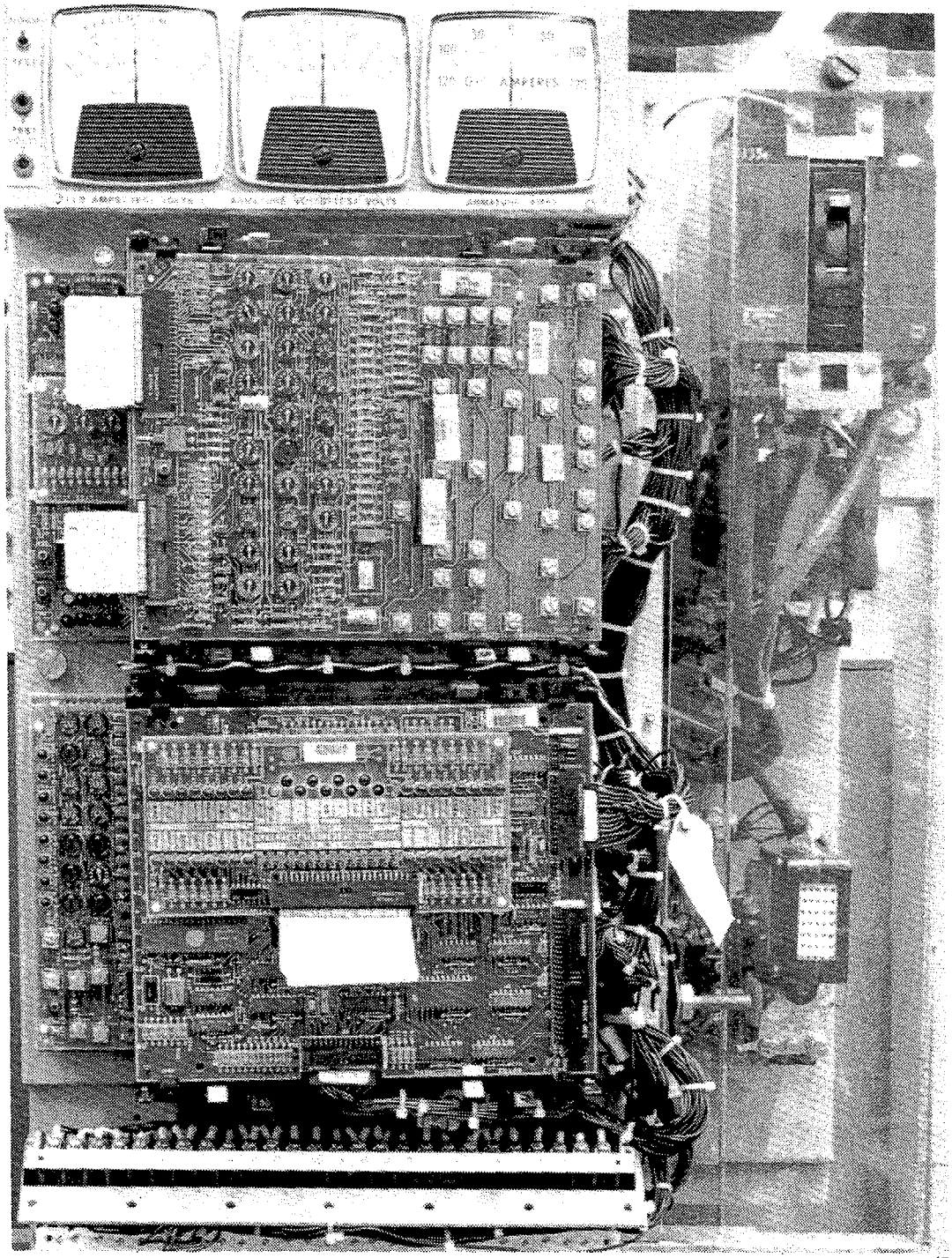
HIGH HORSEPOWER ("C" Version - DS3828C)
- Shown in Figure 1-4.
- Handles DC currents to 1325A and DC voltages to 700V.
- Powers motors rated up to 800HP.
- Operates in six-pulse reversing or non-reversing mode.

SUPERBRIDGE ("C" Version - DS3828C)
- Shown in Figure 1-4.
- Handles DC currents to 2200A and DC voltages to 630V.
- Powers motors rated up to 1500HP.
- Operates in six-pulse non-reversing mode.

The Low Horsepower SILTRON PCM shown in Figure 1-2 is actually a ***drive module***. A drive module consists of a power conversion module plus AC and DC power components such as contactors, power connections, fuses, and so forth.

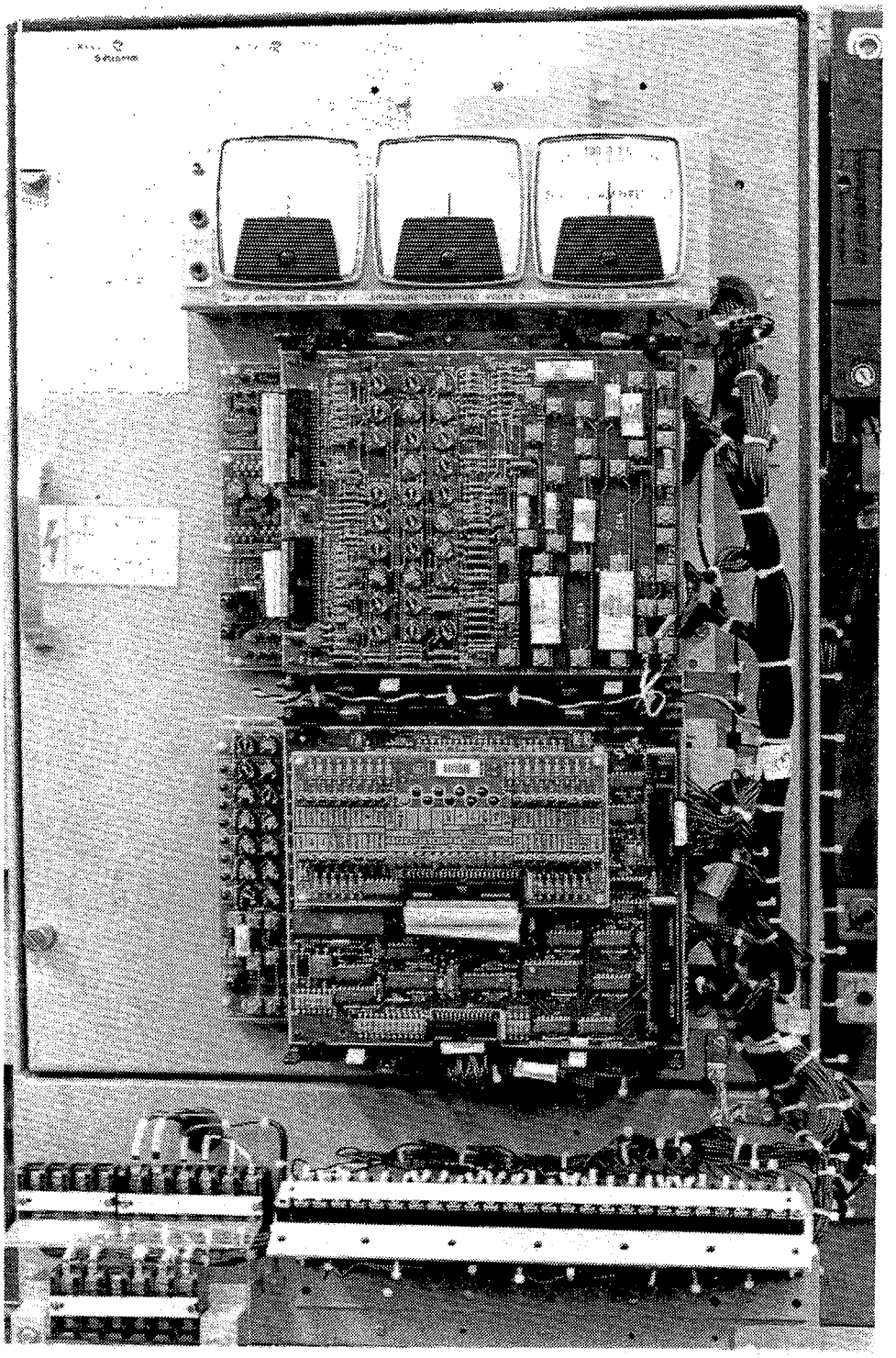
By refering to Figures 1-2, 1-3, and 1-4, you will see that the electronics package of all SILTRON PCM models are similar in appearance. For this reason, most of the photos in this manual will show the Low or Medium Horsepower model. The information will also be applicable to the other models.

SILTRON PCMs are usually mounted in a free-standing panel. Customer power and control connections to the PCM are made via easily-accessible terminal boards.



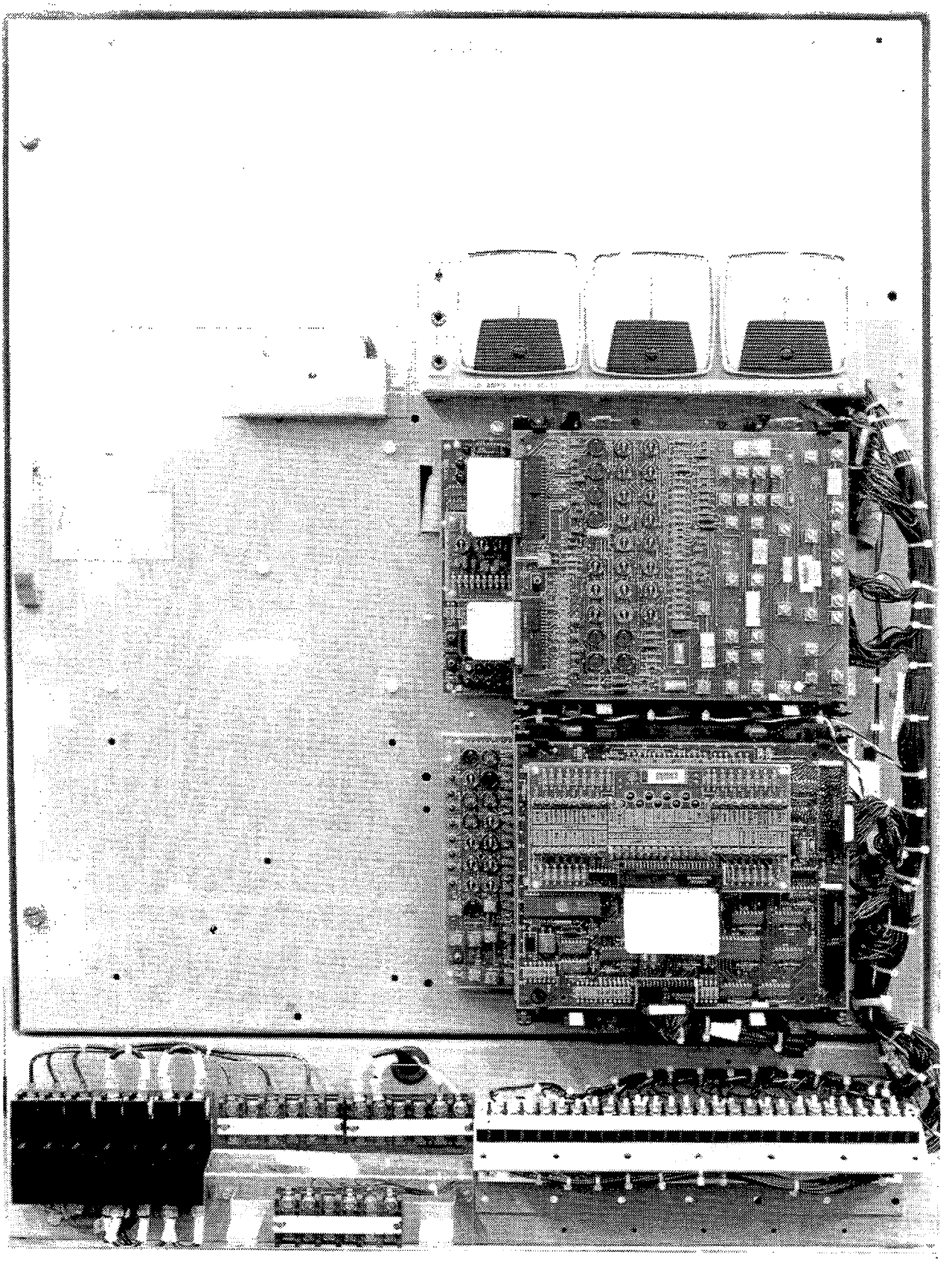
(DSF54844)

Figure 1-2.
SILTRON Low Horsepower Drive Module ("A" or "D" Version)



(DSF54843)

Figure 1-3.
SILTRON Medium Horsepower PCM ("B" Version)
(See Figure 2-8 for Power Components)



(DSF54842)

Figure 1-4.
SILTRON High Horsepower or Superbridge PCM ("C" Version)

EQUIPMENT SPECIFICATIONS

The equipment specifications listed below are of a general nature. For more specific information, see the system specification supplied by General Electric.

Input Power

Voltage: 230 or 460 VAC standard (other input voltages available)
Frequency: 50 or 60 Hz.
Phase: 3

Output Current

Depends on model. See previous section "Equipment Summary".

Cooling

The Low and Medium Horsepower modules can be convection cooled when driving low horsepower motors, but an optional 115 VAC bridge fan is necessary for higher horsepower motors. The High Horsepower and Superbridge modules usually require a 14 or 16 inch panel-mounted fan, such as that shown in Figure 1-5. Whether or not the PCM requires a fan, air louvers at the top and bottom of the enclosure door must remain unblocked to ensure adequate air circulation.

Operating Environment

Ambient Temperature: 0°C to +40°C (+32°F to +104°F)
+50°C (+122°F) maximum temperature with de-rated bridge
Relative Humidity: 5% to 95% (non-condensing)
Altitude: sea level to 1,000 meters (3,300 feet)

When a module has been in operation and is to be shut down, we recommend that you keep environmental conditions the same as during operation. Power supplies, ventilation, and heating or air conditioning (if used) should be left on during the period that the module is down. This will prevent extreme temperature variations that could cause damaging condensation.

Storage Environment

Ambient Temperature: -20°C to +70°C (-4°F to +158°F)
Relative Humidity: 0% to 95% (non-condensing)
Altitude: sea level to 1,000 meters (3,300 feet)

RECEIVING, STORING, AND INSTALLING THE EQUIPMENT

As mentioned, the SILTRON Power Conversion Module is usually mounted in a free-standing panel. You will receive the equipment in a packing case or crate. The packing case is not suitable for storing the equipment outdoors. If the equipment is not to be installed immediately, store it under cover in a clean, dry location away from areas where construction is in progress. Ensure that the storage environment meets the specifications listed under "Equipment Specifications." If the equipment must be stored in an unusually damp area, cover the equipment and provide a heat source to prevent condensation. Should the equipment be exposed to moisture, it should be checked by General Electric service personnel before being put into service.

Examine each shipment carefully upon arrival and report any shortage or damage to the carrier and to the nearest General Electric sales office.

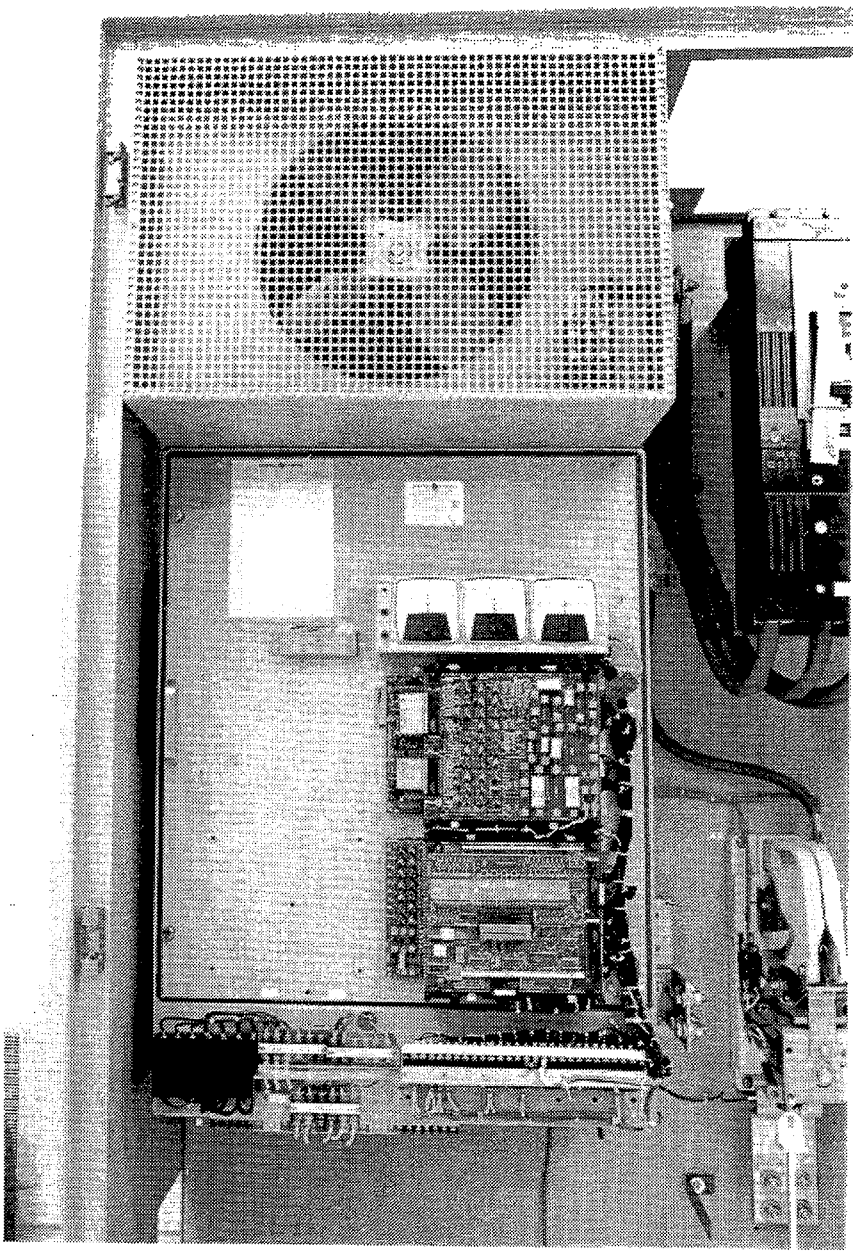


Figure 1-5.
Panel-Mounted Fan for Cooling SILTRON PCM

Be especially careful when lifting or moving the equipment. Refer to the equipment outline drawings for preferred methods of lifting and moving. Before any installation work is done, consult all drawings furnished by General Electric as well as all applicable contract drawings for your particular installation. Pay particular attention to the outline drawings as you plan, install, or modify conduits and busways.

WARNING

Correct grounding of the equipment is essential for safe and proper operation. Connect each panel to a suitable earth ground, such as the structural steel support of your building. Ensure a good mechanical and electrical connection at the panel and the ground source. Be sure to follow the system grounding methods described in the system drawings.

PREVENTIVE MAINTENANCE

Periodic preventive maintenance can add to the operating life of your equipment and prevent down-time. Preventive maintenance for the SILTRON Power Conversion Module involves periodic inspection and cleaning of the unit. Follow these guidelines during preventive maintenance:

1. ***Disconnect all AC power before working on the module.***

2. Clean the exterior and interior of the module using a vacuum cleaner. We do ***not*** suggest using compressed air for this purpose.

3. Visually check all boards, wires, and components for burnt or damaged areas, loose connections, connectors not properly seated, or other signs of potential problems.

4. Check the electrical contacts on the contactors. Discoloration and rough contact surfaces are normal. If beads form due to severe arcing, dress the contact faces with a fine file. ***Never use sandpaper or emery cloth and never oil any part of the power unit.***

5. Do not tamper with adjustment settings unless the module is not working properly and you suspect a faulty adjustment.

6. Printed wiring boards, if extremely dirty, can be cleaned in a lukewarm (100° F) solution of mild household detergent or a laboratory detergent like ALCONOX or NEUTRA-CLEAN. Use a soft-bristled brush such as a paint brush to scrub the board. After cleaning, rinse thoroughly in several baths of lukewarm, fresh distilled water. Shake off excess water, then air dry for a few hours.

SPARE PARTS

General Electric suggests that you maintain a stock of spare parts for your SILTRON PCM. Stocking spare parts will ensure rapid replacement of malfunctioning parts and reduce potential downtime. The Renewal Parts Quotation Booklet, which was part of the documentation shipped with your equipment, contains a list of suggested spare parts.

There are two classes of spare parts: start-up spares and running spares. Start-up spares are critical parts that should be available during installation and tune-up. Running spares should be kept on hand to replace parts that may malfunction during normal operation.

Many parts are interchangeable between the four SILTRON PCM models. For instance, most of the printed circuit boards are identical for all models. Although the SCR cell ratings differ from one model to another, a high-rated SCR cell can replace a lower-rated cell if proper parts are not available, if the snubber circuits are the same, and if the cells are physically similar (i.e., both cells are either stud-type or disk-type and are the same size).

If there are several SILTRON PCMs at one location, it is not necessary to purchase a set of spare parts for each. One or two complete sets of spare parts will adequately cover the replacement part needs of several PCMs.

After you have used a spare part, it is important to replace that part in your stock of spares immediately. This may be done in one of three ways:

- During the warranty period, replace defective parts with General Electric warranty parts.

- After the warranty period, use the General Electric Repair-and-Return service.

- After the warranty period, many parts can be replaced through General Electric's Element Exchange Service. Parts covered by the Element Exchange Service include most printed wiring boards and certain other assemblies. Under this plan, a defective part is returned to General Electric and exchanged for a good part for a percentage of its purchase price.

Expedited delivery of parts is available in emergency situations. Contact your local General Electric sales office for more information.

CHAPTER 2
EQUIPMENT DESCRIPTION

This chapter provides an introduction, block diagram, and theory of operation for the SILTRON Power Conversion Module. It includes a functional description of each major subassembly, including all boards. Since the maintenance philosophy for this product involves troubleshooting to the board level only, a component-level theory of operation is not included.

EQUIPMENT OVERVIEW

The SILTRON PCM is a microprocessor-based SCR system capable of driving one or more DC motors. The PCM will typically be found either in a stand-alone motor control configuration (Figure 2-1) or in a centrally-controlled drive system (Figure 2-2). The SILTRON PCM operates basically the same in either case.

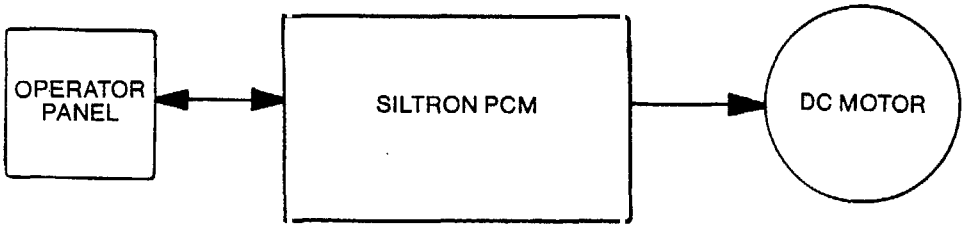


Figure 2-1.
SILTRON PCM in Stand-Alone Configuration

The PCM converts three-phase AC power into adjustable DC power by using SCRs as "adjustable rectifiers". The SCRs control the DC voltage supplied to the motor. An SCR is a solid state device similar to a diode, but with an extra "gate" lead. The SCR will not conduct until a positive pulse or firing signal is applied to the gate. At this point, the SCR behaves like a typical rectifier: it conducts in one direction only. The firing signal needs to be only a momentary pulse; once fired, the SCR latches into a conductive state until the current through the device drops to zero.

As an example, consider the circuit in Figure 2-3. An AC sine wave is applied to the anode of the SCR. The SCR will not conduct until a positive pulse is applied to the SCR's gate. Once the pulse is applied, the SCR will conduct during the positive half-cycle of the AC wave, but will then cut off when the anode goes negative. The SCR remains cut off until the AC voltage swings positive and another firing pulse is applied to the gate. The DC output current and voltage is controlled by varying the point at which the SCR is fired. If the SCR is fired at the very beginning of the AC half-cycle, more current and voltage will be available to the load than if the SCR is fired when the AC wave has reached its positive peak.

The firing circuits in the PCM provide pulses (in the form of pulse trains) to the gate circuits of the SCRs to fire each of them at the correct phase angle. A microprocessor on the Firing Circuit Board determines the correct time to fire the SCRs, based on signals from the Regulator Board, the High Voltage boards, and the cell voltage sensing circuits on the Firing Circuit board. A change in desired speed or motor load will be reflected through the Regulator Board to the Firing Circuit Board and cause a change in the phase angle at which the SCRs are fired. Changing the firing phase angles changes the current and voltage supplied to the motor.

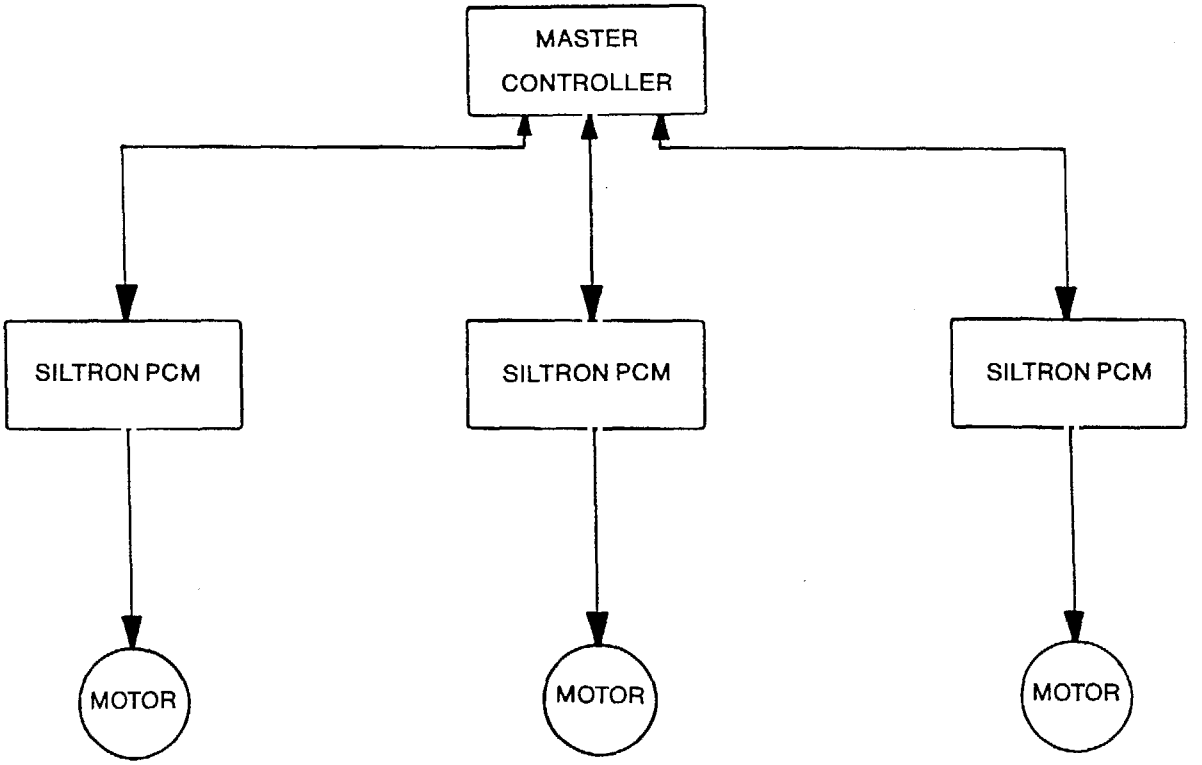


Figure 2-2.
SILTRON PCM in Centrally-Controlled Drive System

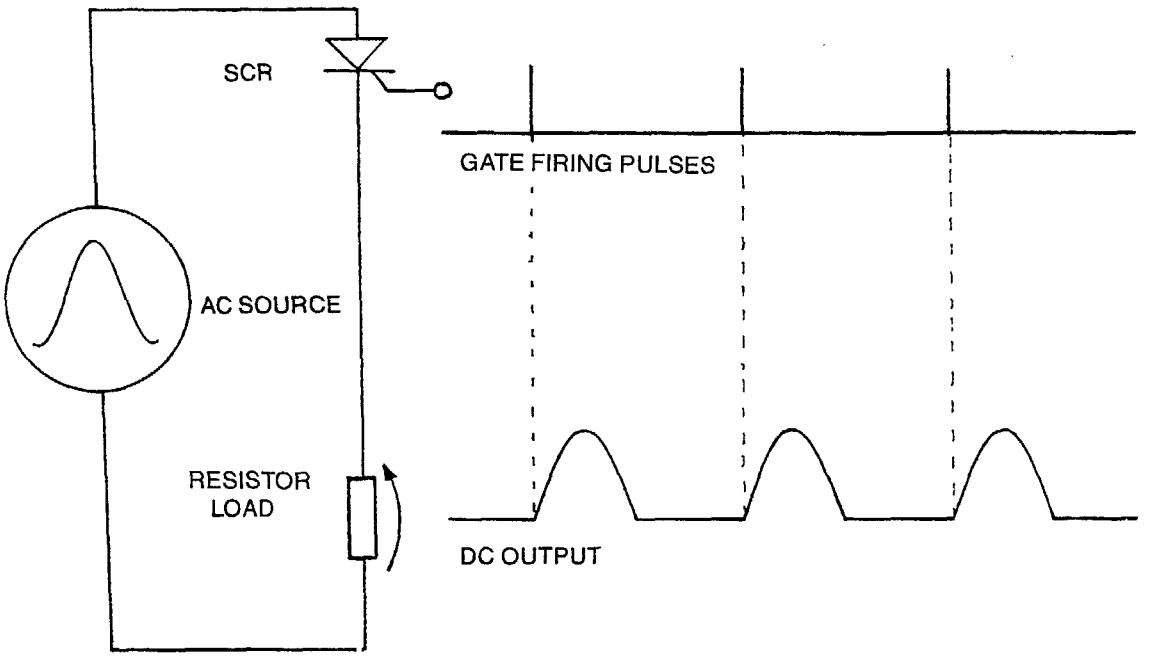


Figure 2-3.
SCR Operation

BLOCK DIAGRAM

Figure 2-4 is a block diagram of the PCM. Refer to this figure during the discussion below.

When the PCM is first powered up, the Sequencer Board goes through a self-test to ensure there are no faults within the Sequencer. This self-test takes about thirty seconds. When the Sequencer board yellow IMOK LED comes on, the PCM is ready to accept commands and drive the motor.

The PCM is controlled by commands sent from one of four possible sources: an operator station, a master controller, a Control Party Line, or a Series Six programmable controller. (A Control Party Line is a data transmission system used for high speed communication between programmable controllers, power conversion modules, operator stations, and other drive system units.) The commands sent from these sources include motor functions such as Jog Forward, Jog Reverse, Emergency Stop, Run, and so forth. In addition, PCM status conditions are sent to the operator station, master controller, Control Party Line, or Series Six controller. Examples of such status conditions are Running, Fault, Ready to Run, and so forth. Incoming commands and outgoing status conditions are buffered by an HIOA, HIOB, or HIOH Input/Output board. The HIOA is used to communicate with an operator station or master controller, the HIOB communicates with the Control Party Line, and the HIOH communicates with the Series Six programmable controller.

Incoming commands are routed to the HSQD Sequencer board. The Sequencer board is responsible for drive sequencing, fault processing, and regulator control. The Sequencer board uses a microprocessor and PROM-based software to interpret the incoming command and control the PCM and motor accordingly. Additional sequencing or logic capacity can be tied into the PCM through an expansion port (via the JC and JD cables) on the HSQD board. In addition, Sequencer boards from several PCMs can be connected to each other through a Drive Coordination Line. The Drive Coordination Line allows several drives to operate together in parallel; if one drive develops a fault, the other drives are notified and all can take coordinated action.

The NGRC Regulator Board takes command signals from the Sequencer board, external references from outside of the PCM, and feedback signals from the motor and determines the proper method of driving the motor. The Regulator Board handles speed, voltage, and current regulation and sends appropriate signals to the Firing Circuit board and Field Exciter Board. An optional HPLA Digital Regulator Board can be interfaced to the Regulator board to process information for digital regulation. In addition, an expansion port on the Regulator board is available (via the JA cable) for adding extra regulating capacity.

The NMEC Field Exciter board controls motor field excitation, field reversing, and motor protection. The field excitation signals are amplified by the NEPB Field Exciter Power board, which powers the motor's shunt field. An expansion port is also available on this board (via the JA cable).

The NFCF Firing Circuit Board controls the firing of the SCR cells and monitors SCR cell voltages and faults. The NFCF contains a microprocessor which generates the SCR firing pulses. The microprocessor is also capable of running a diagnostic routine that checks out the SCR circuitry for faults. The SCR firing pulses from the Firing Circuit board are isolated by the NHV— High Voltage board and then applied to the SCR bridge. (The term NHV— refers to a number of different High Voltage boards used in different SILTRON PCM models; refer to the Firing Circuit board description in the Theory of Operation section for more information.) The output of the SCR bridge is a controlled DC voltage to the motor's armature.

The Control Power Transformer, NPSE Control Power Supply board, and heat sink assembly produce the AC and DC voltages required by the PCM. The DC output includes both regulated and unregulated voltages. In addition, the NPSE board contains a ground point for common signals ACOM, DCOM, PCOM, and safety ground. (Note that for PCMs communicating with a Series Six controller, an NPSS board is used rather than an NPSE.)

The optional Test Meters operate in two switch-selectable modes. In Normal mode, the meters monitor the motor's field current, armature voltage, and armature current. In Test Mode, two of the meters can be used as test voltmeters during adjustments or maintenance of the PCM.

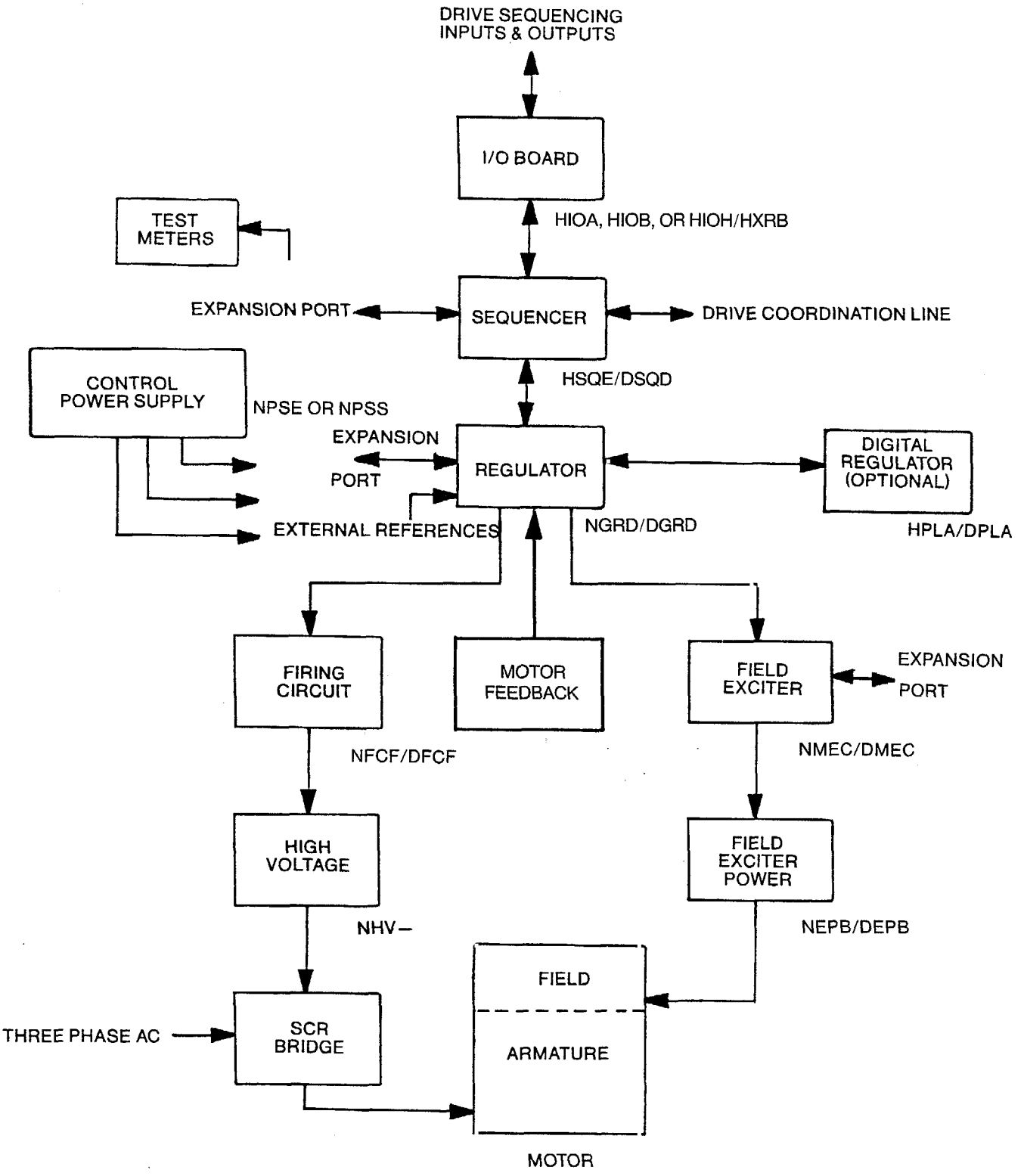
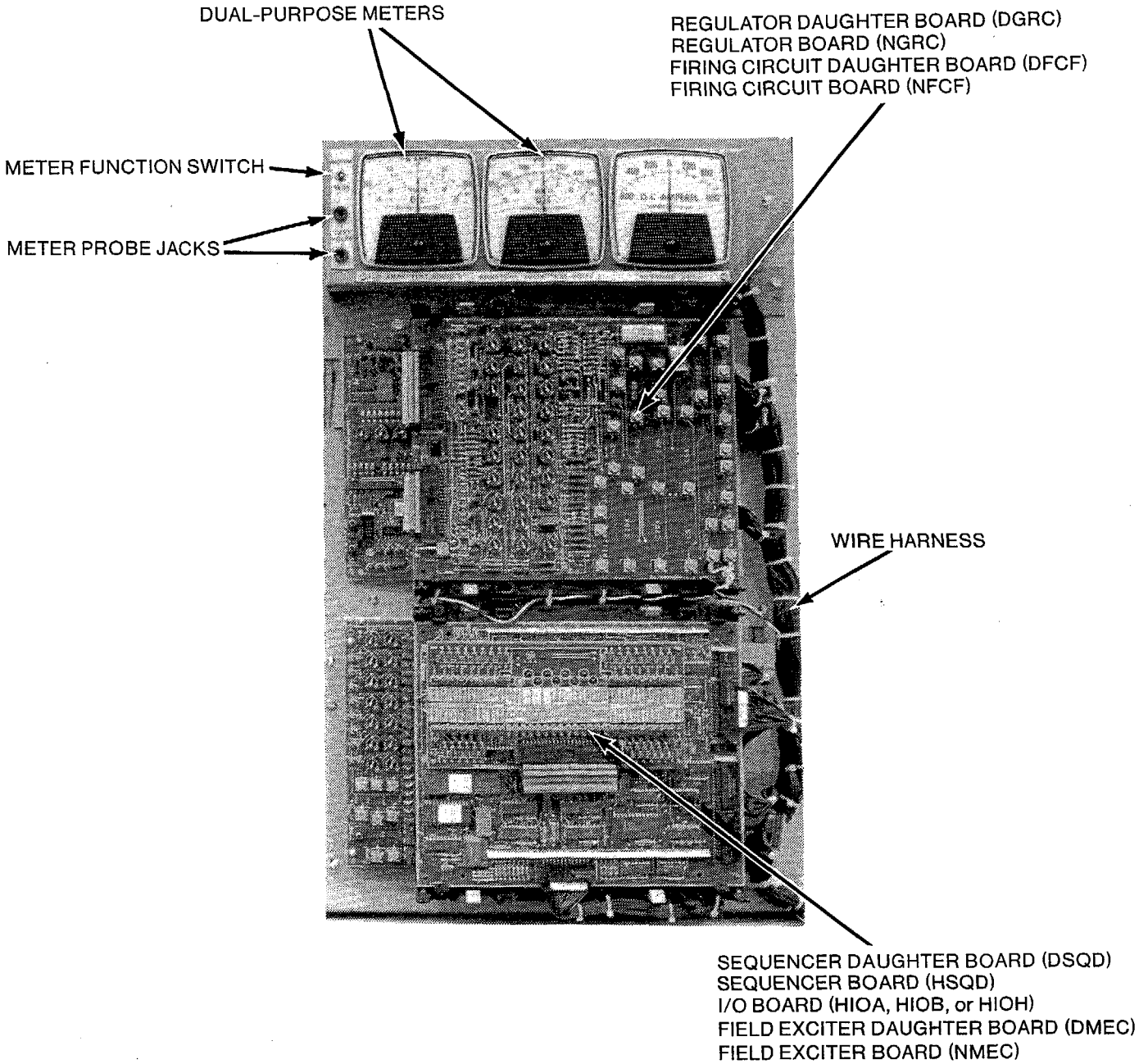


Figure 2-4.
SILTRON PCM Block Diagram

LOCATION OF SUBASSEMBLIES

This section contains photographs that will help you locate and identify all major SILTRON PCM subassemblies. The majority of photographs are of the Low Horsepower model; the Medium and High Horsepower and Superbridge models are similar in appearance and subassembly locations. We have included detailed photographs of printed circuit boards only if those boards contain adjustment potentiometers, indicators, pushbuttons, fuses, or other maintenance-oriented features.



(DSF52970)

Figure 2-5.
Siltron Exterior Electronics Package
(Standard for all Models)

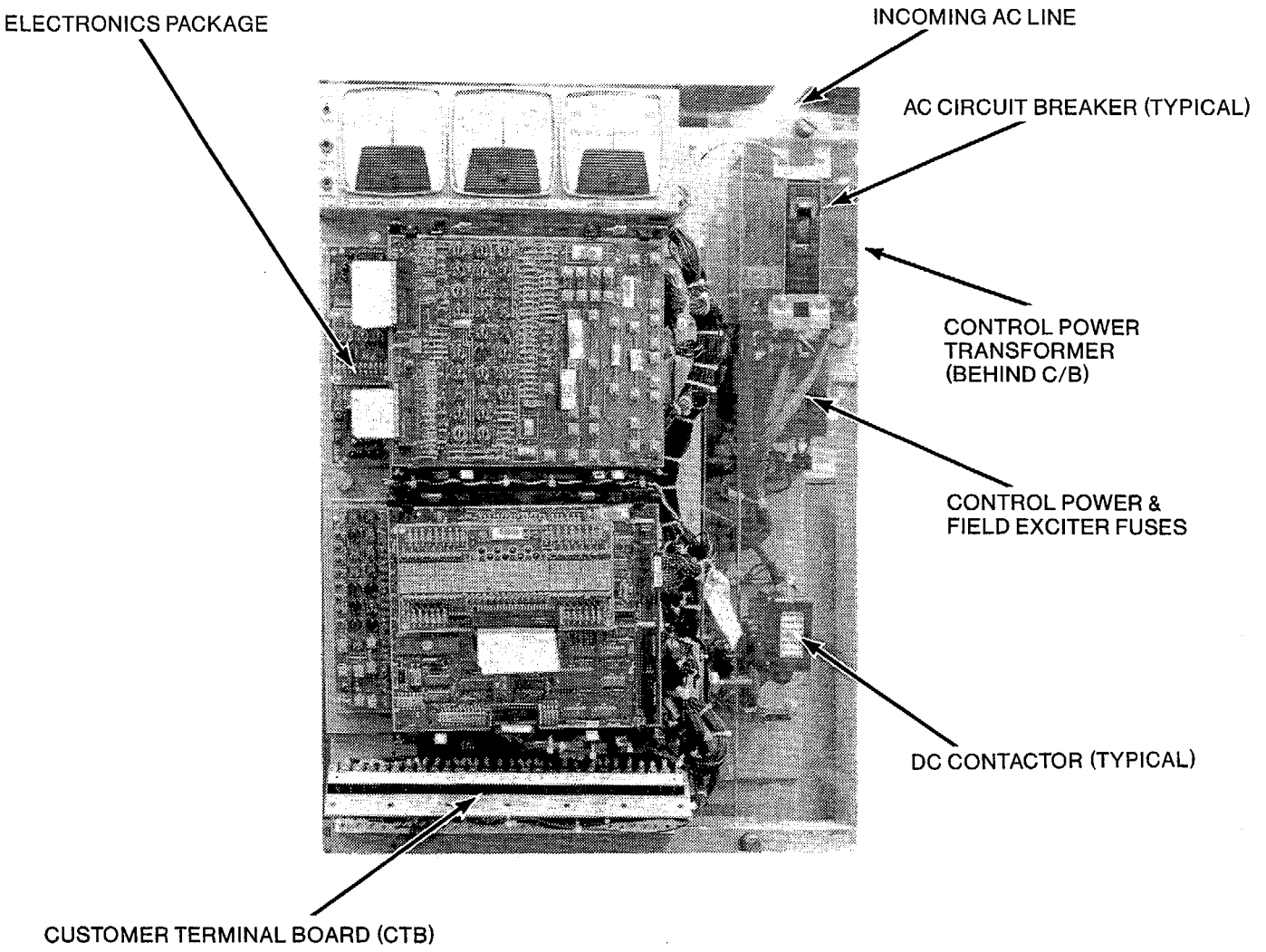
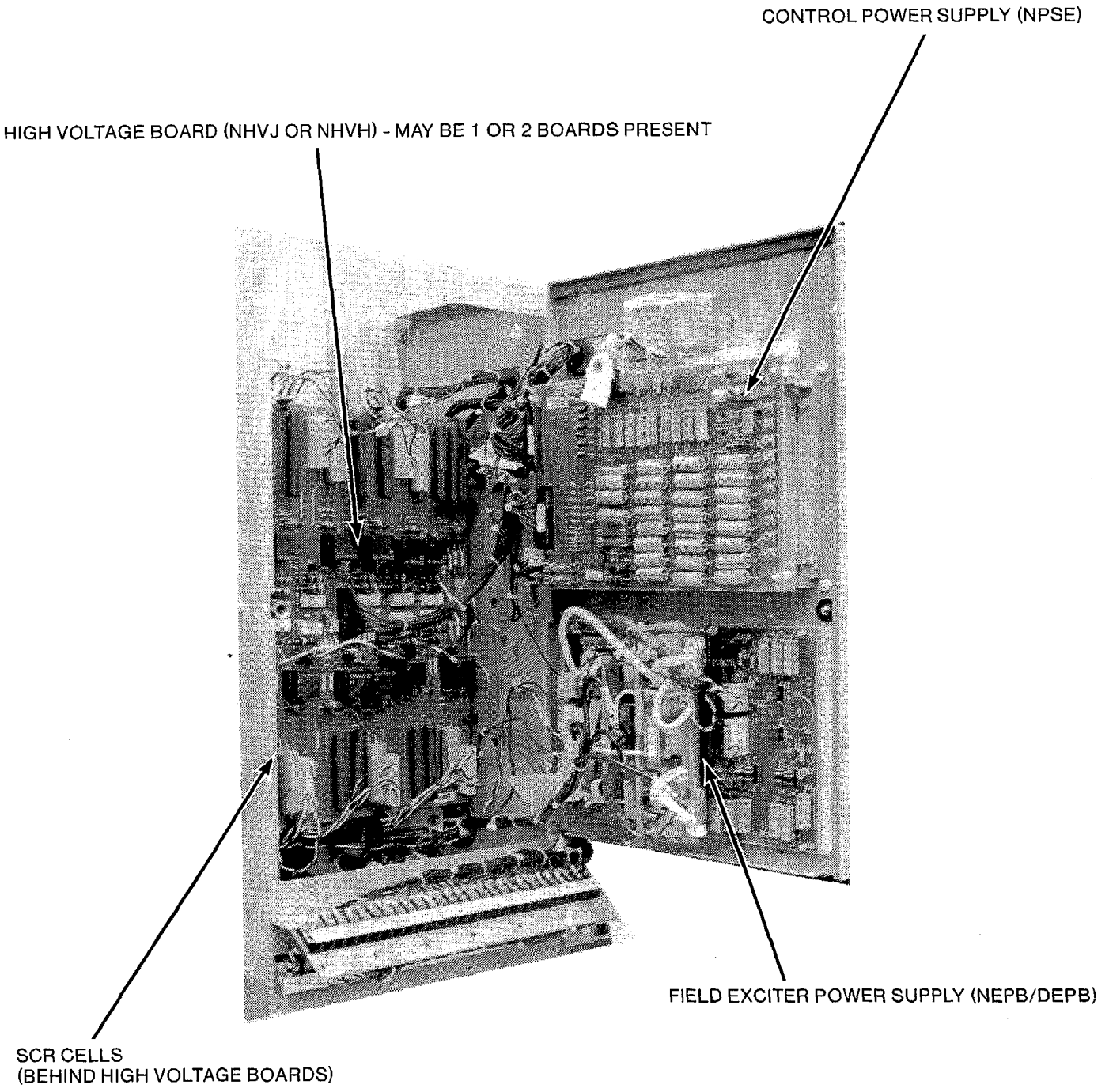


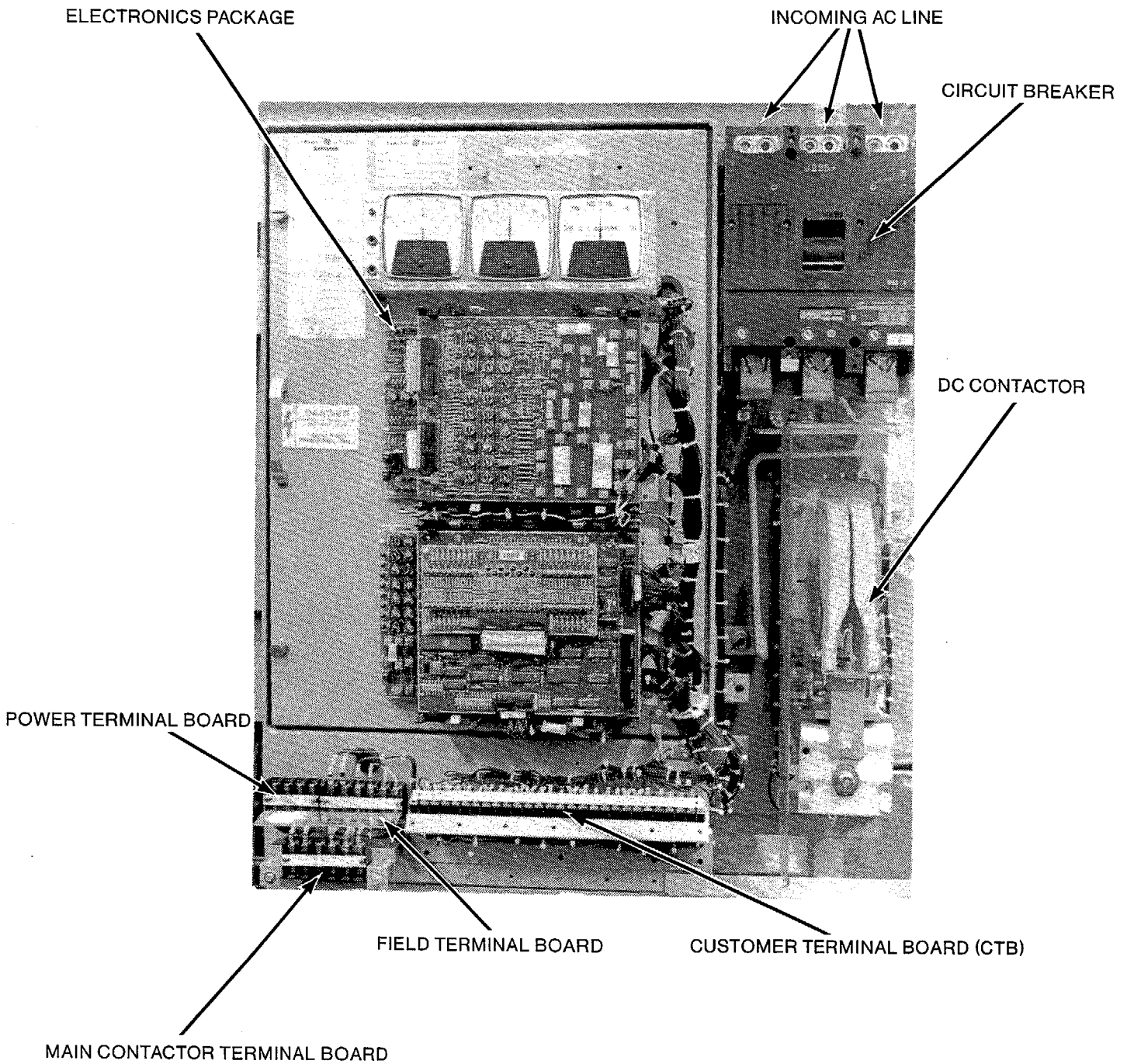
Figure 2-6.
Low Horsepower Model - Exterior View



(DSF54845)

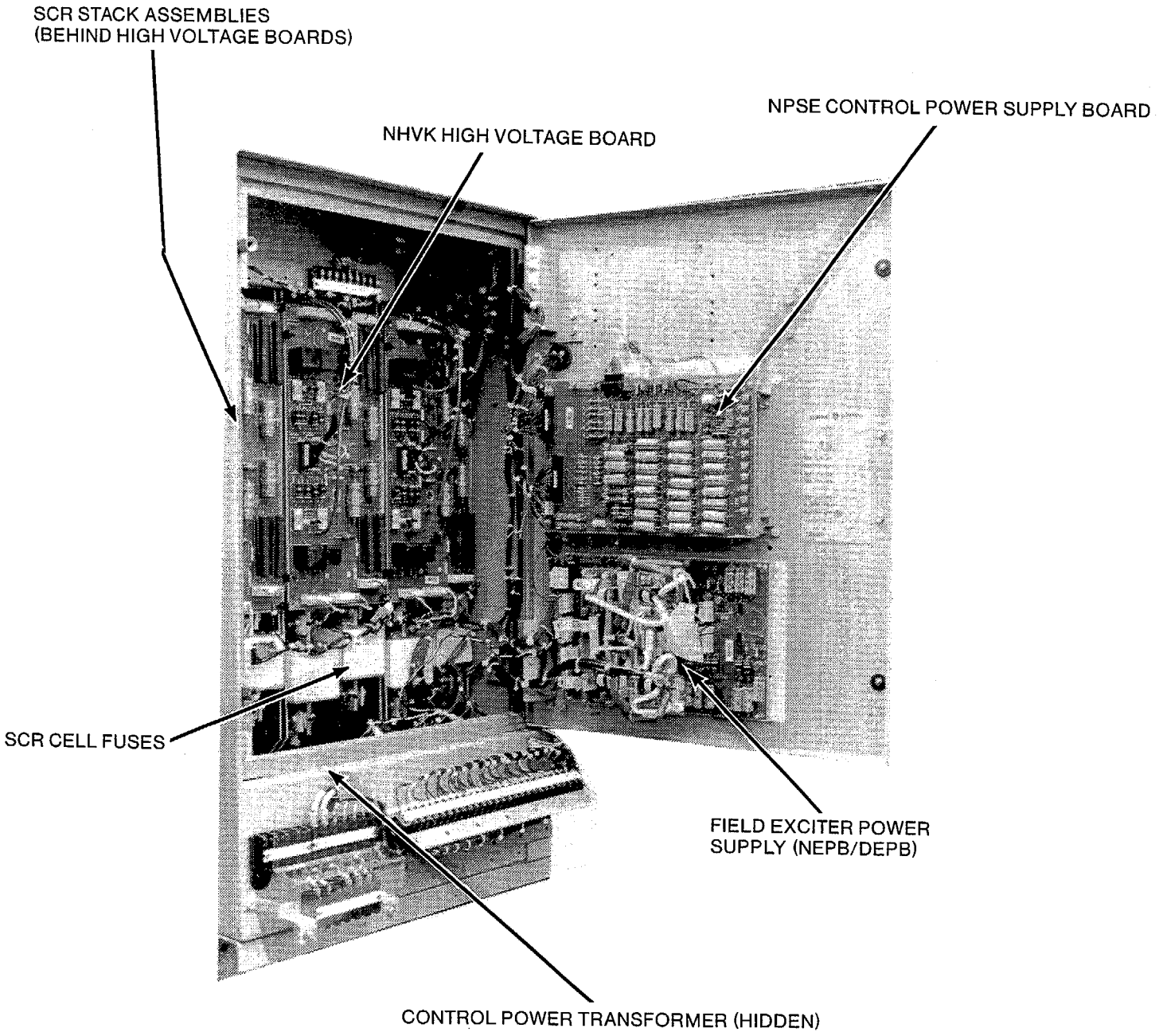
Figure 2-7.
Low Horsepower Model - Interior View

— THESE ARE TYPICAL POWER COMPONENTS —



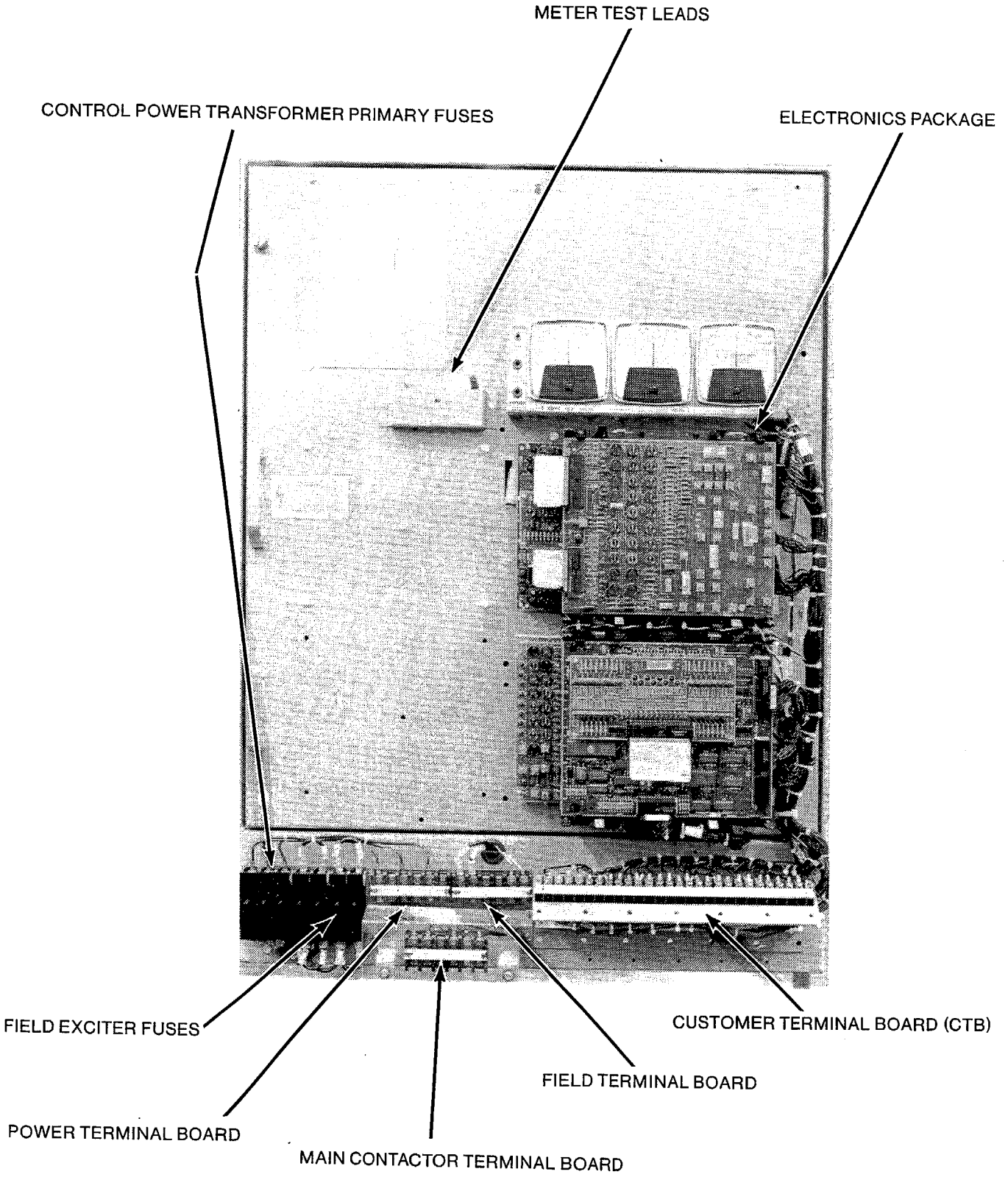
(DSF54843)

Figure 2-8.
Medium Horsepower Model - Exterior View



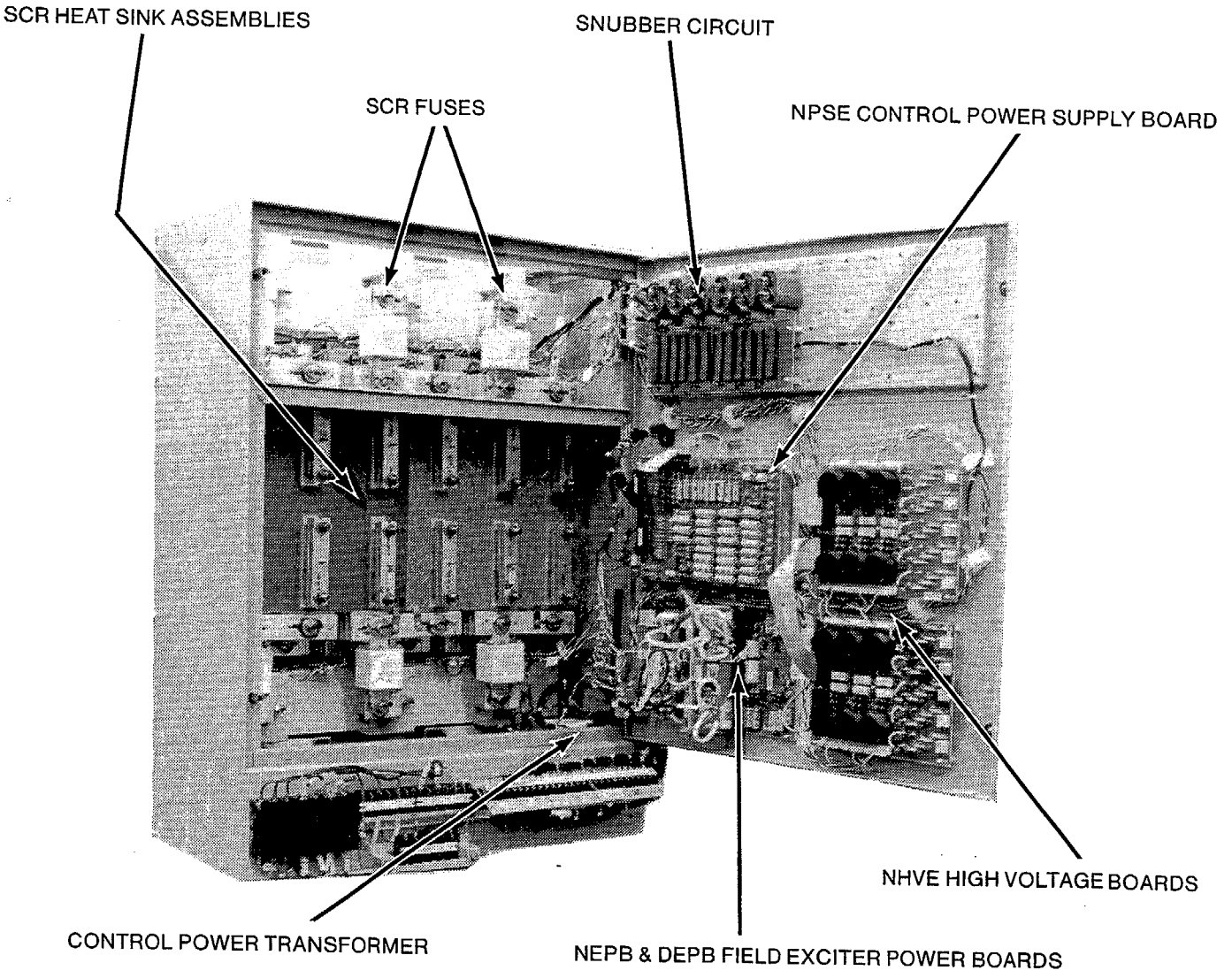
(DSF54846)

Figure 2-9.
Medium Horsepower Model - Interior View



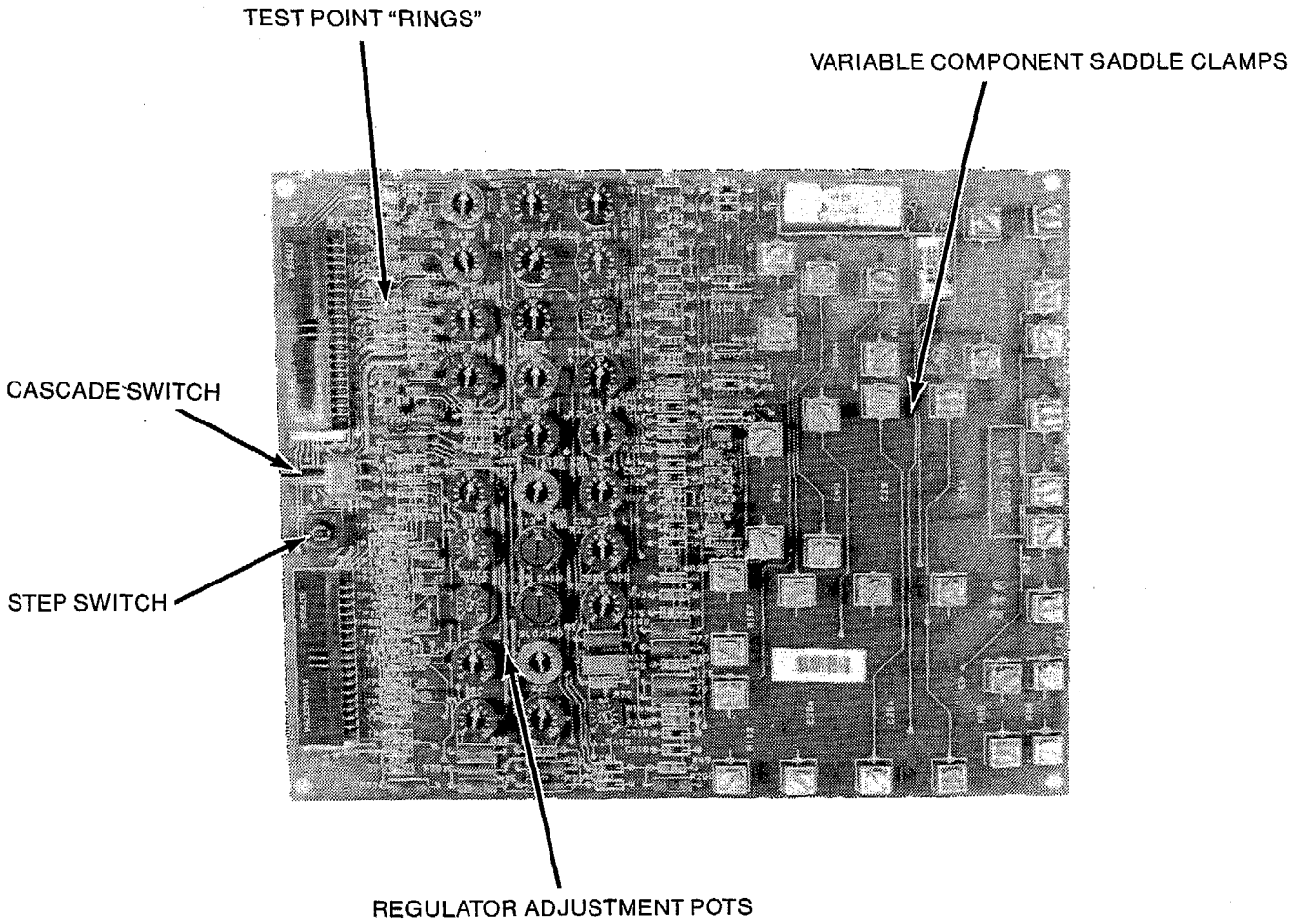
(DSF53547)

Figure 2-10.
High Horsepower Model - Exterior View



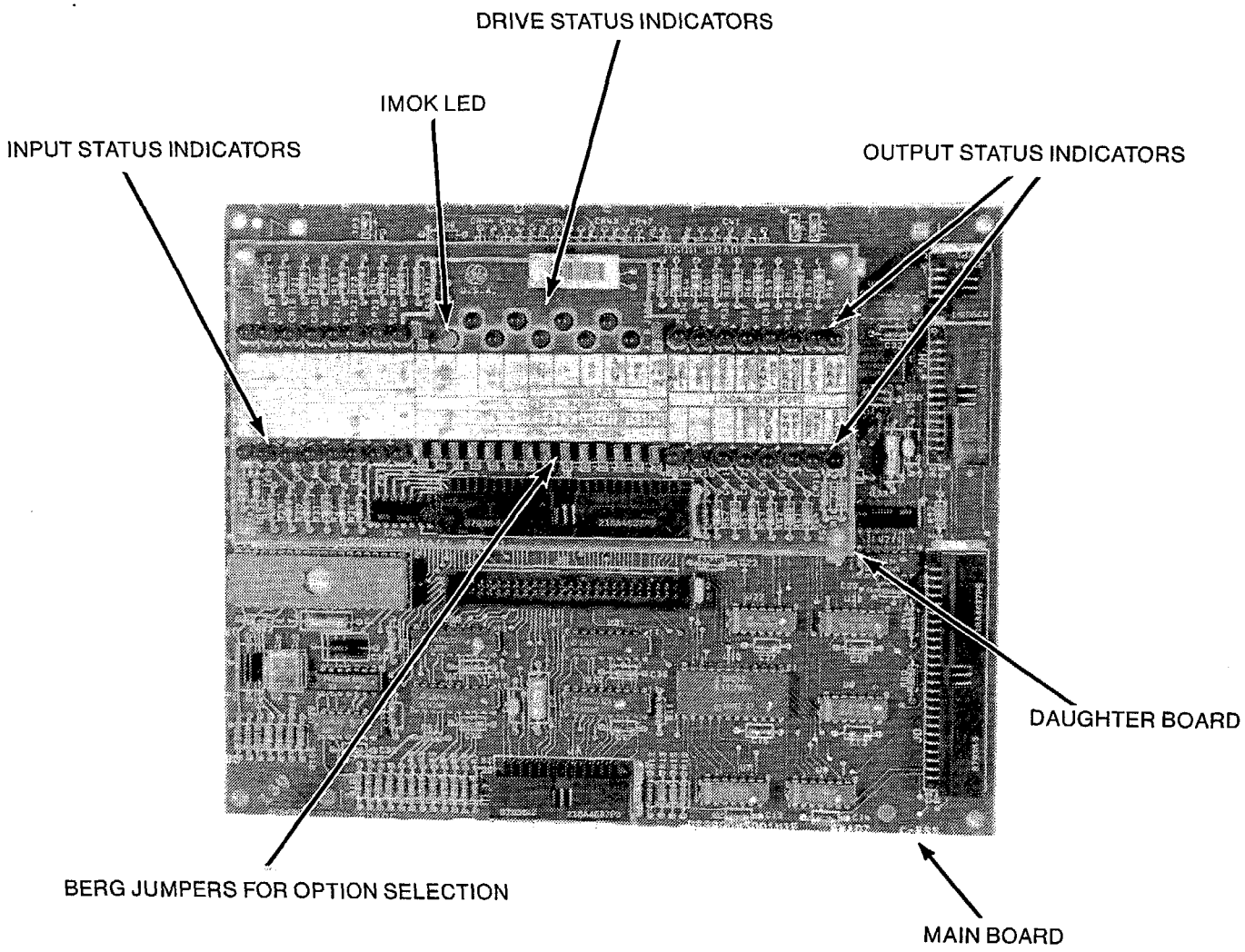
(DSF54847)

Figure 2-11.
High Horsepower Model - Interior View



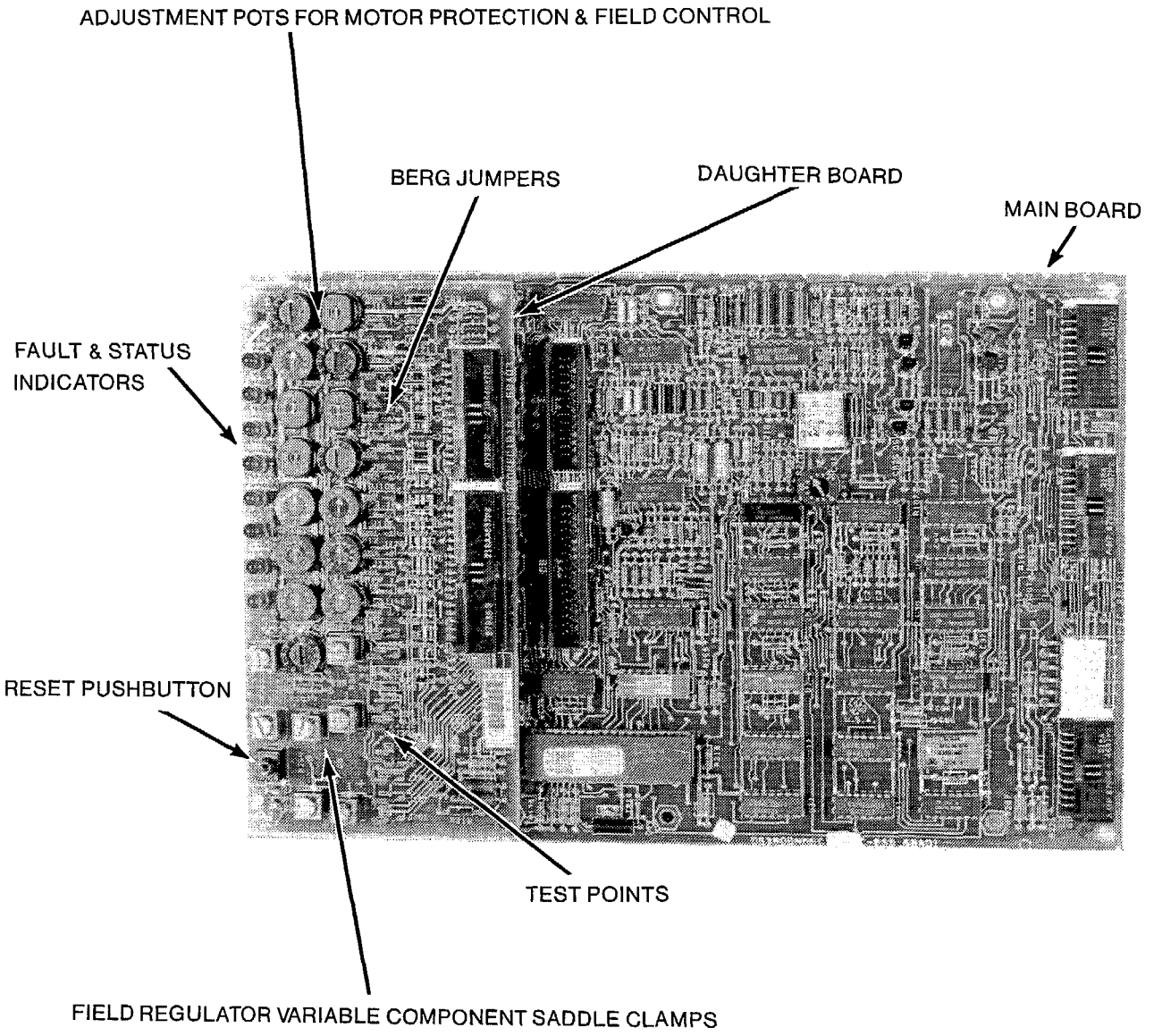
(DSF54918)

Figure 2-12.
Regulator Daughter Board (DGRC)



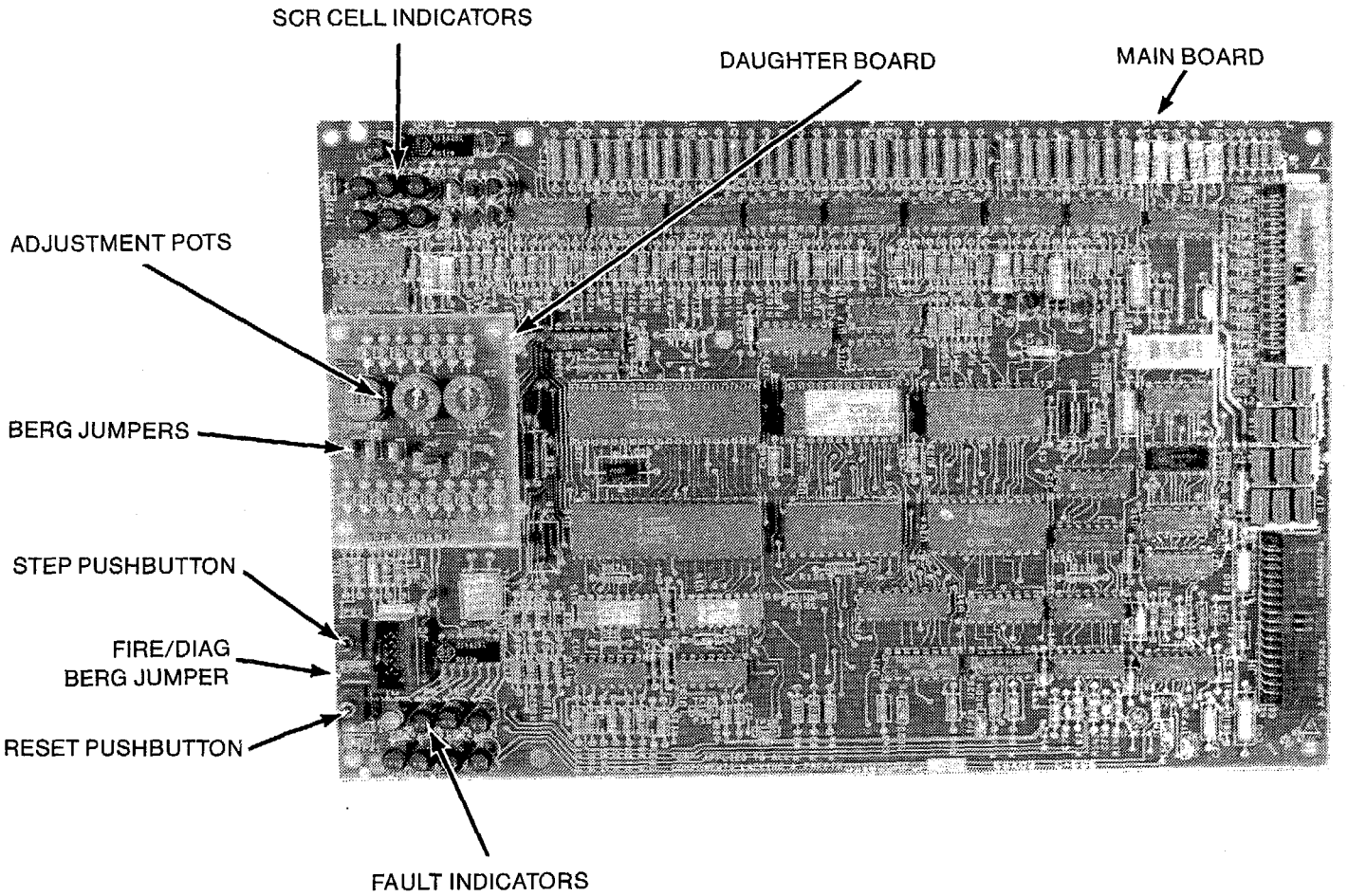
(DSF54915)

Figure 2-13.
Sequencer Boards (HSQD and DSQD)



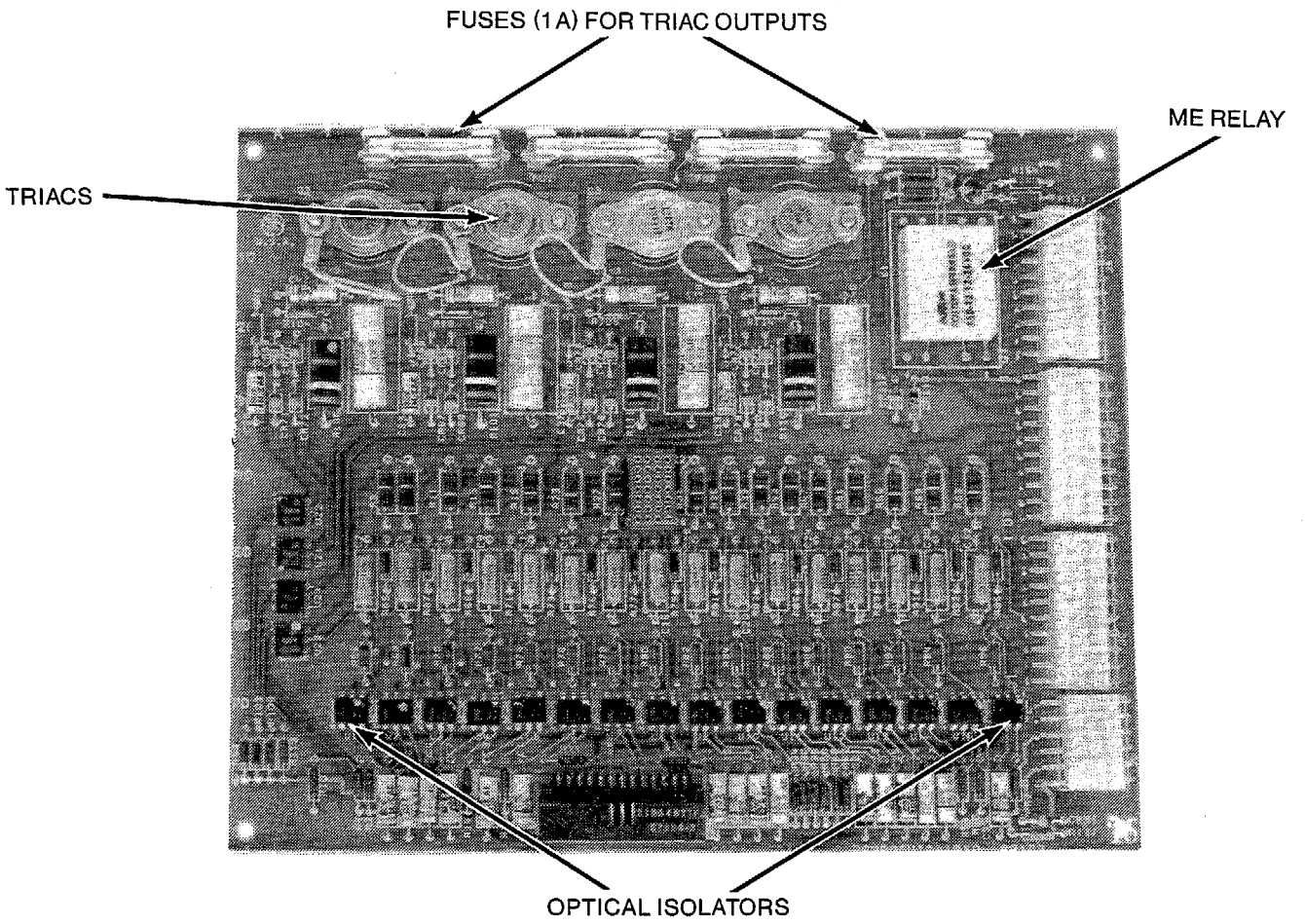
(DSF54914)

Figure 2-14.
Field Exciter Boards (NMEC and DMEC)



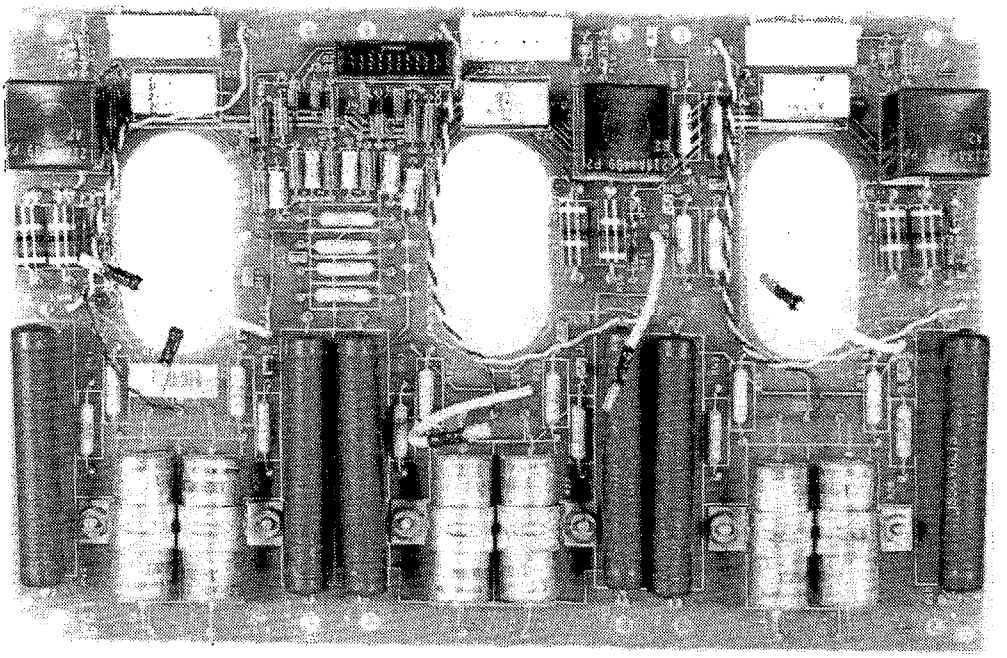
(DSF54914)

Figure 2-15.
Firing Circuit Boards (NFCF and DFCF)



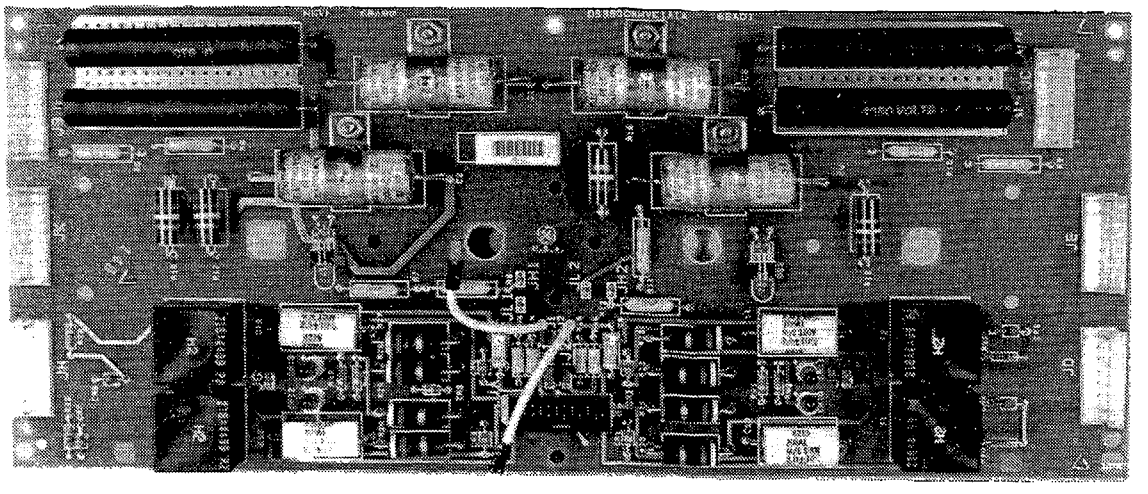
(DSF53348)

Figure 2-16.
Input/Output Board (HIOA)



(DSF54917)

Figure 2-17a.
High Voltage Board (NHVJ) for Siltron "A" Version



(DSF54921)

Figure 2-17b.
High Voltage Board (NHVK) for Siltron "B" Version

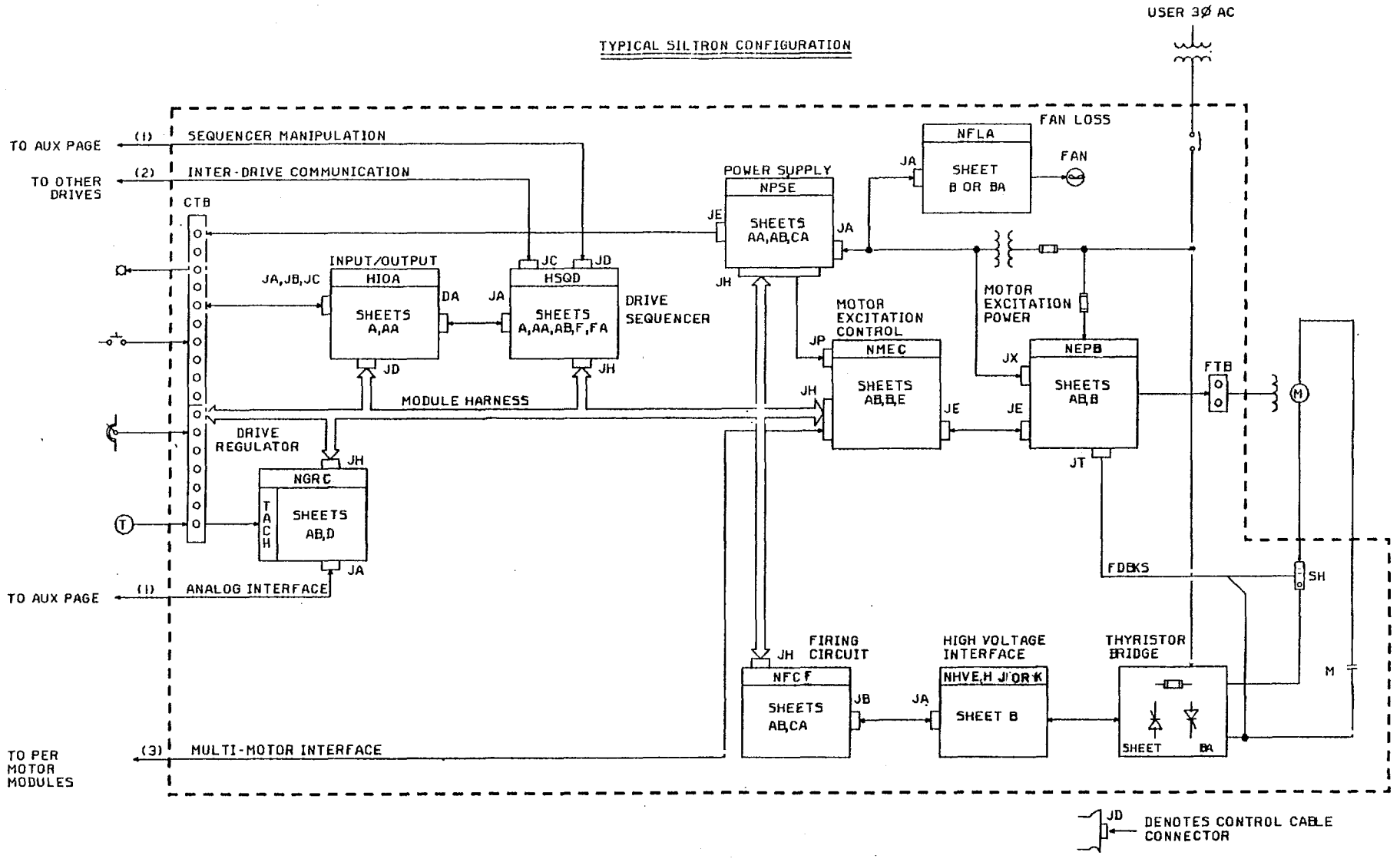


Figure 2-18
SILTRON PCM Detailed Block Diagram

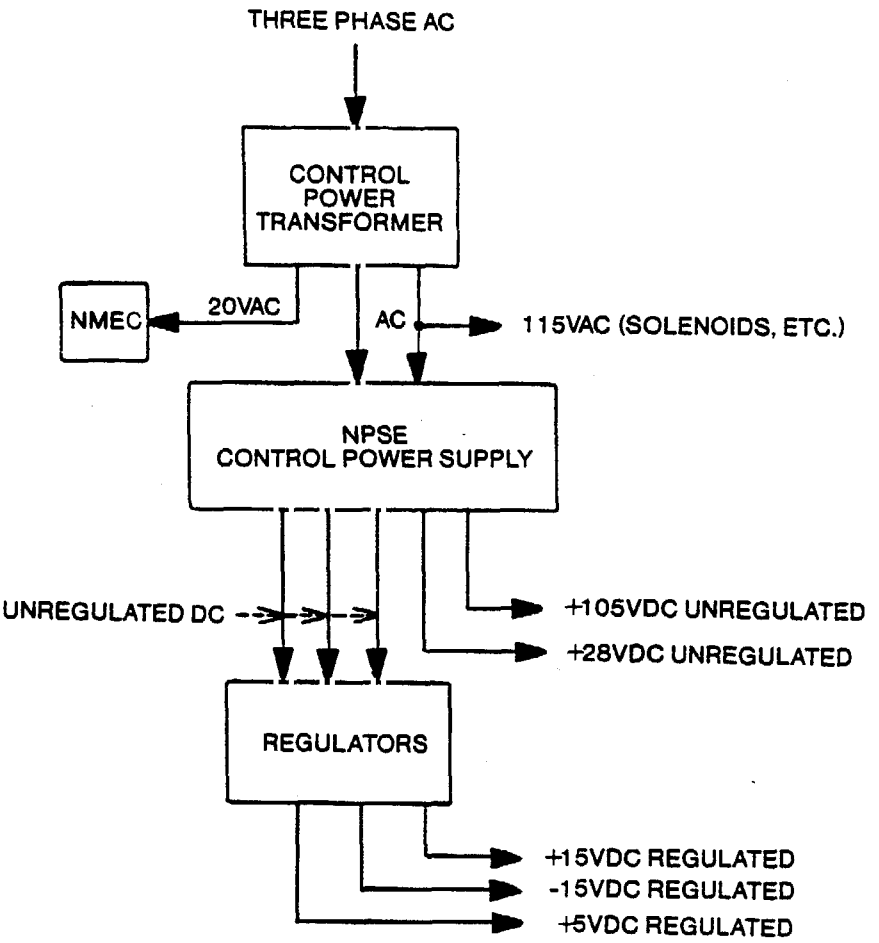


Figure 2-19
Control Power Supply Block Diagram

Proper operation of the PCM depends upon proper phasing of the Control Power Transformer primary leads with the Field Exciter. Three secondary windings of the transformer provide power to the Field Exciter gate supply circuitry. The Control Power Transformer and Field Exciter power must be in phase with each other.

The transformer's secondary consists of 24 windings: three provide 115 VAC, nine provide 20 VAC, six provide 8 VAC, and six provide 17.5 VAC. The 115 VAC output powers lamps within the PCM, as well as an optional cooling fan. If the cooling fan is not used, the 115 VAC power is available for other uses. One leg of the 115 VAC secondary is tied to the NPSE board, where the voltage is rectified and filtered to produce the +105 VDC voltage. Three of the 20 VAC windings provide power for the field exciter gate supply circuitry. The other 20 VAC windings are connected in a "star" configuration which supplies input power to the +28 VDC power supply on the NPSE board. The 8 VAC and 17.5 VAC windings are also connected in a "star" configuration and provide input power to the +5 VDC, +15 VDC, and -15 VDC power supplies on the NPSE board. See Table 2-2 for more information.

The NPSE board rectifies and filters the various AC voltages supplied by the Control Power Transformer. The circuit in Figure 2-20, which shows the +15 VDC and -15 VDC power supplies, is typical of the other supplies on the board. The fuses shown in each of the diode circuits are actually "artwork" fuses (i.e., copper runs on the board itself). These artwork fuses will open to protect the transformer in the unlikely event that a diode fails. If one of these fuses opens, the board must be replaced. Regulator transistors on the heatsink assembly are used to regulate the +5 VDC, +15 VDC, and -15 VDC outputs from the NPSE board. A yellow LED (CR19) verifies that the +5 VDC output is correct; as long as the +5 VDC output remains above +4.75 DC, the LED remains lighted. Should a +5 VDC failure occur, the M contactor will open and drop DC power to the motor. Table 2-3 provides a list of the NPSE components associated with each power supply voltage.

Table 2-2.
Control Power Transformer Secondary Specifications

Wire No.	Phase	Polarity Mark	Destination	Current	Ohms	Voltage	Power
13	1	X	PS-JA5	1.0 A.	5.24	115VAC	+105VDC power supply
14	1		PS-JA9				
15	2	X	PS-JA1	1.0 A.	5.24	115VAC	Misc. PCM AC power
16	2		CTB-46				
17	3	X	FAN PTB-4	1.0 A.	5.24	115VAC	Bridge fan
18	3		FAN PTB-5				
19	1	X	FEP-JX7	0.5 A.	0.96	20VAC	Field exciter phasing power
20	1		FEP-JX6				
21	2	X	FEP-JX12	0.5 A.	0.96	20 VAC	Field exciter phasing power
22	2		FEP-JX11				
23	3	X	FEP-JX1	0.5 A.	0.96	20VAC	Field exciter phasing power
24	3		FEP-JX2				
25	1	X	PS-JA31	1.28 A.	0.22	8VAC	+5VDC power supply
26	N		PS-JA29	--	--	--	+5VDC power supply
27	1		PS-JA33	1.25 A.	0.22	8VAC	+5VDC power supply
28	3	X	PS-JA39	1.25 A.	0.22	8VAC	+5VDC power supply
29	3	X	PS-JA37	1.25 A.	0.22	8VAC	+5VDC power supply
30	2	X	PS-JA35	1.25 A.	0.22	8VAC	+5VDC power supply
31	2		PS-JA38	1.25 A.	0.22	8VAC	+5VDC power supply
32	1	X	PS-JA26	1.25 A.	0.725	17.5VAC	±15VDC power supply
33	N		PS-JA21	--	--	--	±15VDC power supply
34	1		PS-JA30	1.25 A.	0.725	17.5VAC	±15VDC power supply
35	3	X	PS-JA36	1.25 A.	0.725	17.5VAC	±15VDC power supply
36	3		PS-JA40	1.26 A.	0.725	17.5VAC	±15VDC power supply
37	2	X	PS-JA32	1.25 A.	0.725	17.5VAC	±15VDC power supply
38	2		PS-JA34	1.25 A.	0.725	17.5VAC	±15VDC power supply
39	1	X	PS-JA24	0.82 A.	1.32	20VAC	+28VDC power supply
40	N		PS-JA17	--	--	--	+28VDC power supply
41	1		PS-JA22	0.82 A.	1.32	20VAC	+28VDC power supply
42	3	X	PS-JA16	0.82 A.	1.32	20VAC	+28VDC power supply
43	3		PS-JA14	0.82 A.	1.32	20VAC	+28VDC power supply
44	2	X	PS-JA20	0.82 A.	1.32	20VAC	+28VDC power supply
45	2		PS-JA18	0.82 A.	1.32	20VAC	+28VDC power supply

Table 2-3.
Power Supply Components on NPSE Board

Power Supply	Diodes	Fuses	Capacitors	Adjustments	JACKS JA	JCA,B,C	JHS	JHC
+28V	CR1-5	FU1	CI-9	--	18, 16, 14	--	17, 18	9, 10
+5V	CR11-16	FU2	C31-39	R3	24, 22, 20 31, 33, 35 38, 39, 37 26, 30, 32 34, 36, 40	1, 2, 3 11, 13, 15	30-35	11-16
+15V	CR21-26	FU3	C10-12 C16-21	R9	26, 30, 32 34, 36, 40	7, 12, 16	29-34	1-7
-15V	CR27-32	FU4	C13-15 C22-27	R10	11	14, 18, 19 25, 27, 28	19, 21, 23	1-7
+105V	CR41-44	FU5	C41	--	6, 5, 9, 10	--	JE-1	JE-3
115VAC	--	FU6	--	--	1, 2	--	JE-4	--

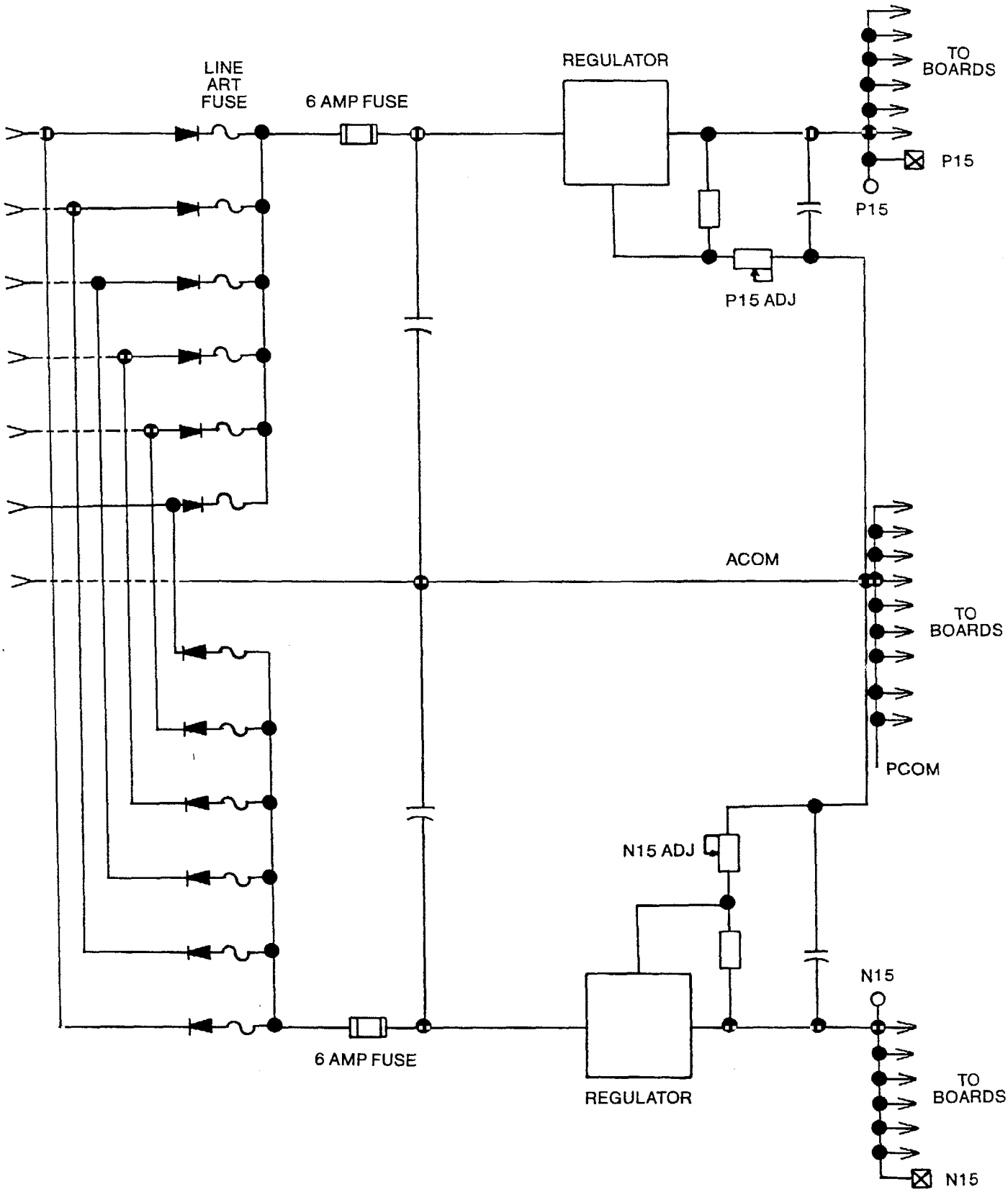


Figure 2-20.
Typical NPSE Power Supply Circuit

Input/Output Boards

Three different I/O boards can be used in the SILTRON PCM. The HIOA board interfaces the PCM with an operator station or other contact-closure device using a control voltage of +105 VDC. The HIOB board is used to interface the PCM with General Electric's Control Party Line (CPL). The HIOH board (in conjunction with an HXRB board) interfaces the PCM with the Series Six programmable controller. Any one of these boards can be used in the PCM. Figure 2-21 is a block diagram of the HIOA board and Figure 2-22 is a block diagram of the HIOB board. For more information on the theory and use of the HIOH board, refer to section 2 of the elementaries for the DS3800HIOH and DS3800HXRA boards.

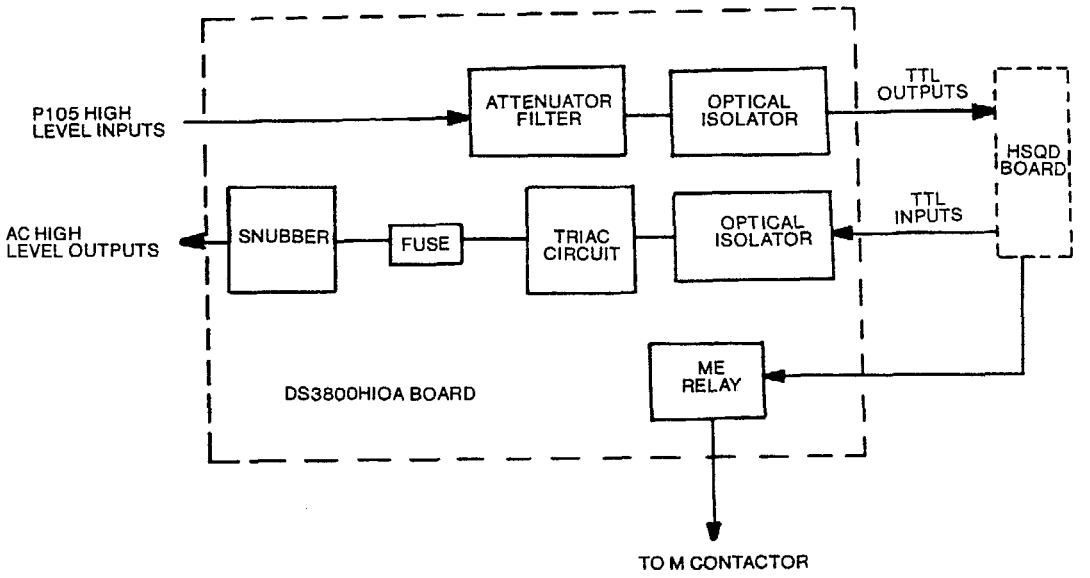


Figure 2-21
HIOA Board Block Diagram

The HIOA board is a high-level parallel interface board that accepts sixteen +105 VDC inputs and provides four 115 VAC outputs. See Table 2-4 for a list of input-output signals. The board inputs are optically isolated and converted to the logic-level signals. When +105 VDC is applied at the input to the board, a logic 0 appears on the corresponding output line to the Sequencer board. The Sequencer board returns four signals to the HIOA board. These signals are optically isolated and fed to a triac and snubber circuitry. This circuitry converts the logic level signal to a 115 VAC signal and provides it to the output side of the card. Each of the 115 VAC outputs is fused. When the circuitry is in an "off" state (i.e., inactive), the equivalent resistance to the 115 VAC output is between 80K ohms and 96K ohms with a current flow of 1 to 2 milliamperes. When the circuitry is in an "on" state (i.e., active), the equivalent resistance to the 115 VAC source is less than 1 ohm. The HIOA board also has an auxiliary pilot relay, the ME relay, which controls the M Contactor. The +105 VDC M Contactor coil is actuated when the ME relay is closed by a signal (logic 0) from the Sequencer board.

The HIOB board is an asynchronous, full duplex data link using a modified SDLC (Synchronous Data Link Control) protocol. The board communicates with an HMHA Control Party Line Master board at rates up to 48K baud. Several HIOB boards may share the same communications link as long as each board has been programmed via jumpers to have a different address. Six wires are used for high speed serial communication between the HIOB and HMHA boards: a transmit pair, a receive pair, and a driver-enable pair. (The driver-enable pair is used only in conjunction with LIMA modules.) Fifteen of the sixteen inputs listed in Table 2-4 are sent to the PCM over the serial link as encoded commands. The M Contactor status signal (M-STAT) is still handled as a high level (+105 VDC) signal, just as in the HIOA board. The +105 VDC signal that closes the M Contactor is also identical to that of the HIOA board. The HIOB card transmits the status of high level outputs (i.e., Running, Fault, etc.) to the HMHA card over the Control Party Line.

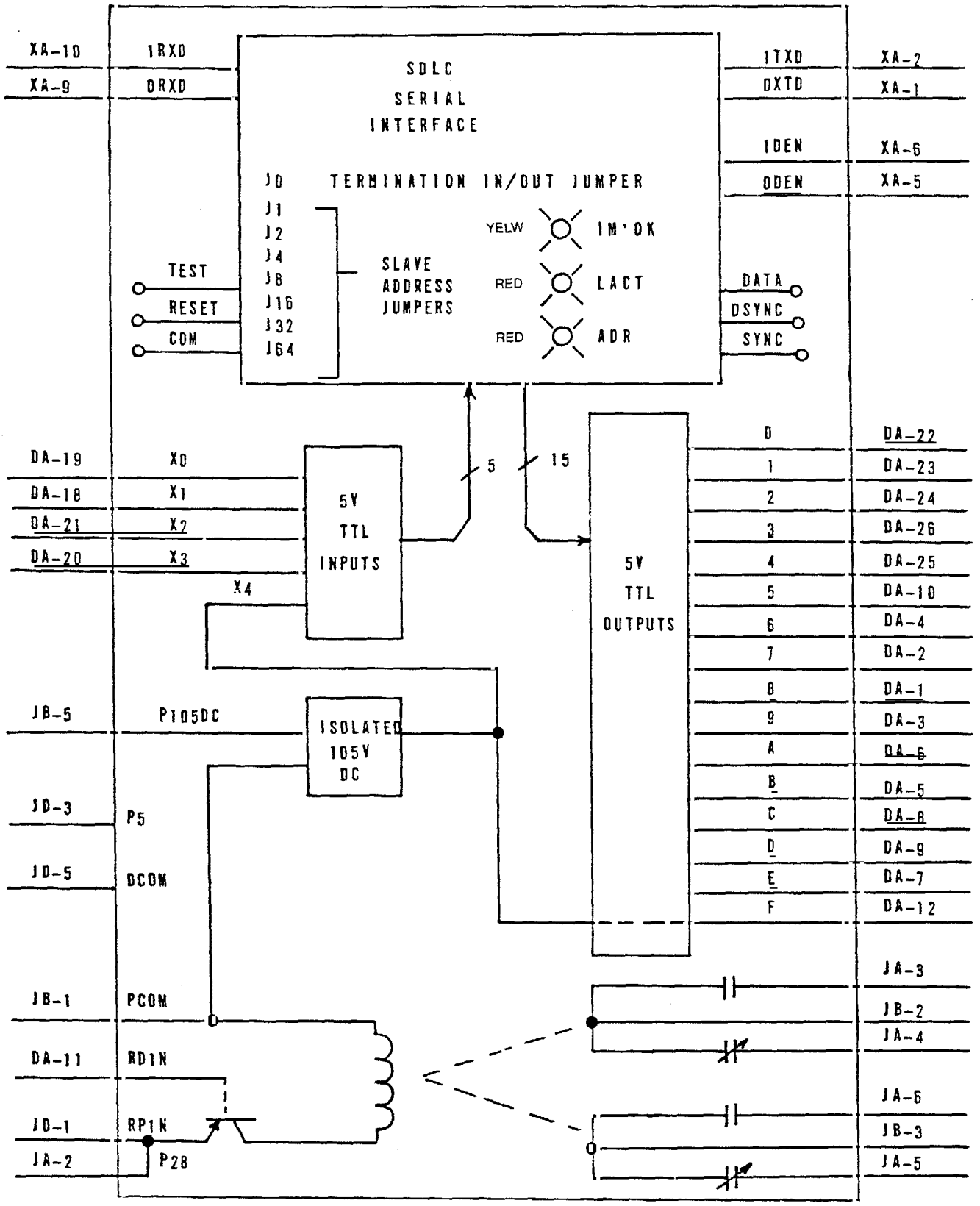


Figure 2-22.
HIOB Board Block Diagram

Table 2-4
HIOA Board Input/Output Signals

HIOA Inputs

Circuit	High Level Input Jack	Normal CTB Connection	Normal Function	Normal Voltage	TTL Level Output Jack
1	JC-8	27	E/Q STOP	+105VDC	DA-26
2	JC-10	28	N STOP	+105VDC	DA-23
3	JC-9	29	RUN PERM	+105VDC	DA-22
4	JC-7	30	START PERM	+105VDC	DA-2
5	JC-3	31	START	0	DA-4
6	JC-4	32	THREAD/SLO	0	DA-10
7	JB-10	33	STU	0	DA-25
8	JB-9	34	JOG FWD	0	DA-5
9	JC-1	35	JOG REV	0	DA-6
10	JC-5	36	CURR MODE	0	DA-3
11	JC-2	37	FLD PRESET	0	DA-1
12	JB-8	38	FWD/REV	0	DA-7
13	JB-7	39	RESET	0	DA-9
14	JB-4	40	FAULT STOP	+105VDC	DA-24
15	JB-6	41	FAULT MASK	0	DA-8
16	JB-5	MTB-5	M STATUS	+105VDC	DA-12

HIOA Outputs

Circuit	Low Level Input Jack	Normal CTB Connection	Normal Function	Normal State	High Level Output Jack	Fuse
1	DA-19	45	TOL ALARM	OPEN	JA-7	FU1
2	DA-18	44	RUNNING	OPEN	JA-8	FU2
3	DA-21	43	FAULT	OPEN	JA-10	FU3
4	DA-20	42	READY RUN	OPEN	JA-1	FU4

Sequencer Boards

The PCM Sequencer boards, consisting of the HSQD main board and DSQD daughter board, perform sequencing and control functions for the entire PCM. The sequencer processes information received from the PCM, the HIOA or HIOB input board, and the drive coordination line input from other PCMs. The Sequencer produces sixteen digital output signals and one analog signal to control the PCM. Figures 2-23A and 2-23B are typical ladder diagrams of the Sequencer's functions. The system elementary will contain information on how to read this type of diagram. Figure 2-24 is a photograph of the customer-oriented display on the Sequencer daughter board. Tables 2-5, 2-6 and 2-7 list inputs to and outputs from the Sequencer board. These tables do not list inputs to the Sequencer board from the HIOA or HIOB board; these inputs are listed in Table 2-4.

The Sequencer board contains a microprocessor and PROM-based software that monitor all input lines and produce the appropriate output signals as shown in the ladder diagrams of Figure 2-23.

Sequencer board inputs and outputs are monitored by LEDs on the front of the daughter board (see Figure 2-24). The LEDs marked INPUTS represent command signals sent from the I/O board (see Table 2-4). The LEDs marked LOCAL OUTPUTS represent the regulator manipulation signals (i.e. releasing suicides and applying references). The LEDs marked OUTPUTS represent drive sequencing signals such as "FAULT" and "READY TO RUN". These signals are routed to the I/O board where they become outputs from the PCM. The IMOK LED is lighted as long as the Sequencer board is operating properly.

A series of jumpers on the Sequencer daughter board is used to vary the configuration of the board. You can determine the purpose of each jumper by referring to Figure 2-23A or 2-23B or the system elementary included with each requisition. Four of the jumpers (J5S, J9S, J10S, and J16S) are used to connect the Sequencer board with the drive coordination line through a cable wired between connector JC of all PCM Sequencer boards. In other words, if one drive in a coordinated system experiences an Emergency Stop (ESTOP), all drives with the appropriate jumper (J9S) will also experience an Emergency Stop. Refer to the system elementary for more information on these jumpers.

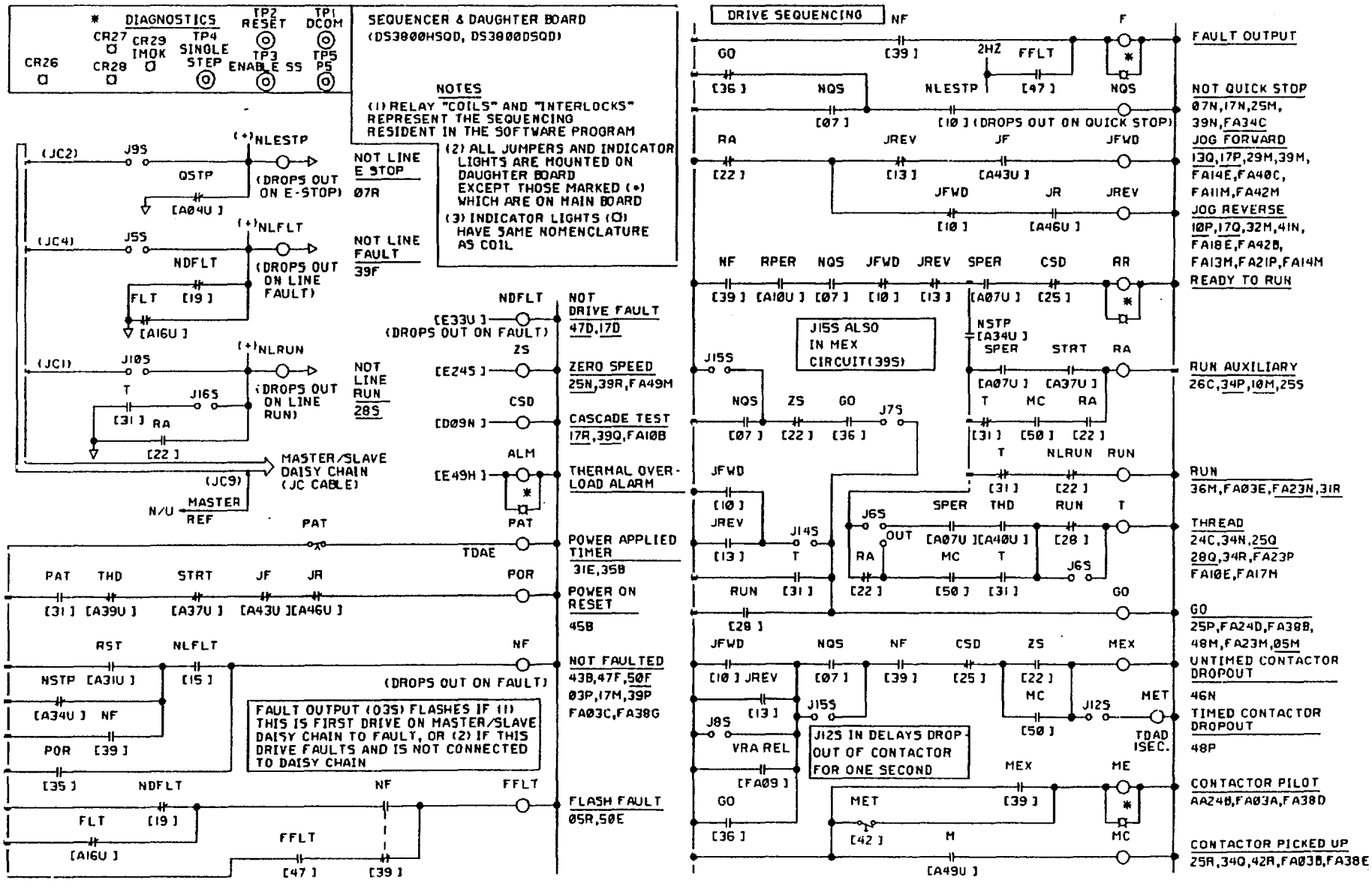


Figure 2-23A.
Typical Sequencer Ladder Diagram (Drive Sequencing Functions)

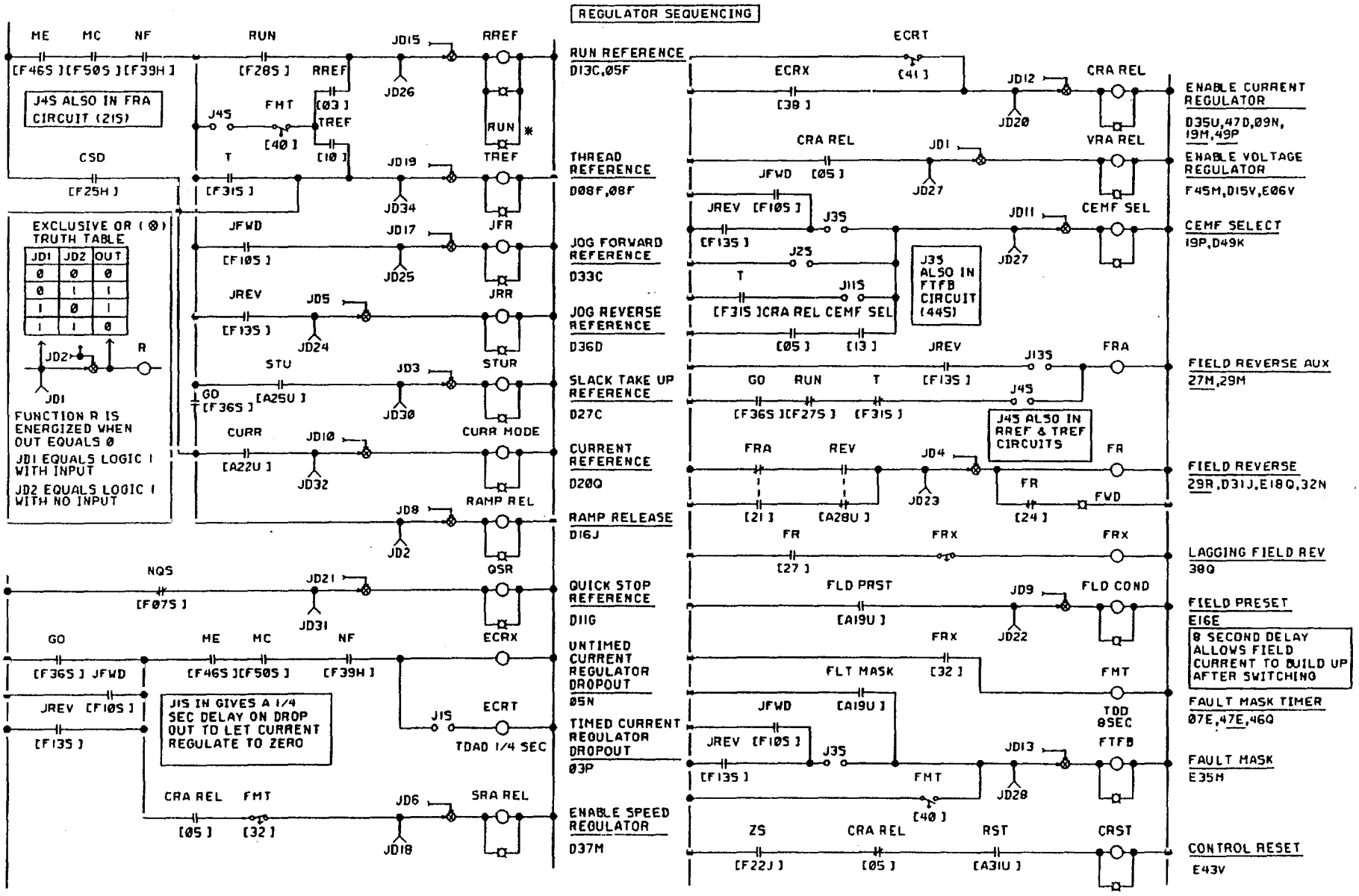
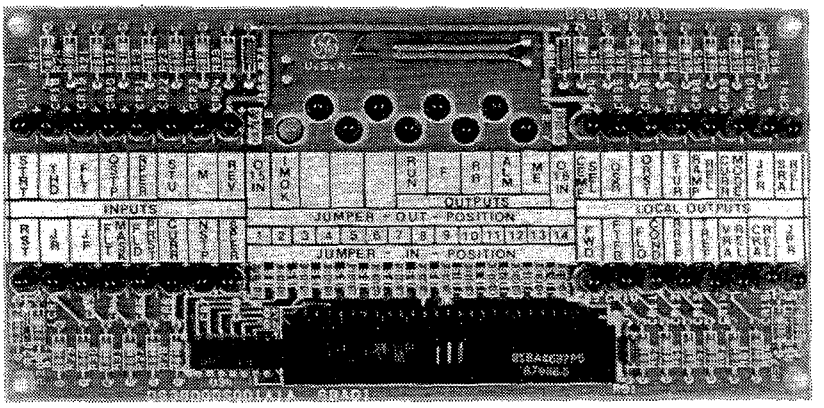


Figure 2-23B.
Typical Sequencer Ladder Diagram (Regulator Sequencing Functions)

Additional sequencing may be added to the PCM through an expansion port at connector JD. This would provide extra control and sequencing functions. The extra sequencing circuitry cannot be mounted within the SILTRON PCM; it must be located in a suitable external location. The expansion port allows the additional circuitry to operate in conjunction with the Sequencer board; all input and output signals are available to both.



(DSF-54709)

Figure 2-24
Detail of Sequencer Daughter Board

Table 2-5.
Inputs to Sequencer Board from PCM

Signal Name	JH Connector Pin Number	Signal Level
FAULT	22	0=Fault
ZSPD	7	0=Moving
TOL ALARM	13	0=Alarm
CSD	9	0=Test

Table 2-6.
Inputs to Sequencer Board from Other PCMs via Daisy Chain

Signal Name	JC Connector Pin Number	Signal Level
RUN	1	0=Run
ESTOP	2	0=Estop
FSTOP	4	0=Fstop
SPD REF	9	0=Stopped +10V=Full Speed (Analog Signal)

Table 2-7.
Outputs from Sequencer Board to PCM

Signal Name	JH Connector Pin Number	Signal Level
CTL RESET	23	0=Reset
CEMF SEL	10	0=CEMF
REV	11	0=Rev
TRR SUICIDE	25	0=Enable
SRA SUICIDE	3	0=Enable
CRA SUICIDE	6	0=Enable
VRA SUICIDE	5	0=Enable
JOG REVERSE	8	0=Jog
JOG FORWARD	4	0=Jog
RUN REF	21	0=Run
THD REF	20	0=Thd
Q-STOP	14	0=Stop
CM	24	0=Curr
STU	18	0=Slack TU
SPD REF	1	0=Stopped +10V=Full Speed (Analog Signal)
FLD PRESET	15	0=Preset
FAULT MASK	17	0=Mask

Regulator Boards

The PCM Regulator boards, consisting of the NGRC main board and the DGRC daughter boards plus the optional HPLA and DPLA digital regulator boards perform a variety of motor regulation functions:

1. Provide run/slow reference control.
2. Process CEMF reference for terminal voltage regulation.
3. Provide a timed-rate reference.
4. Provide slack take-up, plus jog forward and jog reverse functions.
5. Provide adjustable rise or droop.
6. Regulate speed or CEMF.
7. Provide fixed gain on integrators for cascade testing (see the Troubleshooting Guide for more information).
8. Provide speed regulation with current limit or current regulation with speed limit.
9. Provide inertia compensation.

The circuits discussed below are shown in Figure 2-25. Adjustments, jumpers, and variable components are listed in Table 2-8. Note that this board is a general purpose regulator board with a wide variety of possible uses. This section discusses only the most frequently used circuits: the Voltage Regulator, Current Regulator, Speed Regulator, Time Reference Circuit, and Regulator Expansion Port. Any of these regulators may be totally or partially bypassed.

Voltage Regulator. This circuit consists of two amplifiers: the Voltage Summing Amplifier (VSA) and Voltage Regulating Amplifier (VRA).

The VSA sums incoming signals. Normally these signals are from the voltage reference (an optional external input signal), the current regulating amplifier (CRA) or the speed regulating amplifier (SRA). The VSA also includes adjustable limits which limit the output voltage range and, therefore, the armature voltage range. A positive input to the VSA calls for a more positive armature voltage.

The VRA compares the armature voltage feedback signal (VFB) with the VSA output and produces an input signal to the firing circuits (NFCF board).

The armature voltage signal is obtained from the firing circuits via the High Voltage (NHV—) boards and is scaled so that 7 VDC is equal to rated voltage.

The VRA output is interpreted by the firing circuits as a firing angle command. This produces the required changes in armature voltage. A +7 VDC output results in full SCR output voltage. The VRA includes a suicide circuit which is controlled by the sequencer. Upon command, the suicide circuit forces the gain of VRA, and therefore its output, to near zero. Additionally, VRA provides the proper frequency response for different bridge configurations. This is automatically controlled by a signal from the firing circuits.

Current Regulator. This circuit also consists of two amplifiers: the Current Summing Amplifier (CSA) and the Current Regulating Amplifier (CRA). These circuits are quite similar to the Voltage Regulator.

The CSA sums inputs from the speed regulator and/or optional external input signals. Adjustable clamps are provided. The clamps will act as a current limit if the sum of the current references call for excessive current.

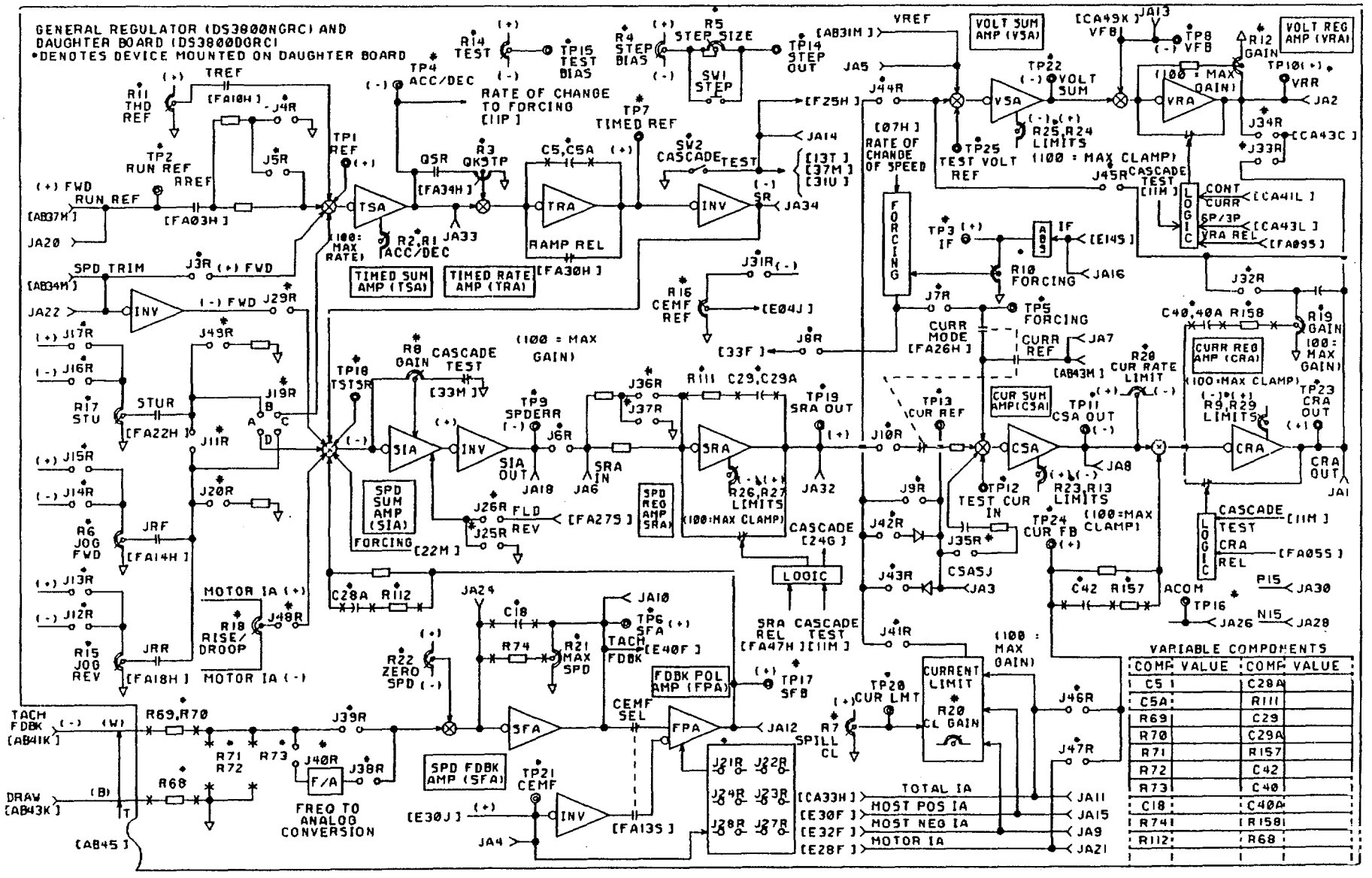


Figure 2-25.
Regulator Board Circuit Diagram

Table 2-8.
Regulator Board Adjustments and Variable Components

Test Point	Signal Name	Variable Components	Adjustment Potentiometers	Jumpers
TP1	REF			
TP2	RUN REF			
TP3	IF			
TP4	ACC/DEC			
TP5	FORCING		R10 - FORCING	J7R, J8R
TP6	SFA	C18, R68 R69,R70,R71 R72,R73,R74	R21 - MX SPD R22 -ZERO SPD	J38R, J39R J40R
TP7	TIMED REF	C5, C5A	R1 ACC R2 DEC R3 QK STP R11 THD REF	J3R, J4R J5R, J19
TP8	VFB			
TP9	SPD ERR	C28A, R112	R6 JOG FWD R8 GAIN R15 JOG REV R17 STU R18 RISE/DROOP	J11R, J12R J13R, J14R J15R, J16R J17R, J19R J20R, J48R, J49R
TP10	VRA		R12 GAIN	J34R
TP11	CSA OUT		R13 - LIM R23 + LIM R28 CUR RATE LIM	
TP12	TEST CUR IN			
TP13	CUR REF			J9R,J42R,J43R,J35R
TP14	STEP OUT		R4 STEP BIAS R5 STEP SIZE SW1 STEP SW	
TP15	TEST BIAS		R14 TEST	
TP16	ACOM			
TP17	SFB			J21R, J22R J34R, J24R J27R, J28R
TP18	TSTSR			
TP19	SRA OUT	C29, C29A R111	R26 - LIM R27 + LIM	J36R, J37R J6R
TP20	CUR LIM		R7 SPILL CL R20 CL GAIN	J41R
TP21	CEMF			
TP22	VOLT SUM		R24 + LIM R25 - LIM	J44R, J45R
TP23	CRA OUT	C40, C40A R158	R9 - LIM R29 + LIM R19 GAIN	J32R
TP24	CUR FB	C42,R157		J46R,J47R
TP25	TEST VOLT REF			

The CRA compares the armature current feedback signal with the output of CSA. The resulting signal is applied to the voltage regulator and causes it to call for changes in armature current. CRA output scaling is such that 7 VDC is equivalent to 100% armature voltage. The armature current feedback signal is derived from a shunt isolator and is scaled so that + or - 10 volts is equivalent to maximum operating overload current. If more than one motor is being controlled, the feedback signal may be equal to the sum of all motor currents. CRA also includes adjustable clamps, provides the desired frequency response characteristics for proper regulator action and a suicide circuit. The suicide circuit is controlled by the sequencer.

A current forcing circuit is provided. This function is used to overcome the effects of inertia during acceleration and deceleration. This circuit uses signals proportional to the rate of change of motor speed to produce a signal which is applied to CSA.

Speed Feedback. The speed feedback can be obtained from an analog tachometer, a digital tachometer or motor CEMF. The tachometer signal is scaled and filtered by the speed feedback amplifier (SFA). Scaling is such that 6.67 volts DC represents 100% motor speed. A frequency to analog converter is provided when a digital tachometer is used. Inverter (INV) inverts the pre-scaled CEMF signal to supply proper polarity to the feedback polarity amplifier (FPA). Digital tachometers do not distinguish between forward and reverse, therefore FPA is used to provide this function. The signal polarity of FPA can be a function of CEMF polarity or the field reversal logic signal. The output of FPA is applied to the speed regulator.

Speed Regulator. There are three amplifiers in the speed regulator. These are: the speed summing amplifier (SIA), an inverting/non-inverting amplifier (INV), and the speed regulating amplifier (SRA).

The SIA sums inputs from selected references with the speed feedback signal and amplifies any differences (error signal).

INV is normally used as a unity gain inverter. In this mode, it inverts the SIA signal to provide the proper polarity to the SRA. When field reversing is used, a signal causes INV to switch to a non-inverting unity gain amplifier. Again, this provides the correct signal polarity to the SRA.

The SRA further amplifies the speed error signal. Its output signal can be applied to the voltage regulator or the current regulator. The SRA includes limit adjustments, stabilization components and a suicide circuit. The limits act to clamp either current, when the speed regulator feeds the current regulator or voltage when feeding the voltage regulator.

Spillover Current Limit. This function can be used to force the output of the SRA in a direction which causes a reduction in output current. This is normally applied when the current regulator is not used or when the highest of motor currents is to be limited.

Timed Acceleration - Deceleration. This function contains three amplifiers: the timed summing amplifier (TSA), the timed rate amplifier (TRA) and an inverter (INV).

The TSA sums the selected input reference signal with the output of INV and amplifies the difference.

The TRA integrates the signal from the TSA. The value of the integrating capacitance, in conjunction with the TSA clamp settings, determine the rate of change of the timed reference signal. Since the TRA is an integrator, a suicide circuit is required.

INV provides the proper signal polarity to the TSA and the speed regulator. Signal scaling is such that 10 VDC equals 100% motor speed. A quick stop circuit is included to speed up deceleration.

Firing Circuit

The firing circuit consists of the main firing circuit board (NFCF), it's daughter board (DFCF) and the high voltage card(s) (NHVJ for Low Horsepower "A" version, NHVK for Medium Horsepower "B" version, NHVE for High Horsepower "C" version, and NHVH for Low Horsepower "D" version). These control the firing of the SCRs in the main bridge. Figure 2-26 shows a simplified block diagram of the firing circuits.

The firing circuit uses a phase angle control signal and line voltage zero-crossing information from a phase-locked loop to shift the phase angle of the SCR gate reference. The firing circuit fires the forward or reverse SCR cells as required to provide higher or lower voltage in the forward or reverse direction. This produces a DC output voltage which is proportional to the voltage reference. If the demand on the SCR bridge is light, it is common for the firing circuit to fire the forward and reverse cells alternately.

The SCR bridge output voltage is sent from the bridge to the NHV— board where it is attenuated. The attenuated voltages are measured differentially by the cell voltage sensing circuits on the NFCF board. This circuit, shown in Figure 2-27, detects the presence or absence of SCR cell output voltage. If all SCRs have sufficient voltage across them (i.e., none are shorted or turned on), the cell voltage sensing circuit allows reversing. Six LEDs located on the upper left corner of the firing circuit board monitor SCR cell voltage; the intensity of the LED is proportional to the percentage of time that the SCR cell has voltage across it. If an LED is completely dark, this indicates a shorted SCR cell or blown fuse.

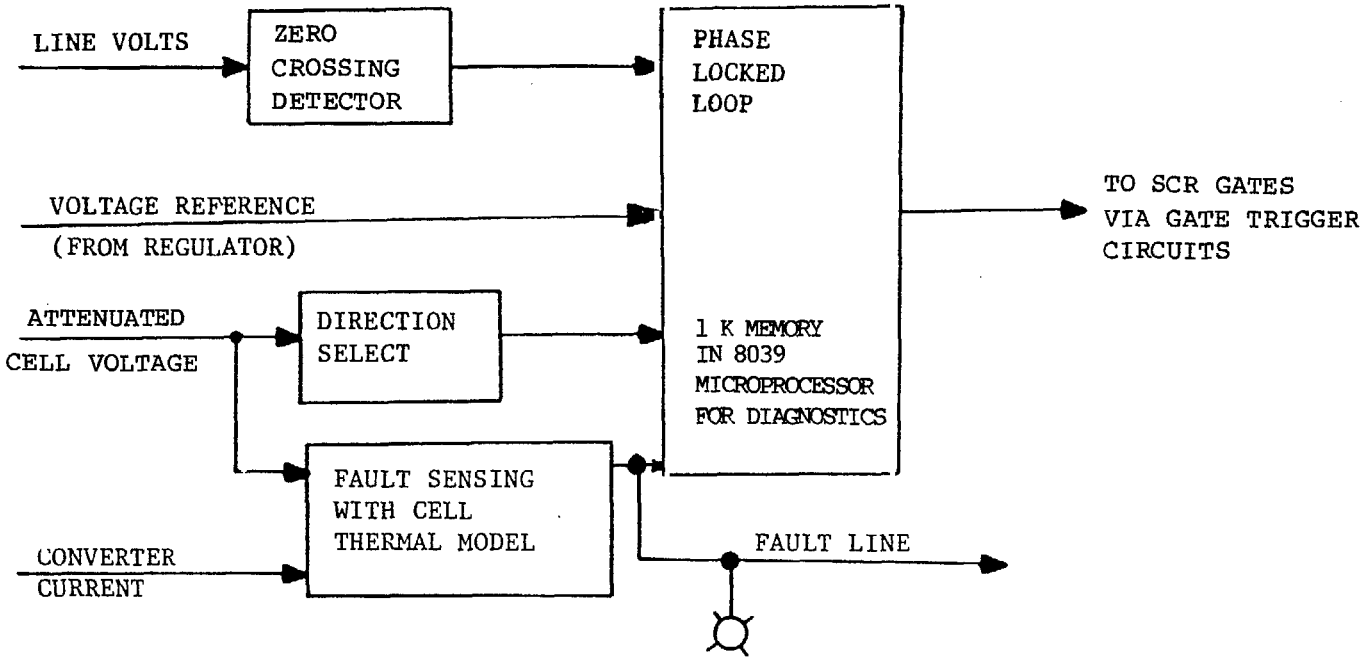


Figure 2-26.
Firing Circuit Block Diagram

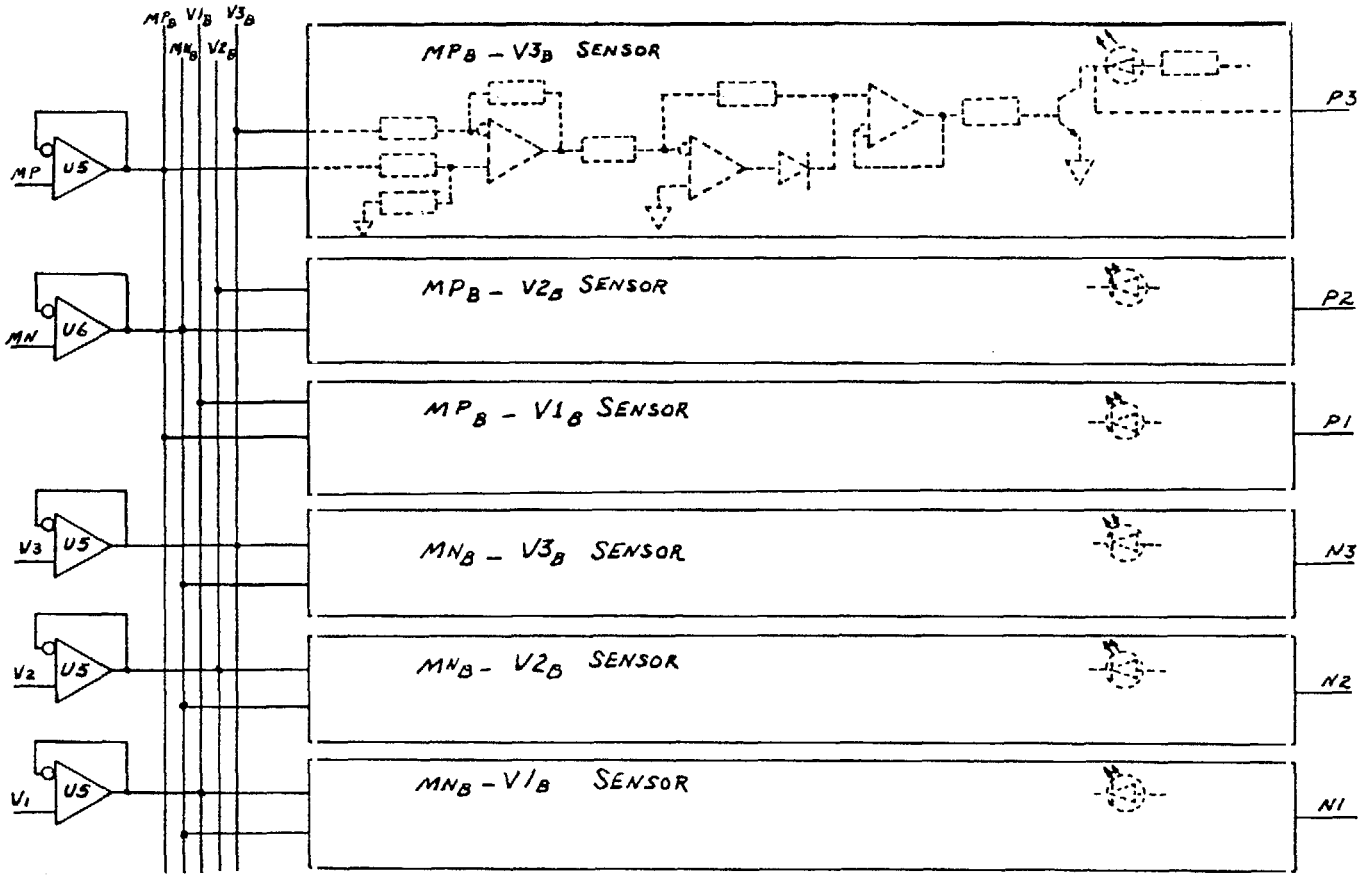


Figure 2-27.
SCR Cell Voltage Sensing Circuit

A group of eight LEDs in the lower left corner of the board monitors the NFCF board's status. The LEDs are driven by outputs from subroutines of the program on the NFCF board. These subroutines monitor possible fault conditions within the PCM. The IMOK LED remains on as long as the NFCF board is operating properly. The other LEDs remain off as long as there are no PCM faults. These LEDs and the fault conditions they monitor are:

- PH - Phase rotation or phase loss fault when initialized
- IEC - Instantaneous SCR bridge overcurrent condition
- TOL - SCR bridge thermal overload condition
- CNDO - Diametric fault
- BRKR - Circulating current fault
- 1FLT - Commutation fault
- SPARE - Used for DIAG program

If several faults occur together, the first fault to occur will cause its corresponding LED to flash while the LEDs indicating subsequent faults remain on and do not flash.

The two pushbuttons located on the left side of the board are used to reset the NFCF microprocessor and step the DIAG program. Neither the RESET nor the STEP pushbutton should be depressed during normal operation. This causes an emergency stop of the drive and would interrupt the process.

The NFCF board contains a built-in diagnostic test procedure that checks the firing circuit and SCR bridge components for proper operation. DO NOT RUN THE TEST PROCEDURE WHILE THE PCM IS IN NORMAL OPERATION. To operate the diagnostic routine, follow these steps:

1. Locate the DIAG/FIRE jumper between the RESET and STEP buttons on the NFCF board. Remove PCM power or hold the RESET button depressed while you move the jumper from the FIRE to the DIAG position (FIRE is the normal position for the jumper).

2. When PCM power is returned or the RESET button released, the diagnostic routine begins executing. The routine uses the fault-monitoring LEDS at the lower left corner of the NFCF board as status indicators. Refer to Figure 2-28 during the rest of this procedure.

3. The diagnostic routine will first be in a standby mode. The LEDs will be lighted as in Pattern A of Figure 2-28. The IMOK LED indicates that the microprocessor is operating. The 1FLT LED indicates that the routine is in a standby mode and waiting to execute Test 1. Press the STEP button to start Test 1. (The LCL LED at the top of the board lights each time the STEP button is pressed.)

4. Test 1 causes the LEDs to light for one second each in the following order: CR41, CR42, CR43, CR44, CR46, CR47, CR48 and CR45. This test allows you to find any defective LEDs or associated components. At the end of TEST 1, the LEDs will be lighted as in Pattern B of Figure 2-28. This indicates that Test 1 is over and Test 2 is ready to execute.

5. Press the STEP button to execute Test 2. This test first causes the firing circuit to check for shorted cells. If the cell is shorted, the IMOK LED does not light, CR46-CR48 are lighted, and the bad cell is identified by CR41-CR44 flashing. See Figure 2-28. The firing circuit attempts to fire each SCR one at a time. The test then checks the cell voltage sensing circuit to see if the cell actually fired. The configuration of the equipment under test (set by jumpers on the DFCF board) determines which SCR cells are tested. For example, a three-pulse, non-reversing PCM would test only cells P1F, P2F, and P3F. As each cell is pulsed, the IMOK LED lights if the cell fires properly. If the cell voltage does not collapse after the SCR is fired, the IMOK LED does not light, CR46-48 are lighted, and the bad SCR is identified by CR41-CR44. The routine will continue to pulse a bad cell and will light an LED on the NHV— board to identify the defective cell or circuit until the STEP button is pressed, at which time the next cell in sequence is pulsed. This continues until all cells have been tested.

After each cell passes the open cell test, the program continues to fire the cell for a short period of time which lights an LED on the NHV— board and allows identification of the cell being fired.

If the STEP pushbutton is pressed after a shorted cell has been found, no other cells are tested and this test step is terminated.

6. After all cells have been successfully tested (or the STEP button has been pressed), the LEDs will light according to Pattern C of Figure 2-28. This signals the end of the diagnostic test routine. Pressing the RESET button allows you to restart the diagnostic routine from Test 1.

7. After all testing is complete, remove power and move the DIAG/FIRE jumper back to the fire position.

CAUTION

The forward and reverse signal lines which connect to CTB4 and CTB5 are logic level (0V or 5V) signals. Damage will occur if CTB4 or CTB5 is connected to any circuit other than DCOM. Be sure to exercise extreme care if using these circuits during servicing.

DIAGNOSTIC TEST CODES

CODE NO.	LED NUMBERS CR44	CR43	CR42	CR41	CELL LEG
0	○	○	○	○	
1	○	○	○	●	P1F
2	○	○	●	○	P2F
3	○	○	●	●	P3F
4	○	●	○	○	N1F
5	○	●	○	●	N2F
6	○	●	●	○	N3F
7	○	●	●	●	P1R
8	●	○	○	○	P2R
9	●	○	○	●	P3R
10	●	○	●	○	N1R
11	●	○	●	●	N2R
12	●	●	○	○	N3R
13	●	●	○	●	
14	●	●	●	○	
15	●	●	●	●	

○ = LED OFF

● = LED ON

PATTERN A

IMOK	PH	IEC	TOL
●	○	○	○
CR45	CR48	CR47	CR46
CR44	CR43	CR42	CR41
○	○	○	●
SPARE	CNDO	BRKR	1FLT

PATTERN B

IMOK	PH	IEC	TOL
●	○	○	○
CR45	CR48	CR47	CR46
CR44	CR43	CR42	CR41
○	○	●	○
SPARE	CNDO	BRKR	1FLT

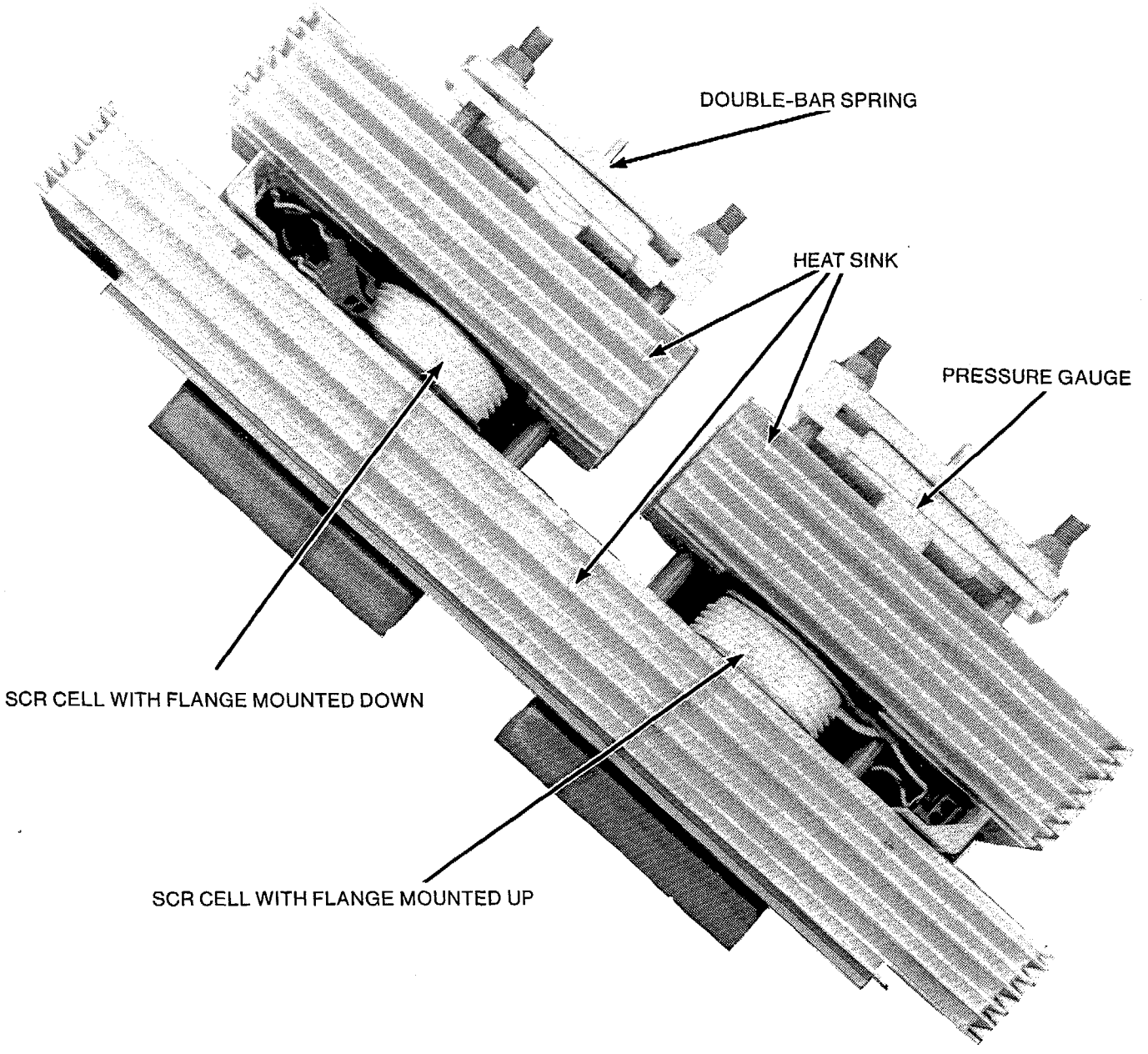
PATTERN C

IMOK	PH	IEC	TOL
●	○	○	○
CR45	CR48	CR47	CR46
CR44	CR43	CR42	CR41
○	○	●	●
SPARE	CNDO	BRKR	1FLT

Figure 2-28.
Fault-Monitoring LED Diagnostic Patterns on NFCF Board

SCR Circuits

Depending on the model, the SILTRON PCM SCR circuits consist of three, six, or twelve SCRs plus fuses and snubber circuits. As shown in Figure 2-29, the SCRs are mounted on heat sinks for effective heat dissipation (this figure shows the High Horsepower Siltron SCR cell assembly; see Appendix B for photographs of the cell assemblies used in the other Siltron models). Fuses protect the other SCR cells in case a cell shorts. The SCR snubber circuits are RC networks that prevent the creation of high voltage line transients when the cells turn off. There are three different types of SCR cells used in the SILTRON PCM. The Low Horsepower model uses either a stud-type or High Line cell. The other models use disk-type (PRESS-PAK) cells of various sizes and ratings. Appendix B contains cell removal and replacement procedures for the SILTRON PCM.



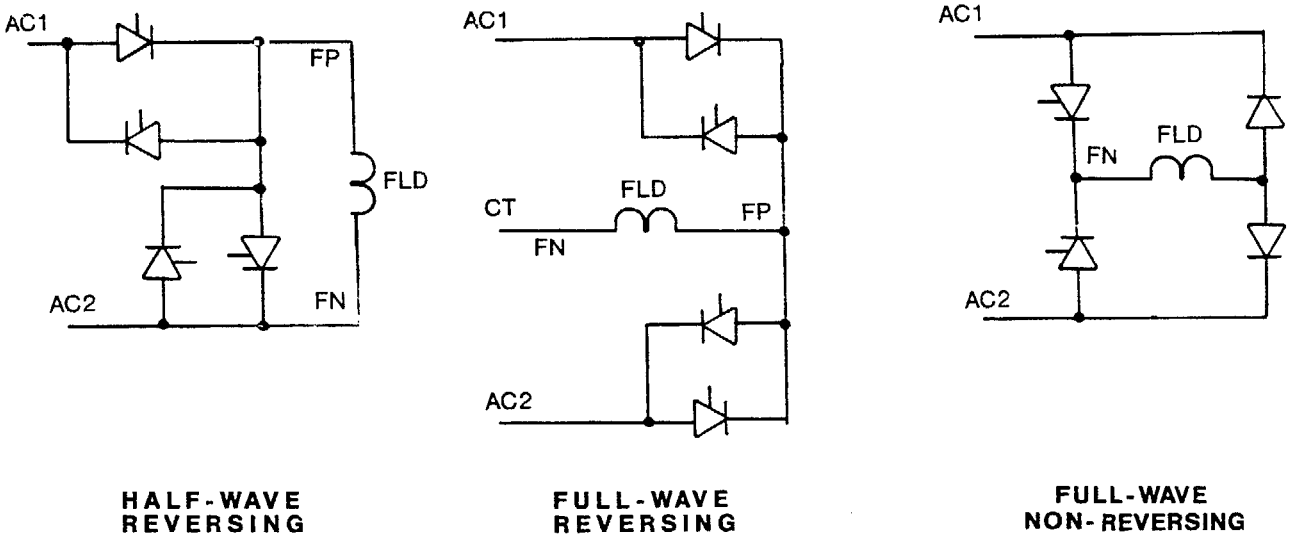
(DSF-53703)

Figure 2-29
High Horsepower SCR Cell Assembly (PRESS-PAK)

Field Exciter Boards

The Field Exciter boards, which include the NMEC Main board, the DMEC daughter board, the NEPB power board, and the DEPB daughter board are responsible for motor field excitation and motor protection. Figure 2-30 illustrates the functions of these boards.

The NMEC board contains synchronizing and firing circuitry for three different types of field excitation power bridges: half-wave reversing, full-wave reversing (which requires an external transformer), and full-wave non-reversing. These different power bridges are shown below:



HALF-WAVE REVERSING

FULL-WAVE REVERSING

FULL-WAVE NON-REVERSING

(REQUIRES EXTERNAL TRANSFORMER)

These circuits convert single-phase AC voltage into a variable DC voltage. The synchronizing signal is provided from the center tap of the 20 VAC control winding of the Control Power Transformer. The field exciter output can be regulated according to field current, terminal voltage, or motor speed.

The motor protection functions include overcurrent, overvoltage, overspeed, thermal overload, tachometer loss, and field loss detection. Several of these protective functions rely on four feedback signals from the motor: armature current, field current, armature voltage, and motor speed. The first three feedback signals are isolated and scaled on the Field Exciter boards, while the speed feedback signal is isolated and scaled on the Regulator boards (for tachometer feedback) or the Field Exciter boards (for CEMF feedback).

An overcurrent condition is detected when the armature current feedback exceeds a preset limit. An overvoltage condition is detected when the armature voltage feedback exceeds a preset limit. An overspeed condition is detected when the speed feedback signal exceeds a preset limit. A field loss condition is detected when the field current feedback drops below a preset limit. All of the fault indications are enabled by the jumpers listed in Table 2-9. The setpoints are adjusted by the potentiometers listed in that same table.The other fault circuits are also represented in Table 2-9.

Table 2-9.
Motor Protection Jumpers and Adjustments

FUNCTION	JUMPER	ADJUSTMENT
Overcurrent (IOC)	J7E	R8
Overvoltage	J4E	R11
Overspeed	J3E	R14
Thermal Overload (TOL)	J8E	R9
Thermal Alarm	J20E	R9
Tachometer Loss	J1E	—
Field Loss	J10E	R2

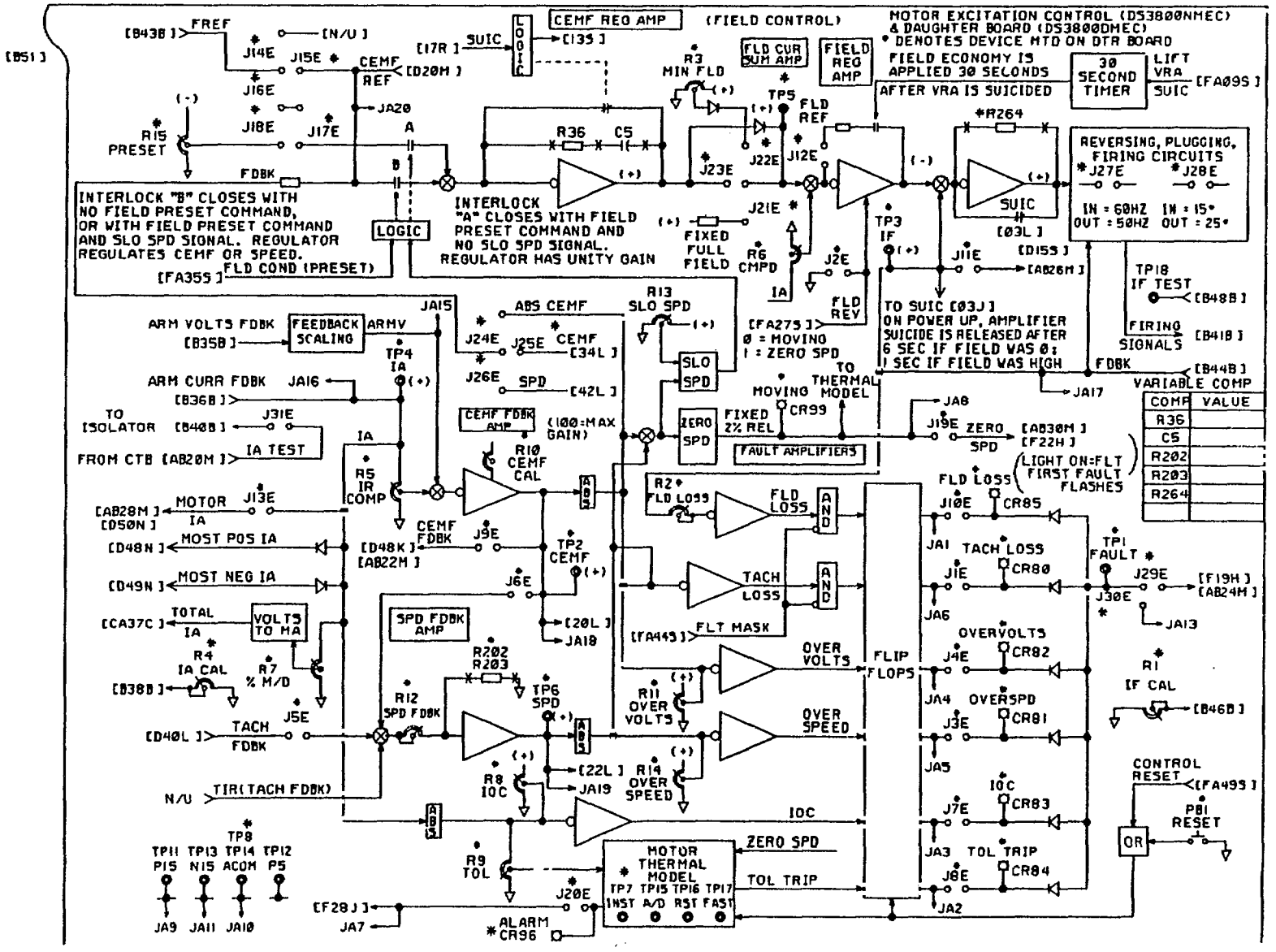


Figure 2-30.
Field Exciter Block Diagram

The thermal overload circuit uses armature current feedback and zero speed indication to detect possible motor overtemperature conditions. The armature current feedback signal is fed into a digital scaling and time constant circuit. The scaling and time constant are modified by a zero speed signal to take into account reduced motor cooling. If the armature current remains at a predetermined level that will raise the motor temperature too high, the TOL ALARM signal is activated and later the TOL TRIP LED lights. The drive will not shut down unless the armature current reaches a trip point approximately 5% above the alarm point.

The tachometer loss circuit compares the absolute value of the CEMF signal and the absolute value of the speed feedback signal with a reference. If the tachometer signal is not present before CEMF reaches 50%, the drive is shut down. The tachometer loss function is not active during a fault mask. A fault mask disables detection of several jumper-selectable faults during field reversal.

Other Functions

Some important PCM functions are handled by circuitry on several cards and by other panel-mounted components such as shunts and switches. These functions include fault detection, PCM reset, feedback processing, M contactor control, and common signal grounding. These functions have been mentioned before, but are covered in detail in this section.

Fault Circuitry. Fault circuitry within the PCM monitors the equipment status and halts or limits operation in the event of abnormal conditions. As shown in Figure 2-31, faults are detected on the NMEC and NFCF boards and processed on the HSQD board. The NMEC board can detect overvoltage, overspeed, overcurrent, motor overload, field loss, and tachometer loss conditions. The NFCF board detects initial phase rotation, SCR instantaneous overcurrent, and loss of operating status (IMOK) conditions. The HSQD board is signalled if any faults occur, and shuts down the PCM in an orderly manner: the Sequencer interrupts applicable reference signals, suicides integrating regulators and the Firing Circuit, and opens the M Contactor. The Sequencer also monitors the drive coordination line input from other PCMs and determines whether its PCM is the first one to fault. If so, the FAULT LED on the HSQD board will flash, indicating a "first fault" by the PCM.

Field Economy. This function will reduce field current to about 70% of its normal value whenever the motor is idle for more than 30 seconds.

Constant Horsepower. These circuits can provide a constant motor horsepower output with a variable motor speed. This is accomplished by maintaining a constant CEMF through field current control.

Reset Circuitry. The primary purpose of the PCM reset circuitry is to clear fault conditions prior to restarting the drive. The PCM can reset from pushbuttons within the PCM or from an operator station. The reset function is illustrated in Figure 2-32.

The two reset pushbuttons within the PCM are located on the NFCF and DMEC boards. The RESET pushbutton on the DMEC board clears faults on the NMEC board only. The RESET pushbutton on the NFCF board clears faults on both the Field Exciter board and the Firing Circuit board. If the PCM is not in a fault condition and either RESET pushbutton is pressed, the Sequencer will react as though a fault had occurred and perform a coast stop. For this reason, the RESET pushbutton on the DMEC board can be used just as the emergency stop operator pushbutton is used. This "fault stop" can be cleared in several different ways. Removing the +105 VDC signal from the N-STOP external input will clear the fault stop, as will applying +105 VDC to the external RESET input.

Depressing the N-STOP button on the operator panel clears external faults seen by the Sequencer through input CTB40; some examples of these faults are local overtemperature conditions and faults signalled from another PCM. Deenergizing the N-STOP will clear these faults (input CTB28). Depressing the operators reset will clear all faults including internal drive faults (input CTB39). **NOTE THAT PRESSING THE NFCF RESET BUTTON DURING DRIVE OPERATION COULD RESULT IN DAMAGE TO THE EQUIPMENT.**

Feedback Circuitry.Several boards within the PCM detect and process current, speed, and voltage feedback from the motor. A shunt in the motor armature circuit provides current feedback through a shunt isolator on the NEPB, DEPB, DMEC and NMEC boards. See Figure 2-33. Since there may be more than one motor powered by the PCM, four separate signals are available at the wiring harness for current regulator feedback: greatest positive current, greatest negative current, sum of all currents, and current from one specific motor. However, all motor protection functions are on a per motor basis and are based on that motor's current. A low-value resistor shunt in the field circuit provides the field current feedback signal.

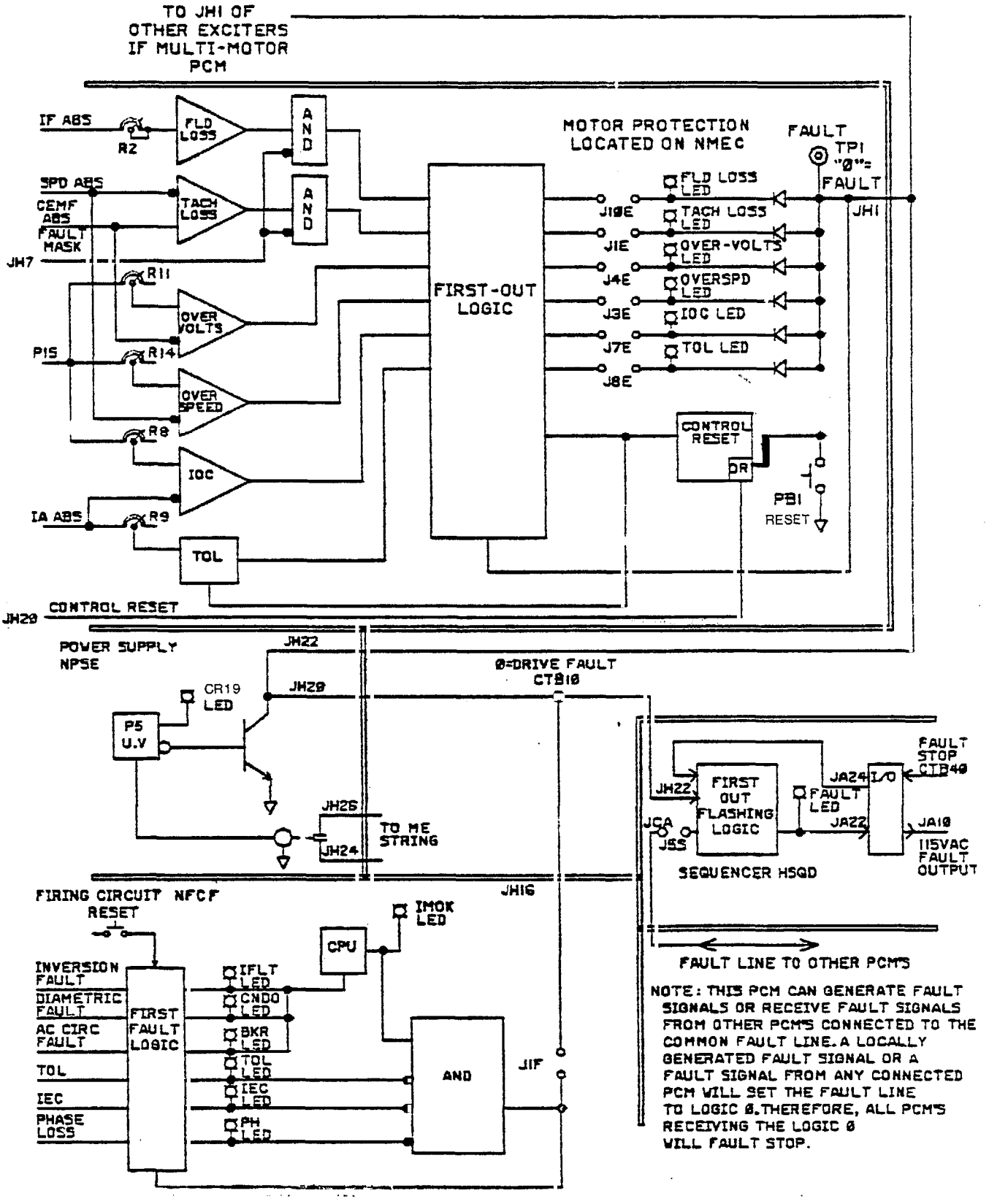


Figure 2-31.
Fault Circuitry

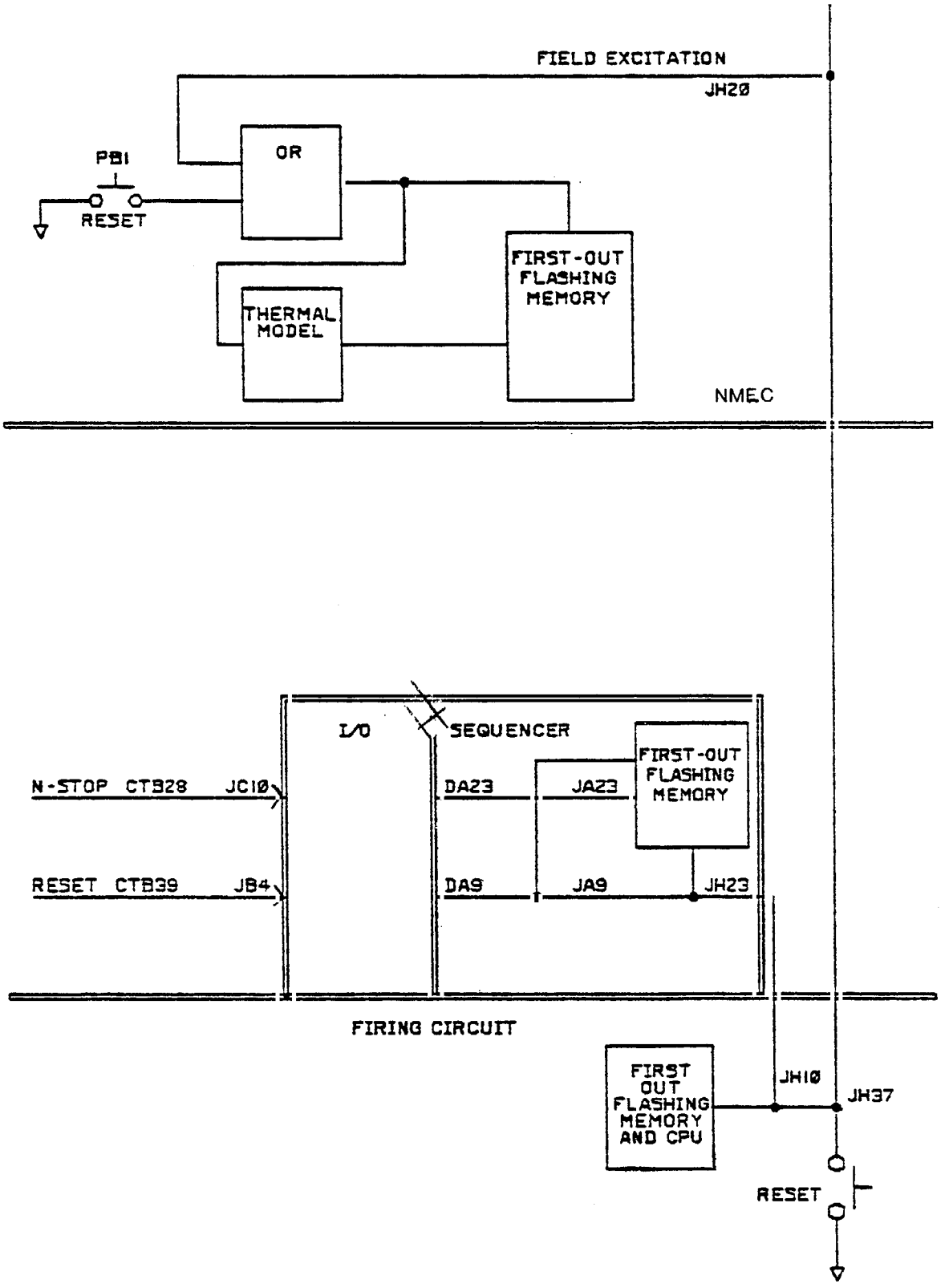


Figure 2-32.
Reset Circuitry

Speed feedback signals may come from one of three sources: an analog tachometer, a digital tachometer, or motor CEMF. See Figure 2-34. The analog tachometer develops a voltage proportional to speed that is scaled and amplified on the NGRC and DGRC boards. If extra motors are being powered by the PCM, each motor will have its own NMEC and DMEC boards; these boards are part of the Per Motor Module (PMM). In either case, the processed feedback signal goes to the NMEC board for motor protection purposes. The NMEC and DMEC boards develop an isolated signal proportional to CEMF that may be used as a speed feedback signal in some circumstances. The speed feedback signal's polarity may be changed via jumper options on the DGRC board.

Two types of voltage feedback are possible within the PCM. The High Voltage and Firing Circuit boards produce a signal that reflects SCR bridge voltage. The Field Exciter boards produce a signal that reflects motor armature voltage. See Figure 2-35.

Current Feedback for Multi-Motor Drives. SCR protection is based on total armature current as determined by the motor-to-drive potentiometer setting(s) (R7E - %MD). The function and setting of this pot is best illustrated by the conditions in the following example:

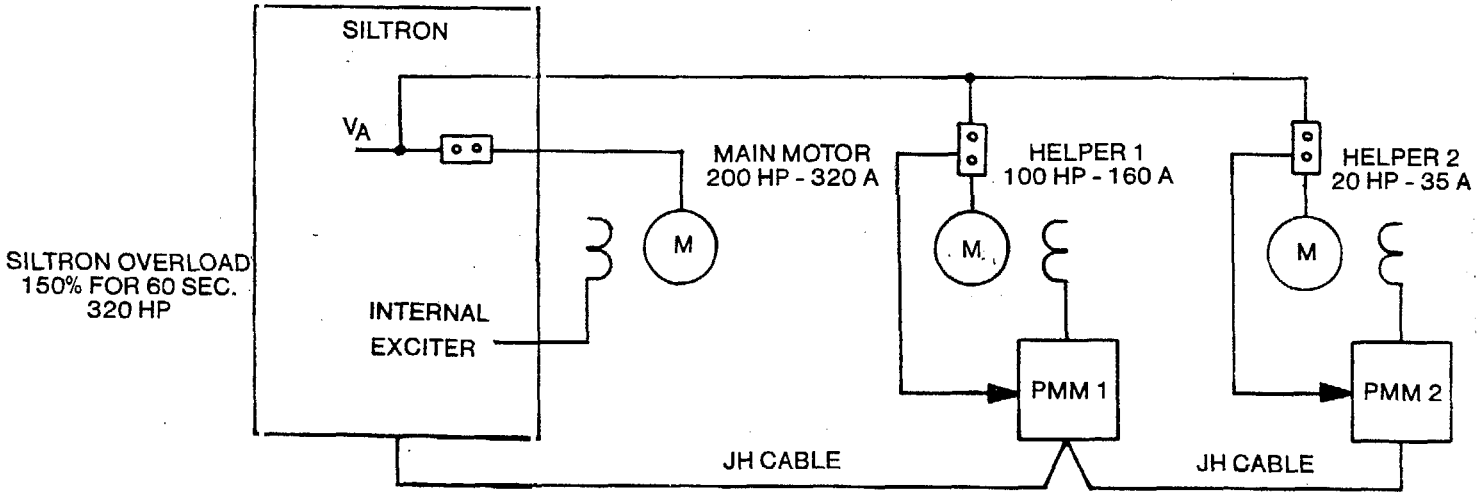


Figure 2-36 illustrates how the JH cable ties the various motor currents together. Assuming we want to regulate the main motor current only, jumpers J13E (DMEC) and J47R (DGRC) have to be in the "IN" position. All individual motor current feedback at TP4E (DMEC) are calibrated the same way: 10 volts = 130% × maximum overload × 100% I_A. The calculated values for 100% currents are shown in the example of Figure 2-36.

Regulator current feedback is determined only by the main motor. Therefore, current feedback to the regulator is 5.12 volts at TP24R (DGRC), which is equal to 320 amperes.

For bridge protection (IEC) the total current from all 3 motors is required. The IEC level is fixed at 4.4 volts at TP22F (DFCF) which is equivalent to the sum of all motor currents times 130% times maximum overload (which is 150% for this example):

$$\text{Volts at TP22F} = 4.4\text{V} = (320\text{A} + 160\text{A} + 35\text{A}) \times 1.3 \times 1.5 = 1004 \text{ amps} = \text{IEC}$$

The individual motor contribution to this voltage is strictly a function of individual motor amps to total amps.

Main motor contribution at TP22F = (200 HP/320 HP) × 4.4V = 2.75V

Helper 1 contribution at TP22F = (100 HP/320 HP) × 4.4V = 1.375V

Helper 2 contribution at TP22F = (20 HP/320 HP) × 4.4V = 0.275V

Total voltage at TP22F for supression = 2.75 + 1.375 + 0.275 = 4.4V

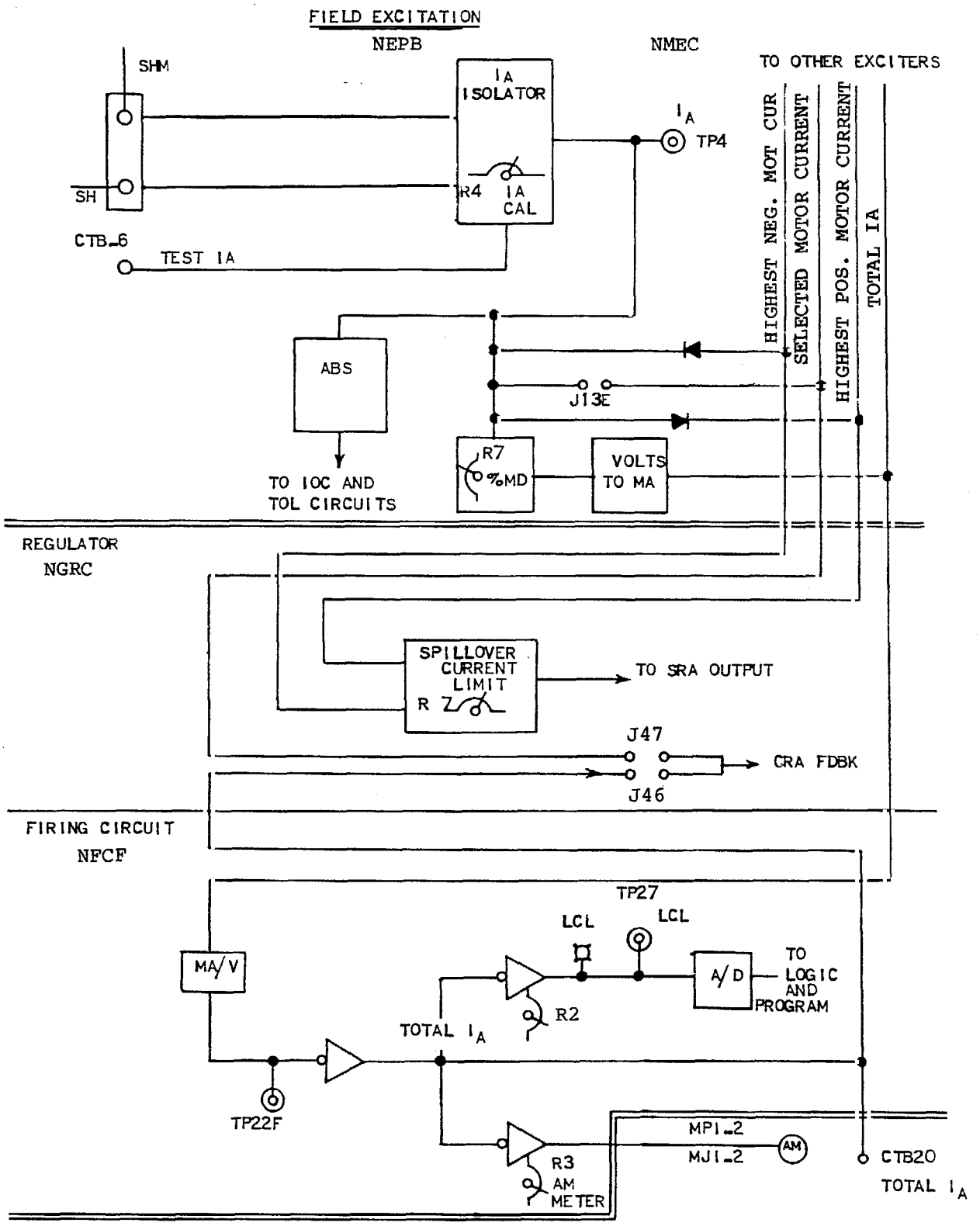


Figure 2-33
Current Feedback Circuitry

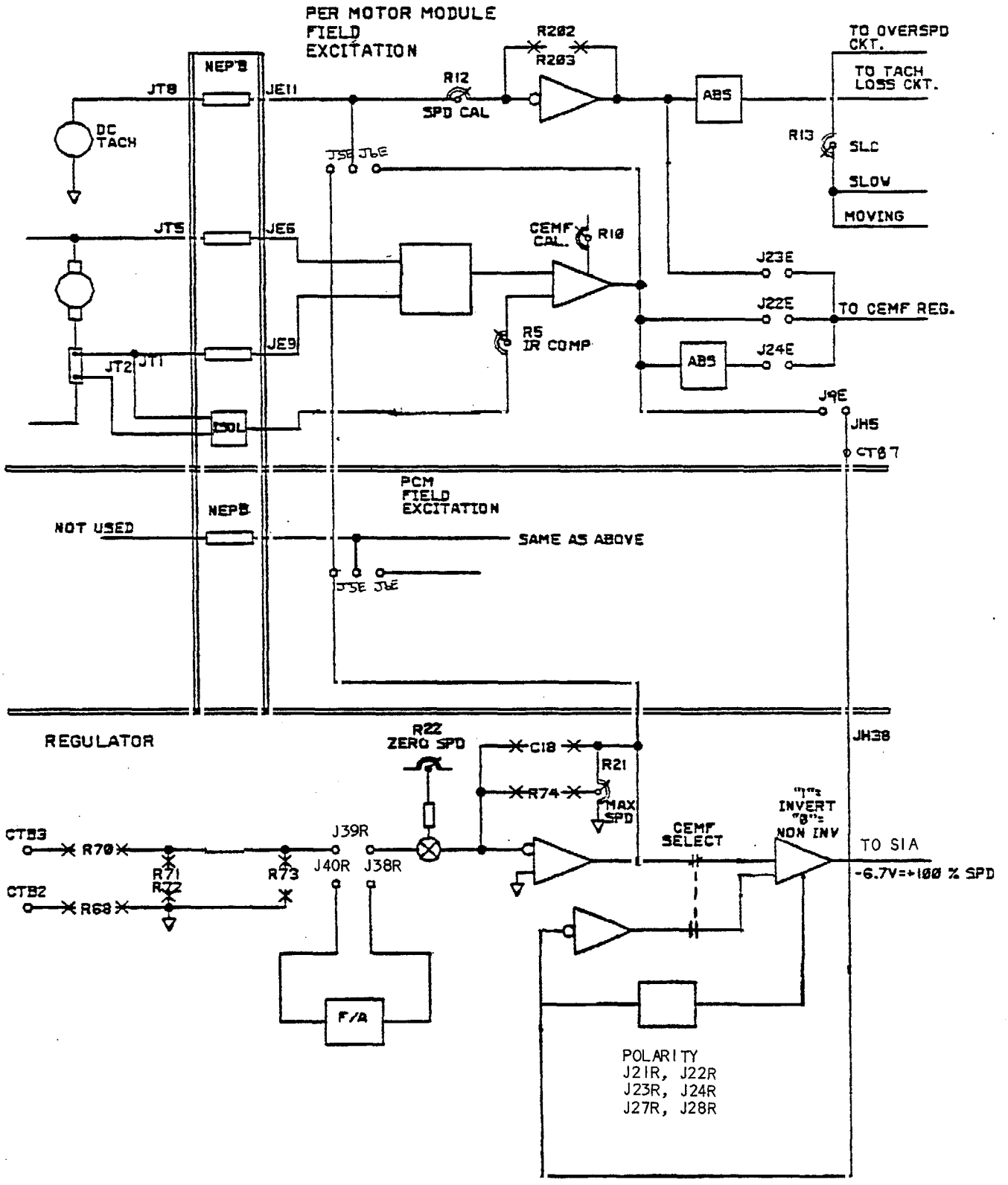


Figure 2-34.
Speed Feedback Circuitry

Adjustment of each motor's contribution is done using the R7 (% **MD** = ratio of **M**otor HP to **D**rive HP) on each drive. This can be adjusted two ways (described using the above example):

A. To adjust one motor at a time, using a millivolt source across shunt and with R4 calibrated for 10V at TP4, adjust R7 for main motor in example for 2.75V at TP22F. Move the millivolt source to second motor shunt and after adjusting R4, adjust R7 (HELPER 1 PMM) for 1.375 volts at TP22F (SILTRON NCFD). Repeat procedure for 3rd motor. Adjust R7 (HELPER 2 PMM) for 0.275 volts.

B. With all test points TP4 simultaneously at 10V (assuming individual I_A calibration previously done) turn all rheostats fully CCW and inject test voltage into CTB6 resulting in a 10 volt signal at each TP4 (SILTRON, PMM1, PMM2). Adjust R7 on main motor (SILTRON) for 2.75 volts on TP22F. Adjust R7 on PMM1 (HELPER 1) until TP22F reads 2.75V + 1.375V = 4.125V. Adjust R7 on PMM2 (HELPER 2) until TP22F reads 2.75V + 1.375V + 0.275 = 4.4V.

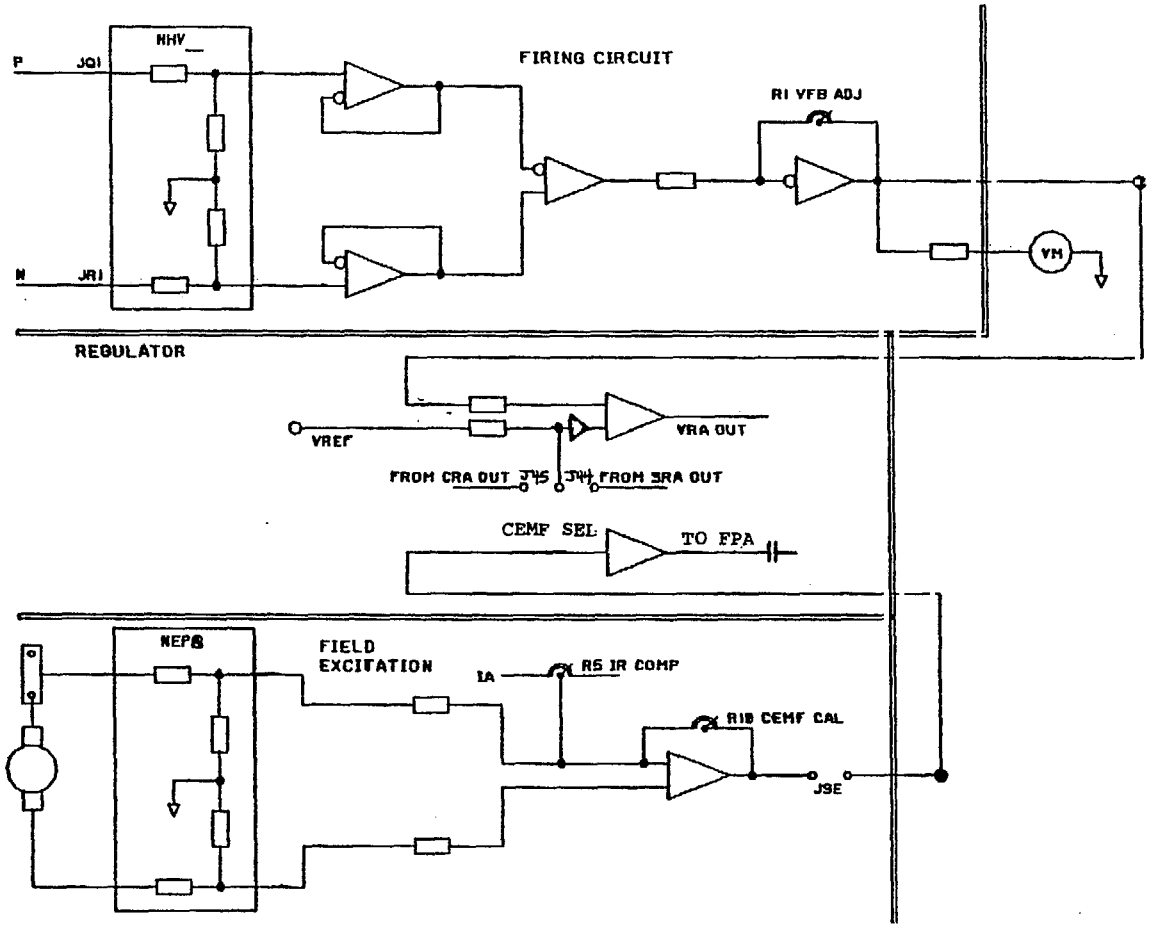


Figure 2-35.
Voltage Feedback Circuitry

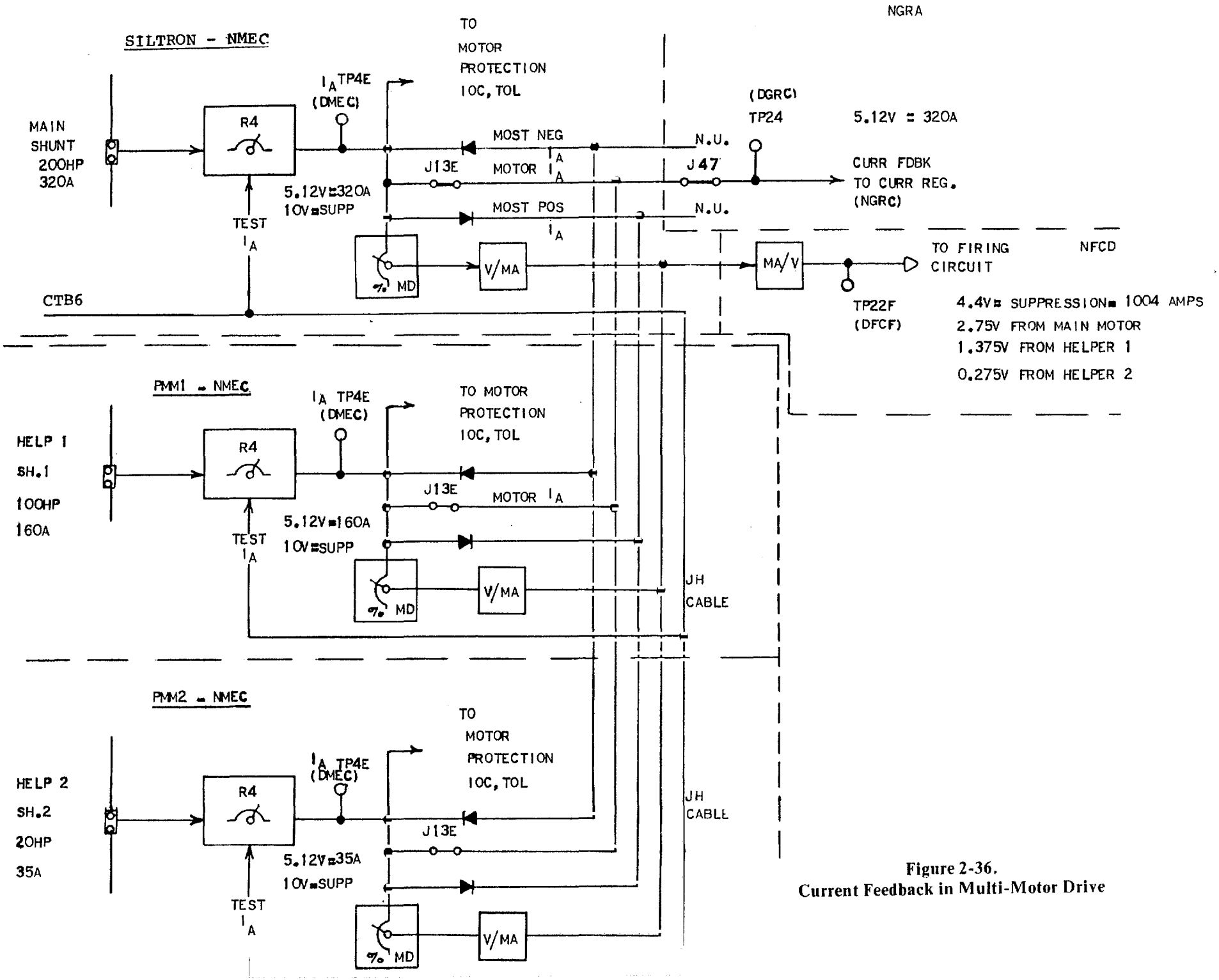


Figure 2-36.
Current Feedback in Multi-Motor Drive

M Contactor Control. The M Contactor control circuits are shown in Figure 2-37. The M Contactor closes when the ME relay on the I/O board is energized. This occurs only when all necessary external interlocks are closed and the Sequencer board sends a signal to the I/O board to close the M Contactor.

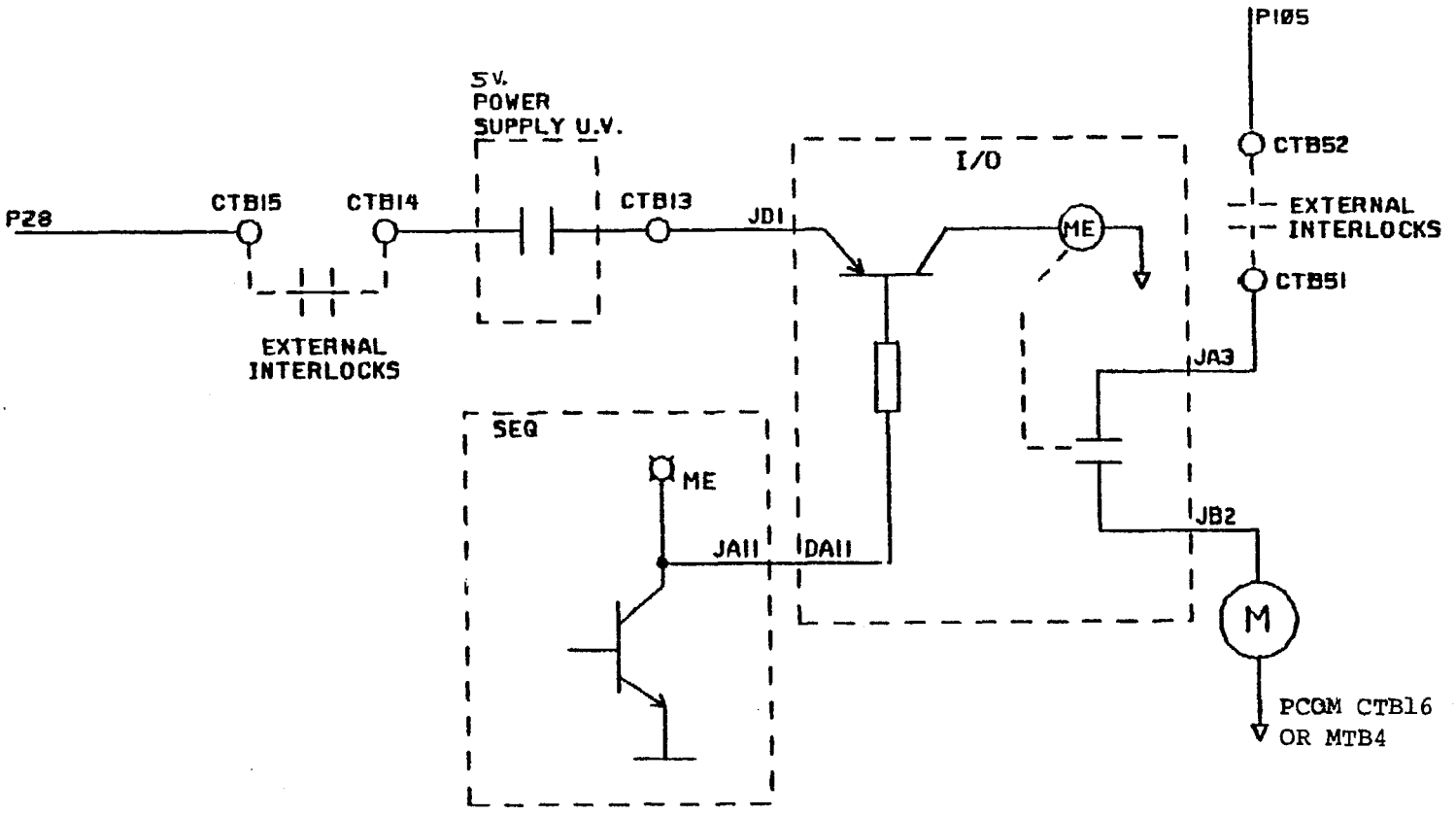


Figure 2-37.
M Contactor Control

Common Grounding. All commons (ACOM, DCOM, and PCOM) are tied together on the NPSE board. A jumper (JGND) bypasses a capacitor and ties those commons to chassis ground. See Figure 2-38. In a system of PCMs tied together by a drive coordination line, no PCM should have this jumper in. The other PCMs should be tied to the first PCM through ACOM/CCOM. All power supplies not jumpered to chassis ground are bypassed to chassis ground through a capacitor. Only individual motor drives may have this jumper in; refer to the system elementary for this information.

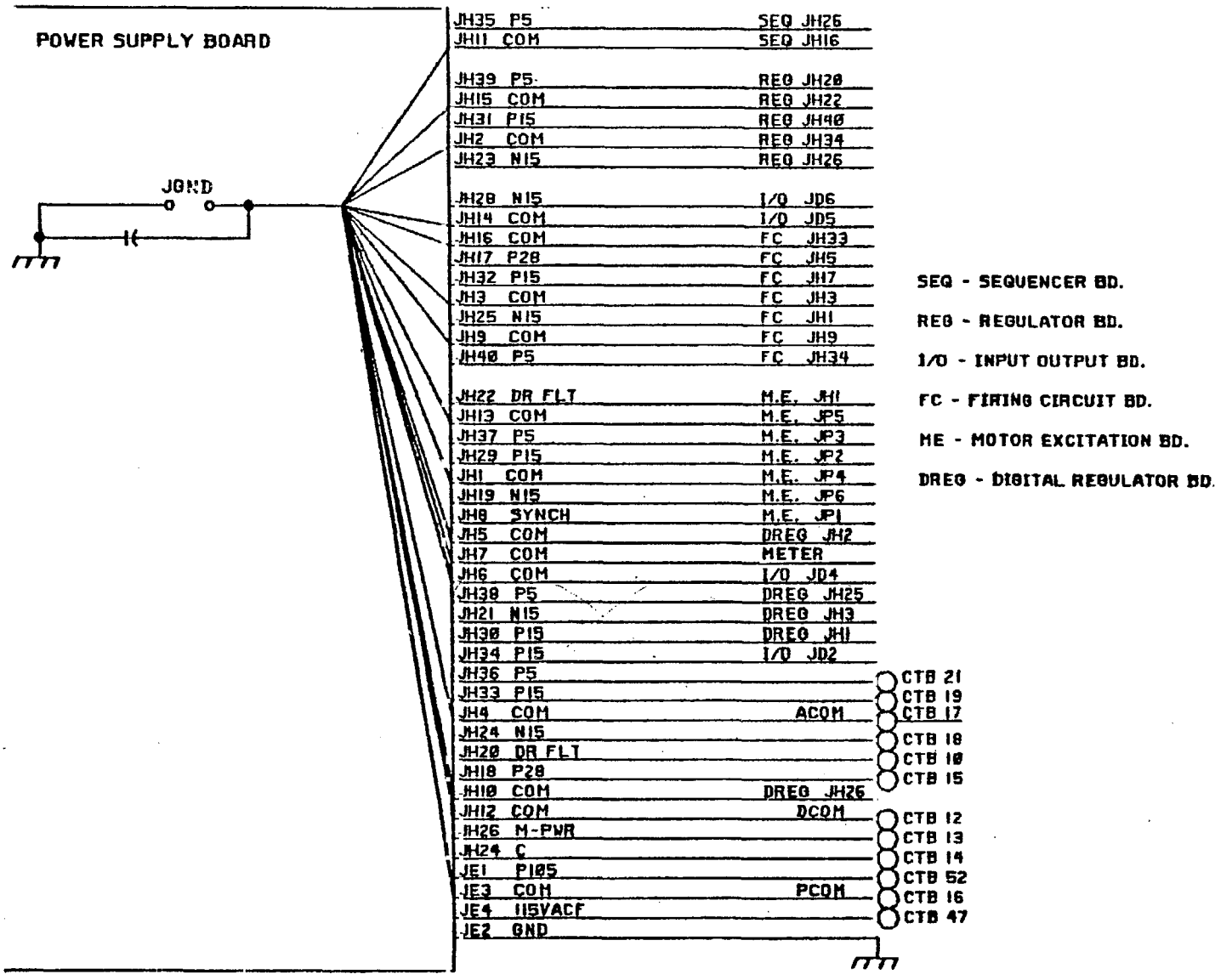


Figure 2-38.
Common Ground Tie Points

— NOTES —

CHAPTER 3
EQUIPMENT SET-UP AND ADJUSTMENTS

This chapter contains set-up and adjustment procedures for the Siltron^{TM} Power Conversion Module. These procedures include setting strapping options and adjusting variable components. Although each module has been tested and adjusted at the factory, you may need to touch up or verify adjustments to suit your specific application. Some adjustments, such as jog and speed points are process-dependent and will change from installation to installation. Process-dependent adjustments should be made during the final stages of installation.

The adjustment procedures in this chapter are for a typical power control module. Some functions or adjustments may not apply to your particular PCM and should be ignored. Refer to the jumper configuration listed in the elementary diagram to see whether a function is used. Information on tuning up other SILTRON configurations can be found in the section titled "Tuning Up Other SILTRON Configurations" which is located directly after the adjustment procedure.

Adjustments not mentioned in this chapter are installation-dependent and should be adjusted as explained in the elementary diagram for the module.

These adjustments are meant to be performed as a single procedure during equipment start-up. Well-trained personnel may perform individual adjustments, but should be familiar with the entire adjustment procedure before doing this type of work. Since one adjustment may affect another, be sure to do the adjustments in the order listed. Always note the position of an adjustment before it is changed so that it can be returned to its original setting if that is desired.

CAUTION

This equipment is designed to operate in a normal industrial atmosphere. Operating radio transmitters nearby may cause the control equipment to misoperate. It is recommended that radio transmitters not be operated within the vicinity of the control equipment when the panel doors are open, nor at any time within the confines of any power control room.

HOW TO USE THE ADJUSTMENT TABLE

The adjustment procedure in this chapter will enable you to properly tune-up the Siltron^{TM} Power Conversion Module. The procedure contains three columns of information:

- COMPONENTS - a list of the potentiometers, jumpers, LEDs, and testpoints involved in the adjustment.
- PROCEDURES - a simple adjustment procedure written in a step-by-step manner for maintenance personnel.
- COMMENTS - a more in-depth explanation of what the adjustment does.

You do not have to read the COMMENTS column to perform the adjustments, although the information in this column will help you understand what the adjustment does. You need only refer to the COMPONENTS and PROCEDURES columns to actually perform the adjustments.

Besides listing the components and telling where they are located, the COMPONENTS column provides other information. Jumpers (components beginning with the letter "J") are further identified by a listing of board coordinates and a reference to the elementary diagram where that jumper can be found. Other components are identified by functional component on the board itself. For instance, an entry in the COMPONENTS column might read J36R (30F D07R). The meaning of this entry is explained below.

J36R (30F D07R)

J36R identifies the component as jumper (J) number 36 located on the Regulator (R) daughter board. The identification **J36R** appears on the board itself next to the jumper. Other components identified by the first letter of the component identification include test points (TP), variable resistors (R), and light emitting diodes (CR). Other boards identified by the last letter of the identification include the Field Exciter (E), Firing Circuit (F) and Sequencer (S) daughter boards.

30F is the coordinate for locating the component on the board. (Coordinates are given only for jumpers (J). Other components can be located by their functional name, which will appear in place of the coordinate. This functional name appears on the board next to the related component.) The numerical coordinates are located along the left side of the board and the letter coordinates are located along the top of the board. The component is located at the intersection of the two coordinates. If the board has no printed coordinates, the symbol - - - will appear.

D07R tells where the component is located within the elementary diagram. (This information is given only for jumpers.) The first letter (D) is the elementary sheet number where the component can be found. The numbers and letter are the coordinates of the component on that sheet.

Unless otherwise specified, all voltages less than 15V should be measured with the built-in test meters. Voltage readings in the adjustment procedures reflect this practice.

HOW TO USE THE SILTRON PCM TEST METERS

Many procedures in this chapter require using the test meters to measure a voltage. The test meters are shown in Figure 3-1. As you read in Chapter 2, the built-in test meters serve two purposes. With the NORM-TEST switch in the NORM position, the meters display field current, armature voltage, and armature current. With the switch in the TEST position, the first two meters become DC voltmeters with a range of -15 VDC to +15 VDC. The input to each meter is through the TEST VOLTS jacks: test jack 1 connects to meter 1 and test jack 2 connects to meter 2. The test jacks accept test leads with banana-type male plugs. Note that all voltages measured on the test meters are with respect to ACOM.

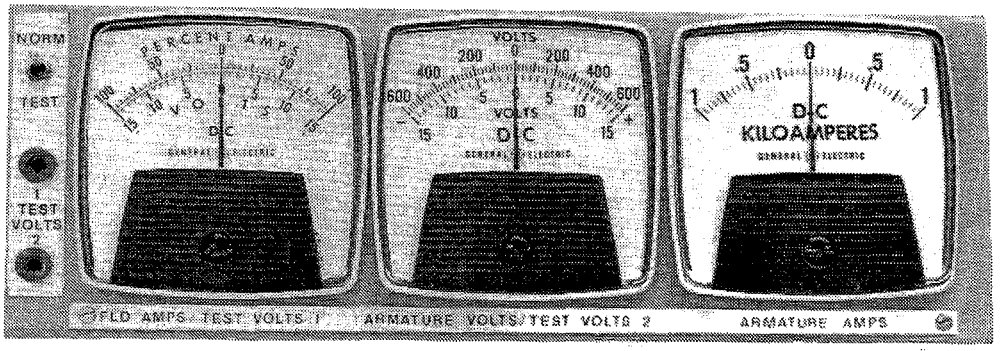


Figure 3-1
Siltron PCM Test Meters

If a procedure requires you to measure a voltage with a test meter and then immediately refer to a value normally displayed on the meter, simply switch the NORM-TEST switch to NORM. It is unnecessary to disconnect the test lead from the test point being monitored.

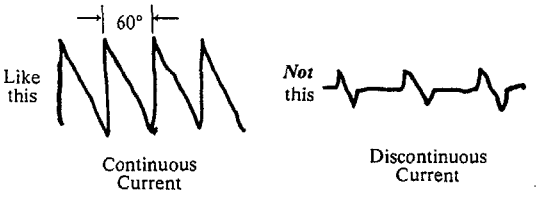
COMPONENTS	PROCEDURES	COMMENTS
	Inhibit reversing on field-reversing type drives until SRA, CRA, and VRA are adjusted.	
		Motor Field Calibration Verification These steps calibrate the field current meter and verify feedback. If an error is found in the feedback, refer to the calibration tables in the elementary furnished with your equipment
J10E (- - - E29R) J12E (- - - E09Q)	Remove J10E and J12E (if used).	Removing these straps temporarily disables field loss & field economy (if used).
R1E - IF Cal	Temporarily jumper capacitor C5 on the DMEC board.	This makes the CEMF regulator straight gain.
R3E - Min Fld	Adjust R1E fully clockwise.	This maximizes current feedback.
	Open field leads and insert ammeter.	
	Adjust R3E until the field meter reads 100%.	Remember that the meter reads in percent. 100% = full field amps.
	Check field current and adjust R1E for rated field current if required.	NOTE: If motor is not to be run for a prolonged period, temporarily adjust R3E until the field amp meter reads 50% of the weak field nameplate rating. This reduces motor heating.
R2E - Field Loss CR85E - Field Loss	Temporarily adjust R3E so that field amps equals 50% of the weak field nameplate reading. For example, if the nameplate reads 1.87/1.3 adjust R3E for 0.5 × 1.3 A = 0.65 A. (If fields are connected in parallel, double all field current settings.)	Field loss setting is 50% of weak field.
J10E (- - - E29R)	Reinsert J10E.	Enables field loss.
	Slowly adjust R2E until CR85E flashes.	
	Readjust R3E for 100% field current.	
	Push SW1 (reset button) to clear the fault.	
J12E (- - - E09Q)	Reinsert J12E (if field economy is used)	Enables field economy (if used).
J44R (- - - D07P) J45R (- - - D11S)	Remove J44R and J45R.	**Convert to Voltage Regulation Only** This disconnects the speed and current regulators coming into the voltage regulator. The purpose is to swing the bridge output voltage with only the voltage regulator.
TP15R - Test Bias TP25R-Test Volt Ref R14R - Test Bias	Jumper TP15R and TP25R. NOTE: Always set the test voltage at zero volts before run and/or stop. Move the pot ***slowly.***	**Connect Test Voltage to Regulator** Allows injection of a variable input directly into the voltage regulator. The input is adjusted by R14R (Test Bias Pot). Clockwise rotation provides positive reference (to +13V) and counterclockwise rotation provides negative reference (to -13V). You can monitor the output at TP15R with the test meters, but the meters may load down the test voltage source. Use a digital voltmeter if an extremely accurate reference voltage is required.
R22R - Zero Spd TP17R - SFB	With motor stopped, adjust R22R for zero volts at TP17R.	**Initialize SFA** This balances any offsets in SFA.

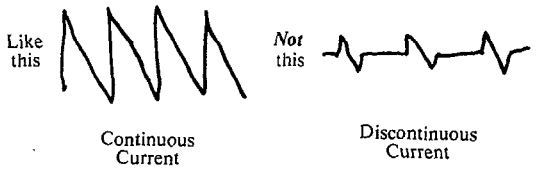
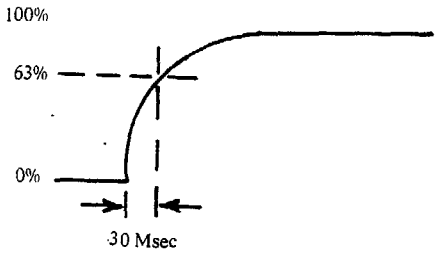
COMPONENTS	PROCEDURES	COMMENTS
		Running Motor Voltage Regulated
R14R - Test Bias R25R-VSA Neg Lim R24R-VSA Pos Lim TP22R- Vlt Sum Out	Adjust R14R for zero volts at TP22R. Adjust R24R and R25R fully clockwise. Monitor the armature voltage with an external meter. Monitor TP22R with the test meter. Pick up run and turn R14R clockwise very slowly. The motor will slowly speed up.	Before starting, adjust the input reference for 0V or the drive might trip on overcurrent. R24R and R25R are voltage limit clamps which must be set out of the way for this test. NOTE: Since the reference goes right into the voltage regulator without current limiting or motor and converter protection, monitor the current while changing R14R ***slowly.***
R12R - VR Gain	Adjust R12R fully clockwise.	
		Armature Voltage Feedback Calibration Calibration is at 1/2 of rated voltage.
R14R - Test Bias	Adjust R14R for -3.5 V at TP22R.	
R1F - Volt FB TP8R - VFB TP22R-Vlt Sum Out	Switch the test meter to normal and compare readings on the external meter and test meter. If adjustments are needed, adjust R1F so that the external meter and the test meter read the same.	This will calibrate voltage feedback. Since the calibration is in the feedback circuit, the built-in voltmeter (which reads feedback) will stay constant while the armature voltage (shown on the external meter) will vary. +7V feedback is equivalent to the following rated armature voltages: 500V for 6 pulse 460VAC input 250V for 6 pulse 230VAC input
		Motor Rotation
TP17R - SFB J21R (- - - D45L) J22R (- - - D45M) J23R (- - - D47M) J24R (- - - D47L) J27R (- - - D49M) J28R (- - - D49L) TP6R - SFA TP21R - CEMF	Check the following: (1) Proper rotation - if not rotate the field leads at FTB (include series field if applicable); (2) Positive polarity speed feedback at TP17R - if incorrect, reverse the tach leads at CTB 2 & 3 or check for correct configuration of J21R, J22R, J23R, J24R, J27R, and J28R per table in elementary.	This procedure runs the motor and verifies proper speed feedback polarity and motor rotation prior to closing the loop later. Changing the jumpers will change the feedback polarity amplifier from inverting to non-inverting or vice versa. This method of changing motor rotation is necessary only if CEMF feedback is used and it is necessary to have a tach feedback polarity at TP6R and CEMF feedback at TP21R. The same applies for field reversal.
		Motor Reversing
R14R - Test Bias J2F (- - - C) TP22R-Vlt Sum Out	Ensure that the process allows reverse rotation of the motor. If J2F is in the REV position, slowly readjust R14R for +3.5V at TP22R to verify reverse operation.	This verifies proper operation of the PCM and motor.
		CEMF Feedback Calibration
R5E - IR Comp TP2E - CEMF R14R - Test Bias R10E - CEMF Cal	Adjust R5E fully counterclockwise. Attach test voltmeter to TP2E. Pick up drive run and while monitoring the motor RPM, increase R14R until the armature voltage is 47.5% of rated voltage (this is 238V for a 500VDC motor and 119V for a 250VDC motor). Adjust R10E until test voltmeter reads 2.5V.	This sets IR compensation to zero. This runs the drive at half the rated CEMF which is equal to V_T - IR (typical IR drop is 5%). With the motor running at no load, V_T = CEMF. R10E will calibrate CEMF feedback (rated CEMF = 5V). This signal is used for comparison to generate signals such as Overvoltage, Tach Loss, Overspeed, and Zero Speed.
		Preliminary Speed Feedback Setting
TP17R - SFB R14R - Test Bias	Move test voltmeter lead to TP17R and adjust R14R until motor runs at about half gear in speed. (To achieve proper speed you may have to adjust the field also.)	Half speed was selected for safety reasons. Top speed calibration will be done later.
R21R - Max Spd TP6E - Spd	Adjust R21R until the test voltmeter reads approximately 3.39V.	Rated speed feedback for the speed regulator is 6.78V at TP6R.
R12E - Spd Adj	Move test voltmeter lead to TP6E and adjust R12E until voltmeter reads 2.5V.	Rated speed feedback for the field exciter board is 5V at TP6E. This signal is used for comparisons to generate signals such as Tach Loss, Overspeed, and Zero Speed.

COMPONENTS	PROCEDURES	COMMENTS
		Overvoltage Adjustment
R14R - Test Bias R11E - Overvolt CR82E - OV	Adjust R14R until the armature voltage is 300V for 250V drives or 600V for 500V drives. Verify that the drive OV trips at this point and that the overvoltage light CR82E comes on. Adjust R11E if necessary.	This sets voltage to 120% of the rated armature voltage.
	Reset the drive.	
R24R - VSA Pos Lim R25R - VSA Neg Lim	Adjust R25R for a positive output just below the trip point.	
	Adjust R25R until the armature voltage is reduced to 275V for 250V drives or 550V for 500V drives.	This sets the voltage limit allowed by the regulator to 110% of the rated armature voltage.
	For reversing drives turn R14R counterclockwise for a negative output just below the trip point. Adjust R24R for -275V for 250V drives or -550V for 500V drives.	
	On drives with field reversal, visually set R24R at the same point as R25R.	
	Slowly turn R14R to zero volts and then NSTOP the drive.	
		Overspeed Adjustment
J4E (- - - E37R)	Remove J4E and reset the drive.	Removing J4E disables overvoltage fault detection while setting overspeed. Overspeed is set at 120% of rated speed unless otherwise limited by mechanical requirements.
R14E - Overspd CR81E - OS TP6E - Spd J3E (- - - E40R) R3E - Min Fld	Pick up run and increase R14R until the test voltmeter reads 6V at TP6E. (NOTE: If unable to reach 6V, slowly decrease R3E until the test voltmeter reads 6V.) If J3E is inserted, verify that the drive trips at this point and that the overspeed light CR81E comes on.	V_{TP6E} = Overspeed % at trip × 5V V_{TP6E} = 120% × 5V = 6V Decreasing R3E weakens the field more. (R3E will be recalibrated later.)
	Reset the drive and reinsert J4E.	
		Set Up for Locked Rotor Tests
	Turn off power.	
	Short field leads together.	
J10E (- - - E29R) J31E (- - - E24G)	Remove J10E and J31E (if in).	Disables field loss detection while field is shorted.
	Remove fuses FU7 and FU8 and turn power back on.	Removes field exciter AC input.
	Lock the rotor. This typically is not necessary if the motor is coupled up but it is recommended that someone be located at the motor to alert you if the motor is moving.	Series fields can cause a drive to rotate if the rotor is not locked.
		Armature Current Feedback Calibration
TP4E - IA R14R - Test Bias R4E - IA Cal	**Method 1:** Attach an accurate millivoltmeter to the shunt. Monitor TP4E with the second meter.	This method uses the actual shunt voltage (equivalent to amps) by circulating current through the motor.
	Pick up run and circulate approximately 50% of rated armature current through the motor by adjusting R14R ***slowly.***	50% current is used to protect the motor.
	Monitor the voltage across the shunt and at TP4E and log the readings.	Never leave current on for more than 15 to 20 seconds or you may damage the motor.
	Drop out run.	

COMPONENTS	PROCEDURES	COMMENTS
	Calculate the required voltage at TP4E per the example below: EXAMPLE: Motor Rating = 64A (40 HP) Shunt Rating = 100MV/75A Max Converter Overload = 200% for 60 sec. Voltage Measured at Shunt = 46 MV Equiv. Amps = 75A/100MV × 46MV = 34.5A Voltage at TP4 should be: V_{TP4E} = 34.5A/64A × 769/200 = 2.07V	769 = 10V/1.3 × 100%
	GENERAL FORMULA: V_{TP4E} = (Shunt Amps/Rated Motor Amps) × (769/Max Overload %) Shunt Amps = Shunt Rating [A/MV] × Reading Across the Shunt [MV] Compare TP4E reading with calculated value and adjust R4E if necessary.	Current feedback at TP4E = 10V at suppression. Suppression (IEC) is fixed 30% above maximum overload. Therefore, rated feedback at TP4 = 10V/(1.3 × Max OL [P.U.]). For 150% overload, 1 P.U. feedback is 5.13V at TP4. For 200% overload, 1 P.U. feedback is 3.84V at TP4.
TP4E - IA R4E - IA Cal J31E (- - - E24G)	**Method 2:** Disconnect both signal wires from the shunt and attach a millivolt source across the signal wires. Make sure that J31E is out.	This method uses a millivolt source to simulate current feedback.
	Calculate MV required at signal wires for 10V at TP4E.	10V at TP4E = 7.69 × 1.3 = 1.3 × Max OL × Rated DC Current.
	Calculate required voltage across shunt wires as follows: EXAMPLE (Same data as above): Curr. which would cause suppression (IEC) = (1.3 × 200% × 64A)/100 = 166.4A Shunt voltage for 166.4A equals: V_{SH} = 100MV/75A × 166.4A = 222MV GENERAL FORMULA: V_{SH} = Shunt Rating [MV/A] × (2×1.3) × Rated I_A	Same comments as for Method 1.
	Set the millivolt source for the calculated value and adjust R4E for 10V at TP4E. NOTE: When you measure the millivolts, measure across the signal wires directly at the shunt connection.	If the feedback is not correct, refer to the calibration tables in the elementary supplied with your equipment.
TP15R - Test Bias J31E - (- - - E24G) R14R - Test Bias TP4E - IA	Temporarily insert J31E (on NMEC). Change jumper from TP15R/TP25R to TP15R/CTB6 (CTB terminal 6). R14 set for 0V at TP15R. NOTE: When using a millivolt source ignore this step and calculate the required voltage across the shunt per the example and general formula in Method 2.	**Motor and Converter Protection Adjustment** This inputs a variable voltage to simulate current feedback and allows setting motor and converter protection by dummying the required voltage. The following five adjustments are motor and converter protection adjustments.
TP7E - INST TP8E - ACOM R5E - IR Comp	Jumper TP7E to ACOM and temporarily adjust R5E fully clockwise to simulate a run condition.	**Thermal Overload Setting Verification** This permits determining the thermal overload trip point for the motor running condition instantaneously.
R14R - Test Bias TP4E - IA R9E - TOL Adj CR99E - MOV CR96E - ALM CR84E - TOL	Slowly adjust R14R until the TOL light (CR84E) flashes. This should occur for 1.04 P.U. current at TP4E. If not, set voltage at TP4E for 1.04 P.U. current and, while holding in RESET (SW1 on DMEC), slowly adjust R9E until the TOL light flashes.	The MOV and ALM lights should come on before the TOL light flashes.

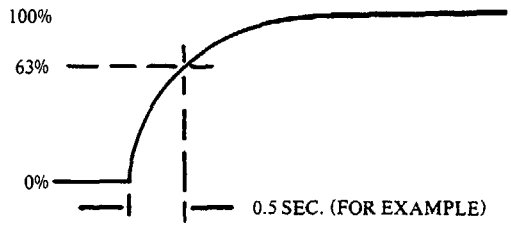
COMPONENTS	PROCEDURES	COMMENTS
	For 150% overload use 5.33V at TP4E for 1.04 PU current. For 200% overload use 4.00V at TP4E for 1.04 PU current. For 250% overload use 3.22V at TP4E for 1.04 PU current.	Set point = 10V × 1.04/1.3 × P.U. max OL current.
	Lower the R14R adjustment for 1 P.U. current at TP4E and push SW1 (Reset on the DMEC board).	The TOL light goes off and ALM light stays on.
	For 150% overload use 5.13V at TP4E for 1 PU current. For 200% overload use 3.85V at TP4E for 1 PU current. For 250% overload use 3.08V at TP4E for 1 PU current.	Set point = 10V × 1.04/1.3 × P.U. max OL current.
	Remove the jumper from CTB6.	The Alarm light goes off because the motor has no current and is cooling.
	Adjust R5E fully counterclockwise.	This simulates a stalled motor, and the MOV LED will stay off.
	Remove the jumper between TP7E and ACOM.	This allows the thermal overload mode to run at the normal rate.
	Reconnect jumper to CTB6 and measure the time for the TOL light to flash. This should be 50 - 60 seconds as previously set by R9E.	R9 is now set properly.
	Disconnect jumper to CTB6.	Verification.
	Adjust R5 full clockwise.	This simulates a running motor and the MOV LED will come on in the next step.
	Reconnect jumper to CTB6 and measure time. Verify that the ALM light comes on after approximately 15 minutes, and that the TOL light does ***not*** trip for at least 60 minutes.	These test times can be decreased by a factor of approximately 10 by connecting TP17 (on NMEC) to ACOM and repeating test.
	Return R5 to original setting or approximately full CCW. Final set-up is made in CEMF calibration step.	
		Motor Overcurrent Setting
	Adjust R8E clockwise.	This moves IOC setpoint out of the way.
R14R - Test Bias R8E - IOC Adj CR83E - IOC TP24R - Curr Fdbk TP4E - IA	Adjust R14R until the test voltmeter reads 9.5V at TP4E. Slowly adjust R8E counterclockwise until the IOC light CR83E flashes.	This simulates the motor overcurrent level (IOC). This sets the IOC setpoint = 0.95 IEC.
		Converter Overcurrent Suppression (IEC)
R14R - Test Bias TP22F - TIA R7E - %M/D CR47F - IEC	Adjust R14R until the test voltmeter reads 10V at TP4E. Monitor TP22F with the second test voltmeter. For single motor drives adjust R7E for 4.4V at TP22F.	Suppression (IEC) = 10V = Max Overload Current × 1.3 P.U. amps at TP4E. Suppression = 4.4V at TP22F.
	For multi-motor drives, refer to Chapter 2 of this manual for further explanation of the R7E adjustment.	For multi-motor drives converter suppression is based on ***total*** armature current. For instance, for two drives with the same horsepower on one Siltron PCM, each motor would contribute 2.2V worth of current to get 4.4V total current feedback.
		Logic Current Limit
R2F - Logic CL	Adjust R2F fully counterclockwise.	This disables logic current limit since it is not effective for this configuration.

COMPONENTS	PROCEDURES	COMMENTS
R14R - Test Bias R3F - Ammeter TP22F - TIA J31E (- - - E24G)	**Ammeter Calibration** Calculate voltage at TP22F equal to current for an easy meter reading as shown below: EXAMPLE: Ammeter scale = 0 - 150A Motor amps = 64A Overload = 200% for 60 seconds Calibration desired at 150A Calculate test voltage for 150A at TP22F. 4.4V at TP22F = 1.3 × 200% × 64A/100 V_{TP22F} for 150A = (4.4V × 150A) / (1.3 × 2 × 64) = 3.97V	
	GENERAL FORMULA: V_{TP22F} = (Amps at Calibration Point/Rated Armature Amps) × (338/Max Overload %)	338 = 440/1.3
	Adjust R14R until test voltmeter reads this value at TP22F. Adjust R3F to read the required amps.	
	NOTE: On multi-motor drives, rated armature current is the sum of currents from all drives.	The ammeter reads ***total*** current, therefore you must look at TP22F and not TP4E.
	Remove J31E.	
TP12R - Test Curr In TP14R - Step Out R5R - Step Size R4R - Step Bias	**Current Rate Regulator Tuneup** Move jumper from TP15R and CTB6 to TP14R and TP12R.	
J9R (- - - D30P) J10R (- - - D27P) J42R (- - - D31P) J43R (- - - D35P)	Temporarily remove J9R, J10R, J42R, and J43R. Also make sure the drive is not in current mode. Temporarily set R28R (Current Rate Limit) fully CW.	This ensures that there will be no other inputs into the CSA amplifier.
J45R (30F D11S)	Insert J45R.	This connects the current regulator to the voltage regulator.
R29R-Pos CRA Lim R9R-Neg CRA Lim R23R-CSA Pos Lim R13R-CSA Neg Lim	Record settings before temporarily adjusting R9R, R13R, R23R, and R29R fully clockwise.	This disables all clamps (limits) on the current regulator by setting them to their maximum value.
TP8R - VFB TP23R - CRA Out R19R - CRA Gain C42, R157	Check that C42 = 0.022 mfd., R157 = 332 KΩ, C40 × R158 = 1/150, and J32R is in.	If different values are used, see elementary for tuneup instructions.
J32R (- - - D17U) C29A, R111	Jumper top of R111 to top of C29A or short C29/29A on DGRC. Pick up run.	This prevents SRA from saturating. Motor should be in locked rotor condition.
	Monitor TP8R (VFBK) with an oscilloscope and slowly adjust R4R for continuous current, usually at 10% to 20% rated current.	For continuous current, voltage feedback at TP8R should be as follows: 
	Monitor TP23R with an oscilloscope and adjust R19R for 0.40 to 0.44V peak-to-peak ripple.	
	Adjust R4R for zero volts at TP14.	This removes current reference.

COMPONENTS	PROCEDURES	COMMENTS
		Current Regulator Response Verification
TP24R - Curr Fdbk TP8R - VFB R4R - Step Bias	Monitor TP24R with a strip chart recorder. Adjust R4R for zero volts at TP14R and pick up the contactor. Adjust R4R for continuous current, which is usually 20% for 6 pulse operation. Confirm by observing on an oscilloscope that the voltage feedback at TP8R looks as follows:  	
R5R - Step Size	Adjust step size R5R for a large current step (100% if possible).	Current must increase and not go through the power converter deadband.
SW1R - Step TP23R - CRA Out	Push STEP button SW1R and run strip chart recorder at high speed (100 or 125 MM per second). NOTE: Never leave current on for more than 15 - 20 seconds or you may damage the motor. The response should look as follows:  	Note that there will be current feedback ripple superimposed on the trace shown at left. No further adjustments should be required for R19R under stall conditions. However under full field and operating conditions it may be necessary to readjust R19R to compensate for interaction between the motor field flux and armature time constant. Also, high current limit settings may affect the armature time constant and thus the stability. These may show up as instability or ringing on current feedback under certain operating conditions. If this occurs, lower the gain by turning R19R counterclockwise until the ringing disappears. Such readjustment is not uncommon. It may be easier to monitor TP23R (CRA Out) with an oscilloscope and adjust R19R for the proper peak-to-peak ripple.
	Drop out contactor.	
R29R-Pos CRA Lim R9R-Neg CRA Lim	Turn R5R fully clockwise.	**CRA Limits & Voltage Limits**
R5R - Step Size R4R - Step Bias J45R (- - - D11S) TP23R - CRA Out	Temporarily remove J45R and pick up the contactor. Monitor TP23R. Turn R4R fully clockwise and adjust R29R for +7.7V. Turn R4R fully counterclockwise and adjust R9R for -7.7V. Drop out contactor and reinsert J45R.	This disconnects the CRA regulator from the VRA regulator. The output of the current regulator is a voltage reference (7.0V for rated voltage). Limits should be set to allow 10% overvoltage (i.e., set just below overvoltage pickup on NMEC board). NOTE: For non-reversing drives with field reversal both clamps will have to be set. ***Do not*** set the negative clamp for zero volts.
	Pick up contactor.	**CEMF Calibration**
TP2E - CEMF R5E - IR Comp R4R - Step Bias	Move test voltmeter leads to TP2E. Adjust R4R for 100% current and verify that TP2E = zero volts. Adjust R5E if needed. Adjust R4R toward zero current and see that TP2E remains at zero. Drop out contactor.	With locked rotor, CEMF = 0. V_T = CEMF + IR = IR The output of the amplifier has to be zero volts at stall.

COMPONENTS	PROCEDURES	COMMENTS
R23R-CSA Pos Lim R13R-CSA Neg Lim TP15R - Test Bias R14R - Test Bias TP11R - CSA Out TP12R-Test Curr In	Move jumper from TP14R to TP15R. This connects TP15R to TP12R. Turn R14R fully clockwise and monitor TP11R. Adjust R13R for -9.3V. Then turn R14R fully counterclockwise and adjust R23R for +9.3V.	**Current Summing Amplifier Limit** This procedure inserts a variable voltage source into the current summing amplifier and sets the current limit on the CSA just below overcurrent (IOC). Overcurrent = Max Overload × 1 P.U. I_A = 9.3V. This value is fixed and is based on design criteria. 9.3V = current limit when using motor I_A as feedback to CRA. When using total I_A as feedback, use .88 × 9.3V = 8.18V.
R28R-Cur Rate Lim R14R - Test Bias TP24R - Curr Fdbk	When R14R is quickly changed from fully clockwise to fully counterclockwise, the current reference swings from -9.3V to +9.3V. This is equivalent to a current change of 300% for 150% overload or 400% for 200% overload. EXAMPLE: Max Overload = 200% Desired Current Rate Limit = 20 per unit per second Calculate the time for 400% excursion or 4 P.U. T = 4 P.U./20 P.U. per second = 0.2 seconds If the current reference changes from +9.3V to -9.3V in 200 Msec, it would be equivalent to 20 per unit per second.	**Current Rate Limit** 18.6V swing = 2 × Max Overload since 9.3V = 1 P.U. × Max Overload
	GENERAL FORMULA: T = 2 × Max Overload [P.U.]/Desired Rate Per Second [P.U. per second]	NOTE: On variable field motors the maximum current rate can be very low. If this value is exceeded, motor flashover can occur.
	Pick up contactor and swing R14R quickly from fully clockwise to fully counterclockwise. Adjust R28R so that the current feedback monitored by the strip chart recorder on TP24R changes in the calculated time.	
	Drop out contactor and remove power.	
J9R (40F D30P) J10R (40F D27P) J42R (42F D31P) J43R (42F D35P)	Reinstall removed jumpers per elementary.	
	With power off, unlock rotor.	
J10E (- - - E29R)	Replace J10E and reconnect field.	This reactivates field loss.
	Reinsert fuses FU7 and FU8.	This applies AC power to the field exciter.
	Remove jumper between R111 and C29A on DGRC board.	
	Turn power on.	
R27R - SRA Pos Lim R26R-SRA Neg Lim TP18R - TST SR R14R - Test Bias TP15R - Test Bias SW2R - Cascade TP19R - SRA	Move jumper from TP12R to TP18R. This connects TP15R to TP18R. Place CASCADE switch SW2R into the cascade position. Monitor TP19R. Adjust R14R fully clockwise and adjust R26R for -9.3V (-8.18V if using total I_A) at TP19R. Adjust R14R fully counterclockwise and adjust R27R for +9.3V (+8.18V if using total I_A) at TP19R.	**Current Limit SRA** This step inserts a reference into SIA to swing amplifier positive and negative. This step lifts the suicide and makes the amplifier straight gain (1:1).

COMPONENTS	PROCEDURES	COMMENTS
	Place CASCADE switch back into the normal position.	
	Remove jumper between TP15R and TP18R	
		Speed Regulator Tune Up
C28A C29 C29A	Remove C28A from DGRC board. Short C29 or C29A on DGRC board..	This will take all electrical time constants out of the speed regulator loop except T_M. This prevents drive from being unstable.
R8R - Gain TP17R - SFB	Monitor TP17R with recorder.	The recorder monitors speed feedback.
TP14R - Step Out TP18R - TST SR	Set R8R at midpoint.	This sets rise/droop to zero.
R4R - Step Bias R18R - Rise/Droop	Jumper TP14R to TP18R.	This repeats a previous procedure of inserting a variable bias voltage and stepping the voltage. This enables you to set a certain minimum speed with R4R and increase speed with R5R via the STEP button.
R11R - SLO/THD	Turn R11R fully counterclockwise.	This temporarily disables the thread reference.
R4R - Step Bias R5R - Step Size SW1R - Step TP19R - SRA	Pick up thread and adjust R4R until the drive runs at about 10% - 20% speed. Adjust R5R for a step when pushing the STEP pushbutton on the regulator daughter board. NOTE: Monitor TP19R to ensure that SRA stays out of saturation while stepping the speed regulator.	This runs the drive at slow speed and steps the SRA.
R8R - Gain	Verify and adjust R8R (gain) for the required response per the elementary. If the response is wrong proceed as follows: EXAMPLE: Required response time = 0.5 sec. Measured response time (with original R111) = 2 sec. or 0.5 radians per second Adjust R8R for required response. If out of range, set R8R at midpoint and re-measure response. Increase R111 by a factor of 4 (4 = 2/0.5). $R111_{NEW}$ = (Actual Response/Required Response) $\times R111_{OLD}$ If response is 0.1 sec. with original R111, decrease the resistance by a factor of 5. After changing R111, recheck response. A typical response curve is shown below:	



Now calculate C28 and C29 from the following formula:
C29 [mfd] = Response in seconds/R111 in MΩ
For example, with R111 determined to be 500 KΩ, a 0.5 sec. response gives:
C29 = 0.5 sec./0.5 MΩ = 1 mfd
Check that R112 = 150 KΩ (fixed)
C28 [mfd] = Response in seconds/R112 in MΩ
For example, a 0.5 sec. response gives:
C28 = 0.5 sec./0.15 MΩ = 3.3 mfd
Insert C28A and remove jumper across C29 or C29A on DGRC board.

COMPONENTS	PROCEDURES	COMMENTS
R12R - VR Gain	Recheck response. Readjustment of R8R should not be necessary.	If a light load instability is encountered, turn R12R counterclockwise to optimize. A slight adjustment of the VSA clamps (R24R, R25R) and CRA clamps (R9R, R29R) may also be necessary.
R11R - SLO/THD R12R - ACC R1R - DEC	Remove jumper from TP14R to TP18R. Pick up drive thread/slow and with R11R fully clockwise adjust R2R for the desired acceleration rate. Push stop and adjust R1R for the desired deceleration rate. NOTE: If a master ramp into RUN REF is used, make sure that the internal ramp is faster than any such master reference.	**Ramp Adjustments** This procedure runs the drive to top speed using the thread reference pot and adjusts acceleration and deceleration rates.
R11R - SLO/THD R13E - Slo Spd R22R - Zero Spd CR99E - Mov	Pick up drive thread/slow and with R11R fully counterclockwise adjust R22R so that the drive moves at its slowest speed or stops moving. Increase R11R to the desired slow/thread speed setting, typically between 1% and 5%. Adjust R13E until light CR99E (MOV) goes from dim to bright. Stop the drive.	**Slow Speed Adjust**
R11R - SLO/THD R6R - Jog Fwd R15R - Jog Rev R17R - STU R3R - Quick Stop R21R - Max Spd	Adjust the following per process requirements: R11R - Thread speed R6R - Jog forward speed R15R - Jog reverse speed R17R - Slack take up R3R - Quick stop Connect the final master reference into the drive and pick up drive run. Adjust the reference for maximum (i.e., operator rheostat fully clockwise). Use a hand tach and adjust R21R for exact surface speed required.	**Machine Speed Adjust** These are typically requisition variables and are set for the drive requirements.
R3E - Min Fld	Adjust R3E so that the armature voltage is 95% of rated voltage. Field amps should not exceed 100%.	**Motor Field Adjustment** This adjusts the field so that the motor runs at no load CEMF (V_T - IR, where IR drop is typically 5%).

— END OF ADJUSTMENT PROCEDURE —

TUNING UP OTHER SILTRON CONFIGURATIONS

The standard adjustment procedure just described is for a typical single motor SILTRON drive with a fixed field and a speed regulator with internal current and voltage loops. The flexibility of the regulator circuits permits many other configurations. Detailed information concerning other configurations and functions is contained in the systems drawing, but some general considerations for these other functions are included below.

Speed by CEMF with Internal Current and Voltage Loops. Tune-up is similar to the standard adjustment procedure except that CEMF rather than a tachometer is used to provide speed feedback.

Speed (Tachometer or CEMF) With Internal Voltage Loop Only. Again, the detailed tune-up is similar to the standard adjustment procedure. But while the current feedback calibration and the current protective functions must still be set up, the current regulator is omitted and the speed regulator feeds the voltage regulator. This means that the SRA clamps (R26R and R27R) should be set as voltage clamps instead of current clamps. Note that this type of regulator usually employs the spillover current limit function (described below).

Spillover Current Limit. Adjustments to R7R (Spillover Curr Limit) and R20R (C L gain) are made in the locked rotor mode (see details of tuning up the current regulator in the standard adjustment procedure for procedures and precautions). Current can only be applied for a short duration to avoid damaging the commutator. Initially, R7R is set at some low value (approximately 5%) and the speed reference is adjusted for rated speed. R7R is increased so that the drive supplies continuous current as observed at TP8R (-VFB); see details in the standard adjustment procedure. Then R20R is set for 0.40 to 0.44 volts peak-to-peak at TP19R. R7R can then be increased until the specified output current is reached or it may be set for a specified voltage at TP20R. Multiple motor systems may require specialized set ups. Refer to the system elementary for details.

Field Regulators. In addition to the normal field current regulator, several specialized field regulator configurations are possible. These regulators are used in combination with those already mentioned and are described below.

Basic Field Current Regulator. This regulator has no adjustments. Its response is critically damped and its time constant is determined by the field characteristics. If field current ripple is excessive (this is expected with fields with short time constants), add filtering in the form of a capacitor in parallel with R264 on the DMEC board. The value of this capacitor in mfd. = 0.02 / value of R264 in megohms. Ripple is excessive if it causes the output of the field current regulator to swing into its clamp. This can be seen with an oscilloscope placed on the output side of R264.

Speed Regulation (Tachometer or CEMF) by Field Current. This regulator has a limited range of control and its specialized set-up is detailed in the system elementary (when used).

Crossover Regulator. This regulator uses CEMF by the field current regulator in combination with the armature speed regulator configuration. Details of set-up depend upon the application and are covered in the system elementary (when used).

Additional Adjustments. The controls listed below are used in special applications. Set-up details are included in the system elementary.

R18R Rise/Droop. This is normally used to balance loading in load sharing applications. Not in use when jumper J48R is out.

R16R CEMF Ref. This is primarily used in crossover regulators to set the crossover point. For other applications it is usually set fully clockwise with jumper J31R out.

R10R Forcing. This is normally used for inertia compensation with a current regulator to maintain constant tension during acceleration and deceleration. The adjustment is application dependent, but in general the drive is run as a speed regulator (tachometer or CEMF) and the output of SRA (TP19R) is monitored while the speed is raised and lowered. R10R is increased until the output at TP19 remains constant during the speed changes.

R6E Compounding. This is used to provide the effect of a compound field motor and is used for special applications, typically with a Per Motor Module (PMM) in a load sharing arrangement. This is similar to the droop function but is accomplished in the speed regulator. R6E is rotated clockwise to achieve the desired effect. When not used, R6E should be set to 0 (fully counterclockwise).

R15E Field Preset. This is used in special applications to set the initial field current reference of the field regulator. Its set-up is described in the system elementary (when used). Jumper J18E is out when this function is not used. Note that the preset function may be implemented from external sources which do not require the use of R15E.

CHAPTER 4
TROUBLESHOOTING

Although the Siltron Power Conversion Module is extremely reliable, there will be times when malfunctions occur. To make finding and correcting such malfunctions easier, General Electric has designed into the Siltron PCM a number of maintenance and diagnostic aids:

- Status LEDs monitor specific drive conditions and faults such as sequencer and PCM status, AC and DC faults, overspeed conditions, and field loss conditions.
- A PROM-resident diagnostic routine within the Firing Circuit board is available to check the firing circuits and SCRs.
- Boards and major assemblies are identified by an assembly location plate attached to the panel.
- Adjustments can be done without disassembling the PCM. Adjustment potentiometers and test points are conveniently located on front of the electronics package.
- Most potentiometers are calibrated for ease in readjustment.
- Adjustments are performed on daughter boards. Replacement of a main board usually does not require readjustment of variable components.
- Two variable voltage sources are available on the Regulator board. These sources can be connected to test points and used to inject voltages into circuits for testing purposes. See Chapter 3 of this manual for more information.
- The PCM configuration can be easily changed by adding or removing jumpers. This simplifies both initial tune-up and troubleshooting.

SILTRON PCM TROUBLESHOOTING GUIDE

In addition to these hardware features, General Electric has provided a detailed troubleshooting guide, ***GEK-83815***, for the Siltron™ PCM. The troubleshooting guide consists of a series of flowcharts that guide you step-by-step through a fault analysis of the PCM. The flowcharts help you identify the failing LRU (Lowest Replaceable Unit). Examples of LRUs are relays, SCR cells, and printed circuit boards. The troubleshooting guide ***does not*** support isolation of faults to components on a printed circuit board.

The troubleshooting guide requires you to observe status LEDs, use the test meter to measure voltages, change board jumpers, and so forth. Generally, you will need only a multimeter to troubleshoot the Siltron PCM. The troubleshooting guide takes into account most of the options available on the equipment. If your PCM does not have an option specified in the flowcharts, simply bypass that step. If you don't know where a component or assembly is located, refer to Chapter 2.

TROUBLESHOOTING GUIDELINES

The troubleshooting guide mentioned above should be used only on PCMs that have been properly installed and have operated correctly for some period of time. We assume that the adjustment procedures in Chapter 3 were followed during installation and initial tune-up. If readjustments or jumper changes are made to a correctly operating PCM and then problems develop, go through the adjustment procedures in Chapter 3 first before using the troubleshooting flowcharts.

As you troubleshoot the PCM, be aware that some faults appear differently on an operating PCM than on a non-operating one. For instance, if a power supply fuse blows during operation, the PCM may signal an overcurrent, overvoltage, or field loss condition. The drive fault light would also be flashing. On a non-operating system, the failure would be obvious only when an operator attempts to start the drive: the drive would not run.

An important part of the troubleshooting test procedures is running the Firing Circuit board diagnostic routine. The description of the Firing Circuit board in Chapter 2 contains complete information on how to operate the routine.

RUNNING DRIVE VOLTAGE REGULATED FOR TEST PURPOSES

Very often it is desirable to run the motor open loop either for test purposes or during troubleshooting. It is sometimes very hard to judge the source of the problem because the regulator can either cause the problem or try to correct for it. By opening the loop, you can determine whether your problem is in the drive regulator, power converter, or motor.

During troubleshooting, when it is desirable to swing the bridge output voltage, is is recommmended that the regulator be re-configured as a manual input voltage regulator only. In this mode, the VRA gain pot must be set full CCW (minimum gain) to provide better (less sensitivity) control of the output in case of voltage feedback loss. This will help in locating such problems as bolted faults, broken wires, etc., without causing undue stress on the drive equipment.

Listed below is a quick and easy routine to do this. Refer to Figure 4-1 while reading the following steps:

1. Remove jumper J44R or J45R, which opens the reference from either the speed regulator or the current regulator.

2. If a reference comes into CTB25, disconnect this input.

3. Record the setting of pot R12R (VRA GAIN), then reset full CCW to provide better manual control.

4. Jumper test point TP15R to TP25R to inject a variable reference into the voltage regulator. Set R14R for zero volts at TP15R.

5. Start the drive through the Sequencer using the normal START/RUN pushbutton, which will unsuicide the regulator and pick up the drive contactor.

6. Slowly turn test pot R14R clockwise to increase armature voltage. (Counterclockwise decreases armature voltage and therefore speed.) If the drive is fully reversing, pot R14R can actually reverse motor rotation.

7. If the motor is now running correctly, the problem is not the firing circuit, bridge, or motor. Therefore, start troubleshooting at the tachometer output voltage, check speed feedback at TP17R, or replace the Regulator board.

8. If the motor is not running correctly, the problem could be in the firing circuit (in which case go to the diagnostic routine) or could be a motor problem.

9. After completing the test, place jumper J44R or J45R in the correct position as indicated in the elementary. Also, disconnect test jumper TP15R to TP25R, and reset pot R12R to the original setting.

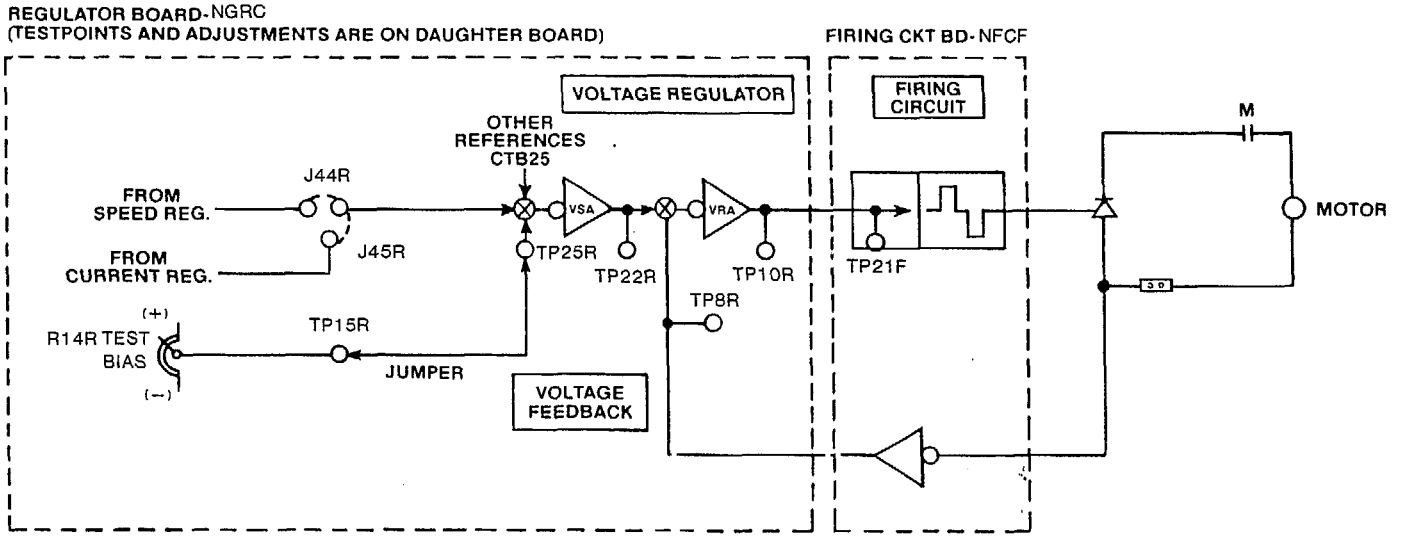


Figure 4-1.
Regulator Board and Firing Circuit Board Circuitry
For Running the Motor Open Loop

— NOTES —

APPENDIX A
POWER CONVERSION MODULE TESTPOINTS

This appendix lists board test points and their values in a four column table. Column 1 (Card TP) lists the test point number. Column 2 (Name) lists the name of the test point; this name is printed on the board next to the test point. Column 3 (Voltage (Design Value)) lists the theoretical voltage level at that test point under the conditions specified. Column 4 (Voltage (Test Value)) is left blank; after the PCM has been tuned up, check each test point and fill in this column with the correct value for your particular installation.

In the following tables, the abbreviation "RAV" stands for "Rated Armature Voltage." This value will be found on the motor nameplate.

REGULATOR BOARD TEST POINTS

Card TP	Name	Voltage (Design Value)	Voltage (Test Value)
TP1R	REF	Test input	_____
TP2R	RUNREF	+4V, +10V, or +15V at top line speed	_____
TP3R	IF	+10V at full field amps	_____
TP4R	ACC/DEC	0V at steady speed	_____
TP5R	FORCING	As set for process	_____
TP6R	SFA	6.79V at top speed	_____
TP7R	TIMED REF	+10V at max ref	_____
TP8R	VFB	+7V at RAV	_____
TP9R	SPEED ERROR	0V at steady speed	_____
TP10R	CONV REF	7.5 to 8V at RAV	_____
TP11R	CSA OUT	Depends on motor load	_____
TP12R	TEST CURR IN	Test current input	_____
TP13R	CURR REF	Depends on motor load	_____
TP14R	STEP OUT	Variable test voltage	_____
TP15R	TEST BIAS	Variable test voltage	_____
TP16R	ACOM	0 Volts	_____
TP17R	SFB	+6.79V at top speed	_____
TP18R	TST SR	Test speed input	_____
TP19R	SRA	≡9.3V regulation range	_____
TP20R	SPILLOVER	As set for process, typically 7.2V	_____
TP21R	CEMF	+5V at rated CEMF	_____
TP22R	VOLT SUM OUT	7.5 to 8V at RAV	_____
TP23R	CRA OUT	7.5 to 8V at RAV	_____
TP24R	CURR FDBK	+10V at suppression	_____
TP25R	TEST VOLT REF	+7V at RAV	_____
TP22F	TIA	+4.4V at maximum overload (IEC)	_____

FIRING CIRCUIT BOARD TEST POINTS

Card TP	Name	Voltage (Design Value)	Voltage (Test Value)
TP1F	P1F	+25V to +35V	
TP2F	P2F	+25V to +35V	
TP3F	P3F	+25V to +35V	
TP4F	N1F	+25V to +35V	
TP5F	N2F	+25V to +35V	
TP6F	N3F	+25V to +35V	
TP7F	P1R	+25V to +35V	
TP8F	P2R	+25V to +35V	
TP9F	P3R	+25V to +35V	
TP10F	N1R	+25V to +35V	
TP11F	N2R	+25V to +35V	
TP12F	N3R	+25V to +35V	
TP13F	0PC	+25V to +35V	
TP14F	0NC	+25V to +35V	
TP15F	C	+25V to +35V	
TP16F	WR	+25V to +35V	
TP17F	RD	+25V to +35V	
TP18F	INT	+25V to +35V	
TP21F	TVR	+4V at RAV	
TP22F	TIA	+4.4V at maximum overload	
TP27F	LCL	Logic "1" at inactive state	

FIELD EXCITER BOARD TEST POINTS

Card TP	Name	Voltage (Design Value)	Voltage (Test Value)
TP1E	FAULT	0V = Fault	
TP2E	CEMF	+5V at full CEMF	
TP3E	IF	+10V at full field	
TP4E	IA	+10V at suppression	
TP5E	FLD REF	+10V at full field	
TP6E	SPD	+5V at full speed	
TP7E	INST	+5V at normal operation	
TP8E	ACOM	0V	

APPENDIX B
SCR CELL AND CELL STACK REMOVAL AND REPLACEMENT

This appendix contains removal and replacement procedures for SCR cell stacks and individual SCR cells. The Low Horsepower Siltron PCM contains either stud-mounted or modular SCR packages, while the Medium and High Horsepower units use the PRESS-PAK SCR.

WARNING

Always remove all power from the PCM before working on the SCR stack. Use care in handling medium and high horsepower stacks because high clamping forces are used.

LOW HORSEPOWER, STUD-MOUNTED SCR CELLS

Stack Removal and Replacement (See Figure B-1)

Required Tools - Standard hand tools and soldering equipment.

1. After removing power, disconnect plugs and tab connections from the high voltage card. Place harness wires out of the way.
2. Push the retainer tabs outward and remove the high voltage card. This provides access to the modular SCR cells.
3. Carefully note connection points. Unsolder the gate lead at the SCR.
4. Disconnect the cathode lead at the SCR and, for the upper heat sink, the lead from the lower SCR to upper heat sink.
5. Lay wires out of the way. Remove the heat sink mounting screws and lift out the heat sink assembly.
6. Reassemble in reverse order.

SCR Replacement

Required Tools and Materials - Torque wrench (30, 150, 300 inch-pound ranges) and GE G641 Silicon grease (or equivalent)

1. Note the orientation of the cathode connection tab with respect to the heat sink and remove the old SCR. Use care - do not damage the heat sink surfaces.
2. Clean the heat sink surfaces and inspect for rough spots and burrs. Smooth up or replace any parts with bad surfaces.
3. Apply a very thin coat of G641 grease to the heat sink/SCR interface surfaces. Wipe off excess grease.
4. Mount the SCR cell, orienting the connection tab as noted in Step 1.

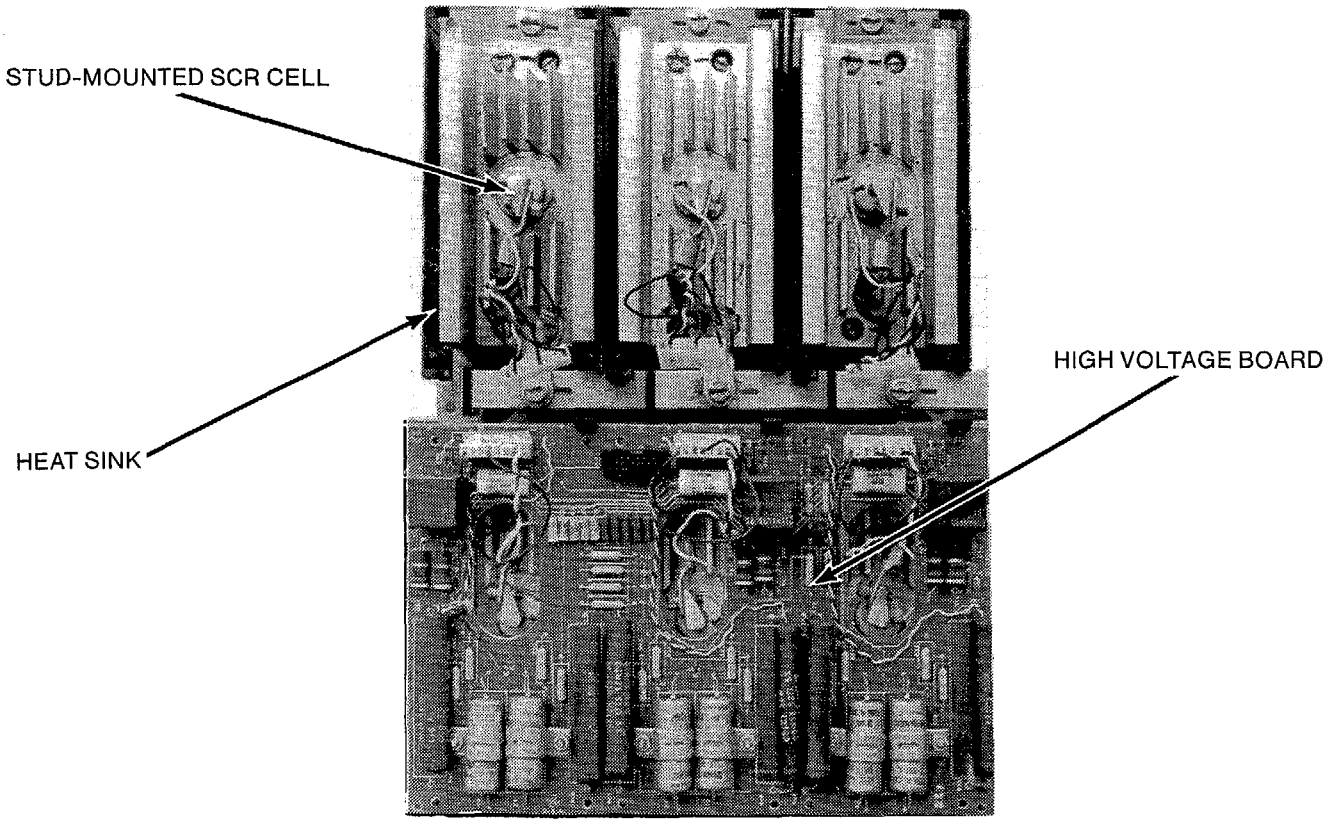


Figure B-1.
Low Horsepower, Stud-Mounted SCR Cell Assembly

5. Torque per the following table. Use care and do not stress the connection tabs.

SCR Number	Torque (Inch-Pounds)
68A7218- - -	30 +5, -0
68A7212- - -	140 ±5
68A7636- - -	280 ±10

6. Mount the stack and make connections.

7. Check all connections before applying power.

LOW HORSEPOWER, MODULAR SCR CELLS

Stack Removal and Replacement (See Figure B-2)

The stack assembly does not have to be removed to service modular SCRs.

SCR Replacement

Required Tools - Standard hand tools (including phillips-head screwdriver), torque wrench (35 inch-pound range), and GE G641 silicone grease or equivalent.

1. After removing power, disconnect plugs and tab connections from the high voltage card. Place the harness out of the way.

2. Push the retainer tabs outward and remove the high voltage card. This provides access to the SCR modules.

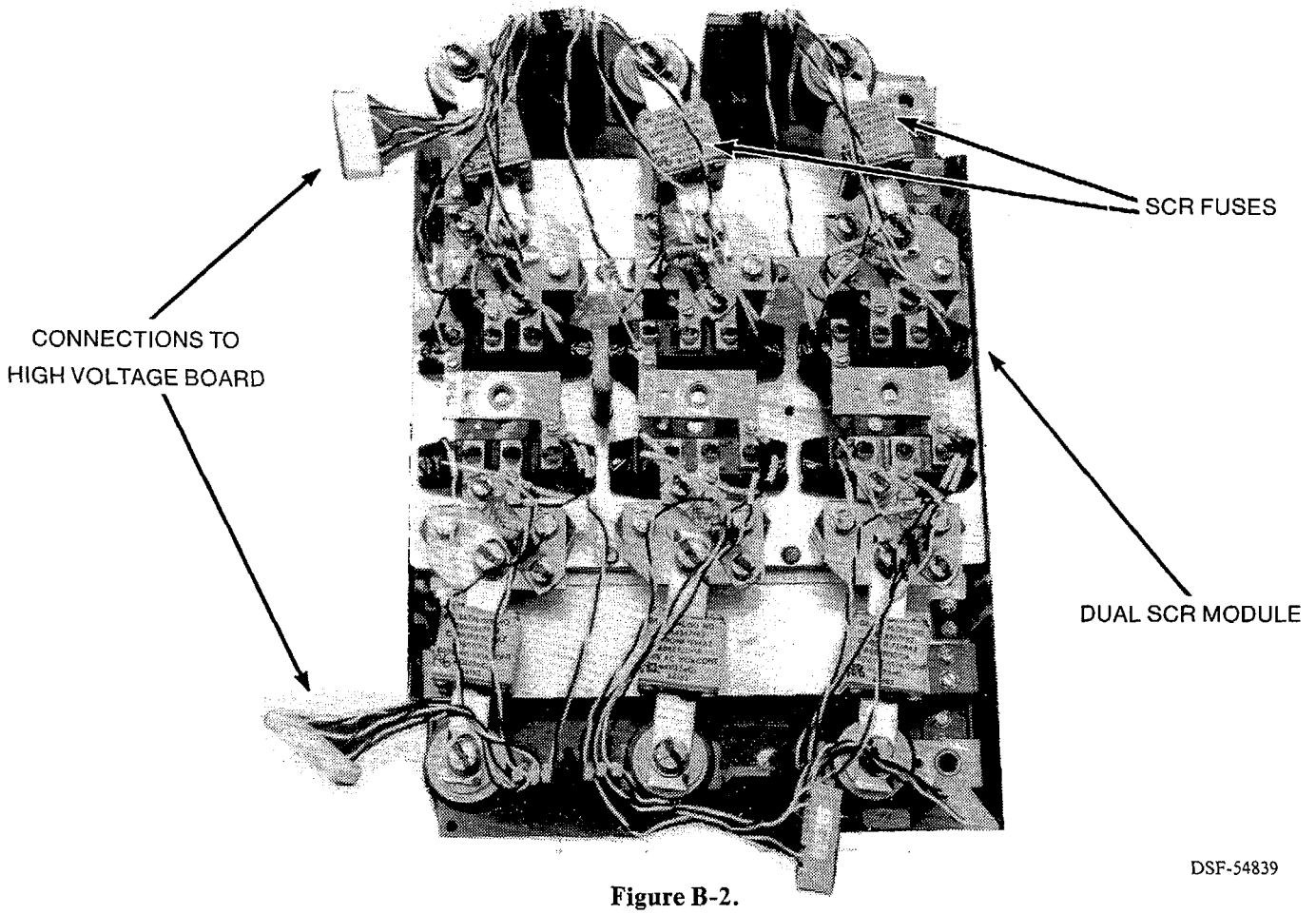


Figure B-2.
Low Horsepower, Modular SCR Cell Assembly

3. Note orientation of the SCR module and the location of its connections.
4. Remove both gate tab connections, noting the terminal designation.
5. Remove phillips-head screws (3) and power connections. Use care with the hardware - these screws have MS metric caps. Do not try to replace with any other type.
6. Remove mounting screws (2) and the SCR module.
7. Clean the mounting surfaces of the heat sink and new SCR module; check for smoothness.
8. Apply a thin coat of G641 grease to the SCR and heat sink surfaces.
9. Mount the new SCR module using the orientation noted in step 3.
10. Torque the mounting screws to 25-35 inch-pounds and remove excess grease.
11. Attach the power connections and gate tabs.
12. Mount and connect the high voltage card.
13. Check all connections before applying power.

MEDIUM HORSEPOWER, PRESS-PAK SCR CELLS

Fuse Removal and Replacement

Required Tools -

1 9/16" combination 12-point box and open end wrench (a 9/16" ratcheting box-end wrench similar to the Craftsman 9HT43683 is an alternate tool which simplifies the job)

1 Nut driver socket set (11/32", 3/8" socket sizes)

1 Screw driver (for early models)

1. Make certain that all power is disconnected from the panel.

2. While not absolutely necessary, removing the high voltage cards from the stack assembly can simplify fuse removal. The high voltage cards can be removed by using the following procedure:

    a. Disconnect the plugs and tab connections from the high voltage card.

    b. Press the retainer tabs outward and lift the high voltage card out of the retainer.

    c. When reassembling some models make sure the pins on the stack line up with the holes in the high voltage card. (NOTE: On early models the close fit between the retainer pin and card mounting hole can cause difficulties. Enlarging the card mounting holes can simplify removal and assembly.)

3. Loosen and remove the fuse mounting bolts.

4. Lift out the fuse. (Refer to the next step for removing the far right-hand fuse).

5. If the module uses a physically larger fuse size, it is desirable to remove the adjacent fuse and plastic barriers to gain better access to the far right-hand fuse. Use the appropriate nut driver. (NOTE: On early models, the plastic barrier was mounted with a screw head at the front. Assembly and future disassembly will be facilitated by placing the screw head inside and nutting it to the plastic base to form a stud mount for the plastic barrier. Then attach the barrier with a second nut.)

6. Reverse the above procedure when mounting a new fuse. Be sure to replace the plastic barrier between fuses if it has been removed.

Stack Removal and Replacement (See Figure B-3)

Required Tools - Same as for fuse removal, plus 1 socket set (3/8" or 1/2" drive) with 9/16" socket and 9" extension and a pair of diagonal cutting pliers

1. Remove the high voltage card. See fuse removal steps 1 and 2 above.

2. Remove both fuses on the stack.

3. Locate the SCR gate leads (red and white leads which were plugged to high voltage card) at the top and bottom of stack. Carefully wrap or tuck these leads into the SCR clamps on the stack.

4. Locate the 2 wires (black) which attach to the stack on the lower busses. Follow them to the red-orange connector. Disconnect and tuck wires into the stack clamps. (NOTE: In some cases this connector is tied in with the panel wire harness. If this is the case, carefully cut the cord ties until the wires/connector from the stack are free. When reassembling, it is not necessary to re-tie these wires.)

5. Using the 9/16" socket with extension remove the bolt adjacent to the stack mounting nut on the upper stack bus.

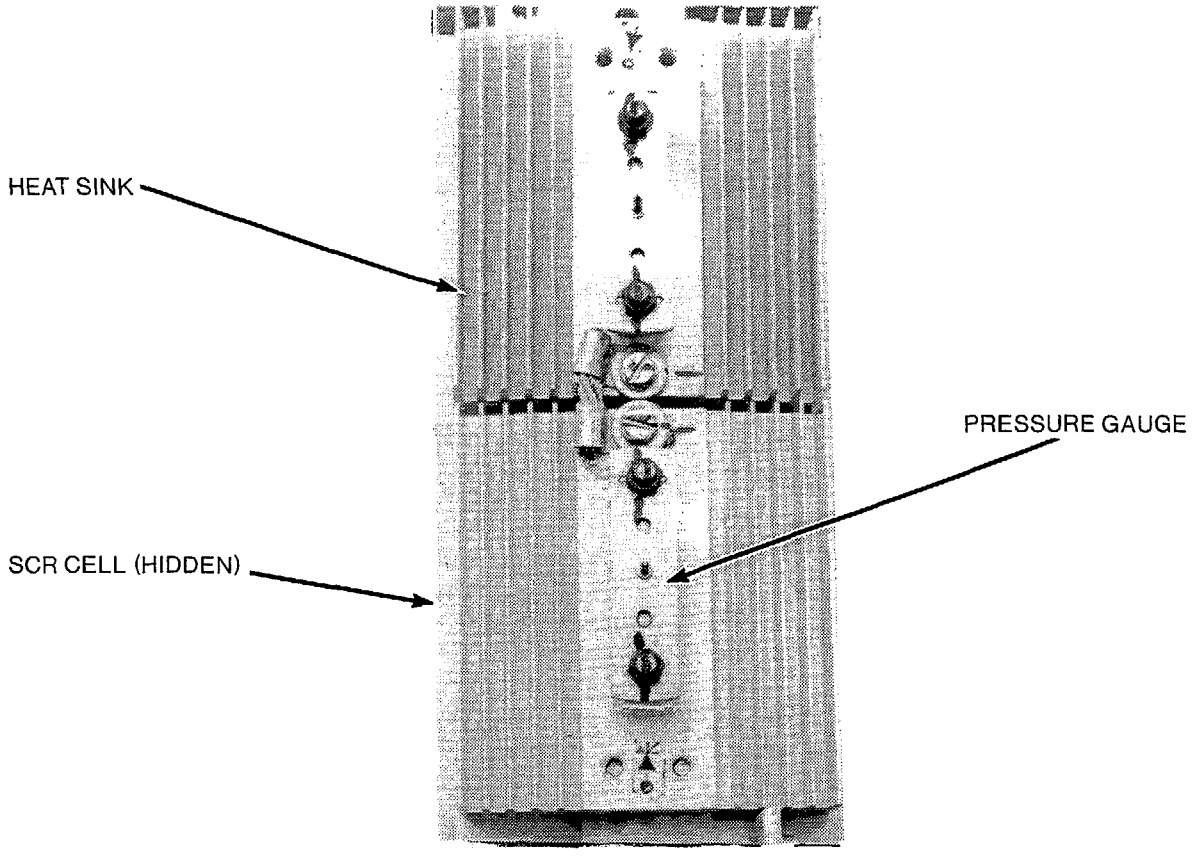


Figure B-3.
Medium Horsepower, Press-Pak SCR Cell Assembly

6. Using the same socket, remove the upper and lower stack mounting nuts and hardware.

7. Check that wires from stack are tucked out of the way. Also tuck lower panel harness out of the way.

8. When all wires are clear, grasp the stack at bottom and top and gently slide off of the studs. Lift the stack out of mount, using a gentle rocking motion. Outer stacks must be turned slightly inward as they clear the mount.

9. Prior to reassembly, be sure to check the stack wires to make sure they are out of the way. Also check the lower harness for clearance and check near the upper bus to make sure that the upper wire harness is clear of the mounting surface. Tuck or tie wires out of the way as required.

10. To reassemble, reverse the above removal procedure.

11. Carefully check the work before applying power.

    a. Check connections to the high voltage card.

    b. Check that all wires are clear and that none are pinched.

    c. Check that the bus and fuse connections are tight.

PRESS-PAK SCR Cell Removal and Replacement

Since this type of SCR is used in both medium and large HP stacks, refer to the following section for details.

HIGH HORSEPOWER, PRESS-PAK SCR CELLS

Stack Removal and Replacement

Required Tools - Standard hand tools, 3/8" or 1/2" drive socket set with 9/16" socket and 9" extension.

1. Remove the upper insulating panels by loosening the screws and sliding the panel toward the center until the keyhole permits removal.

2. Although not absolutely required, removing the cross bracket which supports the upper insulating panels can give additional working space. Loosen the screws and lift up and out to remove.

3. Remove the stack insulating covers by loosening the clamp screws at each corner on each panel and pivoting the clamp out of the way. Then lift out the panel.

4. Remove the fuse bolts and fuses, placing the fuse bus (with sensor wire connection) out of the way. Use care with the bus link to avoid wire breakage.

5. Disconnect the plugs from the stack and tuck the wires out of the way.

6. While supporting the stack, remove the upper and lower bolts and lift the stack out of the unit.

7. Reassemble in reverse order. Be sure to reassemble all insulating panels or cooling efficiency may be reduced.

PRESS-PAK SCR Cell Removal and Replacement

Required Tools - 9/16" box-end or socket wrench, SF1154 silicone fluid, and terminals. (Terminals and tools are available from Drive Systems Renewal Parts Sales in Salem, Virginia as part number 278A4780G-)

1. After removing the stack assembly from the PCM, place it on a clean, dry work surface. Orient with the clamp studs up. Handle carefully since high clamping forces are used. See Figure B-5.

2. Before loosening the clamping nuts, carefully observe and record the pressure gauge indicator. When working with several assemblies, use care not to intermix gauges or any other parts. See Figure B-6

3. Alternately loosen the clamping nuts about 1/6 turn at a time until the pressure is relieved, then remove the hardware, clamp and gauge. To ensure accuracy of reassembly, record the order of assembly and orientation of parts.

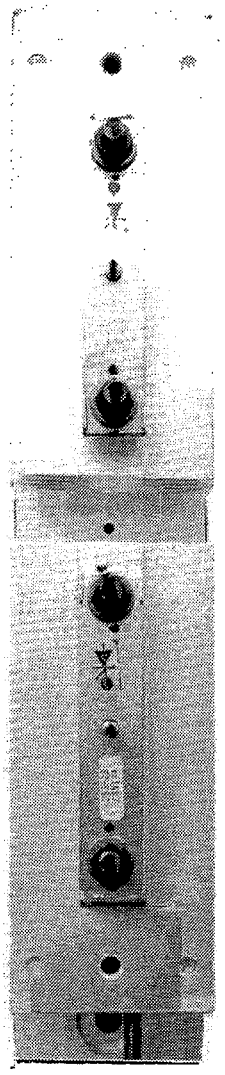
4. Carefully remove the upper heat sink and SCR. If required, remove the center heat sink and lower SCR. Record the location and orientation of SCR(s) and parts. If required, identify the upper and lower SCRs for termination. Handle heat sinks carefully to avoid scratching or denting the contact surfaces.

5. If required, remove the SCR connector from the bracket on heat sink.

6. Terminate the new SCR(s) using same the lead length and terminal type as the old unit. Select terminals from the table below:

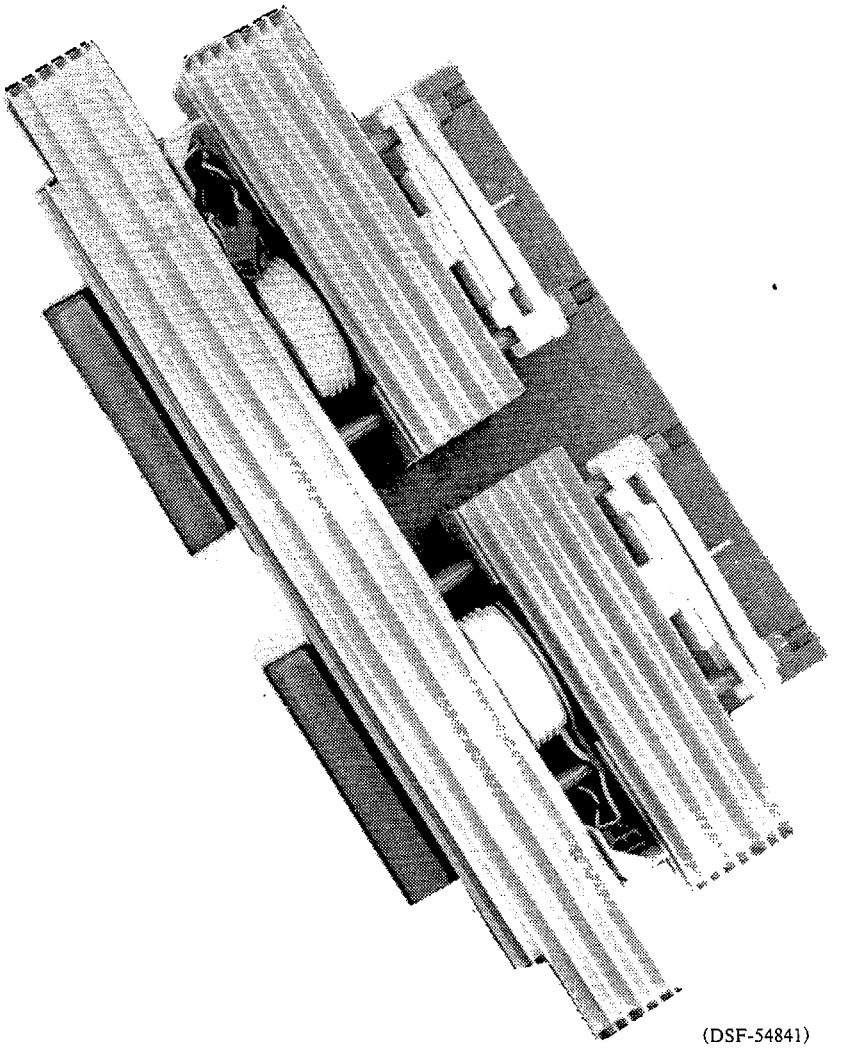
MEDIUM HORSEPOWER CELL CONNECTIONS
Use terminal 218A4602P1 and connector body 218A4602P2
Lower cell (PD/PH Connection)
Red cell lead (C) to pin (1)
White cell lead (G) to pin (2)
Upper cell (PE/PK Connection)
Red cell lead (C) to pin (7)
White cell lead (G) to pin (8)

HIGH HORSEPOWER CELL CONNECTIONS
Use terminal 218A4609P4 and connector body 218A4629P2
Red cell lead (C) to pin 2
White Cell lead (G) to pin 1



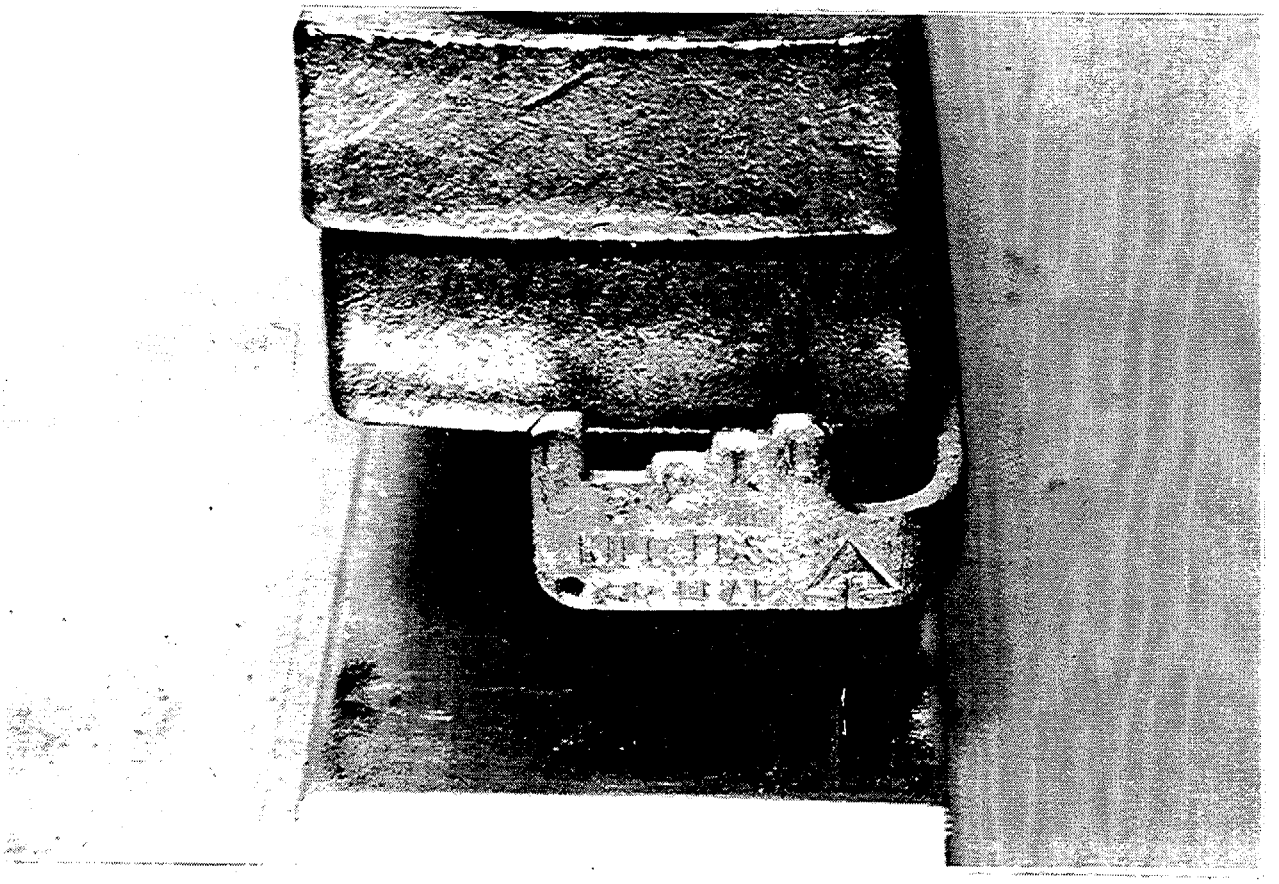
(DSF-54840)

Figure B-4.
High Horsepower, Press-Pak SCR Cell Assembly



(DSF-54841)

Figure B-5.
Side View of High Horsepower, Press-Pak SCR Cell Assembly
(Medium Horsepower SCR Cell Assembly is Similar)



(DSF-53704)

Figure B-6.
Detail of Press-Pak SCR Cell Assembly Pressure Gauge

7. Release the old SCR leads from the connector body, noting lead color and pin location. Also note whether this is an upper or lower SCR.

8. Slide the new SCR leads into the connector body until the pin locks in place. Do not mix upper and lower units since terminations are to different pin locations.

9. Using a clean, lint-free cloth, lightly wipe the contact surfaces of the heat sinks to remove all dirt and grit. Wipe but do not overwipe.

10. Wipe the SCR cell contact surfaces to remove all dirt and grit.

11. Lubricate both SCR cell contact surfaces with a thin coat of GE SF1154 silicone fluid.

12. Install the lower SCR cell while observing original orientation. Position on the pin with the leads approximately 30° off center to clear the clamping rod. If required, attach the connector bracket and set the central heat sink in place. Rotate the SCR cell slightly back and forth to seat.

13. Install the upper SCR cell (if used) and upper heat sink, observing orientation. Again, rotate slightly to seat and position leads as noted in step 12 above.

14. Replace the gauge, clamps and hardware. Observe proper orientation and order of assembly. (NOTE: The gauge is located nearest the heat sink. If swivel washers are used, the curved side goes toward the countersinks in the clamp-spring bar. Check to see that the SCR cells are seated properly by rotating them slightly.
15. Turn nuts down evenly finger tight, making certain that the distances from each nut to the top of the stud are equal within one thread (1/24″).

16. Alternately tighten clamping nuts 1/6 turn each until the gauge has the same reading as before disassembly. Use care since high clamping forces are being applied.

17. If the gauge setting is lost, use this alternate method of clamping:
    a. Follow the initial procedure to step 15.

    b. Take up all slack in the clamp by evenly tightening each clamp nut to 12 ±2 inch-pounds.

    c. Tighten each nut by the full or fractional turns shown in the table below. Make the required rotation on each nut in one smooth operation.

MEDIUM HORSEPOWER CELL - All 33mm cells - 2000 lbs - Turn each nut 1 full turn.

HIGH HORSEPOWER CELLS

Stack Model No. DS3828CS —	
02A, 03A, 04A	33mm cell - 2000 lbs - Turn each nut 1/2 turn.
06A, 08A, 09A	40mm cell - 4000 lbs - Turn each nut 7/8 turn.
11, 12, 13, 15, 16, 17, 19, 23	53mm cell - 6000 lbs - Turn each nut 1 1/4 turns.

18. Carefully check all work and remount the stack.